



Pearson New International Edition

Using Multivariate Statistics

Barbara G. Tabachnick Linda S. Fidell

Sixth Edition



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APPENDIX: Statistical Tables

From Appendix C of *Using Multivariate Statistics*, Sixth Edition. Barbara G. Tabachnick, Linda S. Fidell.
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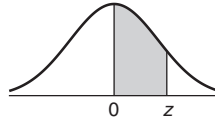
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Statistical Tables

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- Table 2** Critical Values of the t Distribution for $\alpha = .05$ and $.01$, Two-Tailed Test
- Table 3** Critical Values of the F Distribution
- Table 4** Critical Values of Chi Square (χ^2)
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- Table 6** Critical Values for $F_{\max} (S^2_{\max}/S^2_{\min})$ Distribution for $\alpha = .05$ and $.01$

APPENDIX: Statistical Tables

TABLE 1 Normal Curve Areas



<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990

Source: Adapted/abridged from A. Hald, *Statistical Tables and Formulas*, Table 1. John Wiley, 1952.

TABLE 2 Critical Values of the t Distribution for $\alpha = .05$ and $.01$, Two-Tailed Test

Degrees of Freedom	.05	.01
1	12.706	63.657
2	4.303	9.925
3	3.182	5.841
4	2.776	4.604
5	2.571	4.032
6	2.447	3.707
7	2.365	3.499
8	2.306	3.355
9	2.262	3.250
10	2.228	3.169
11	2.201	3.106
12	2.179	3.055
13	2.160	3.012
14	2.145	2.977
15	2.131	2.947
16	2.120	2.921
17	2.110	2.898
18	2.101	2.878
19	2.093	2.861
20	2.086	2.845
21	2.080	2.831
22	2.074	2.819
23	2.069	2.807
24	2.064	2.797
25	2.060	2.787
26	2.056	2.779
27	2.052	2.771
28	2.048	2.763
29	2.045	2.756
30	2.042	2.750
40	2.021	2.704
60	2.000	2.660
120	1.980	2.617
∞	1.960	2.576

Source: Adapted from Table 9 in *Biometrika Tables for Statisticians*, vol. 1, 3d ed., edited by E. S. Pearson and H. O. Hartley (New York: Cambridge University Press, 1958).

APPENDIX: Statistical Tables

TABLE 3 Critical Values of the F Distribution

df_1 df_2		1	2	3	4	5	6	8	12	24	∞
1	0.1%	405284	500000	540379	562500	576405	585937	598144	610667	623497	636619
	0.5%	16211	20000	21615	22500	23056	23437	23925	24426	24940	25465
	1%	4052	4999	5403	5625	5764	5859	5981	6106	6234	6366
	2.5%	647.79	799.50	864.16	899.58	921.85	937.11	956.66	976.71	997.25	1018.30
	5%	161.45	199.50	215.71	224.58	230.16	233.99	238.88	243.91	249.05	254.32
	10%	39.86	49.50	53.59	55.83	57.24	58.20	59.44	60.70	62.00	63.33
2	0.1	998.5	999.0	999.2	999.2	999.3	999.3	999.4	999.4	999.5	999.5
	0.5	198.50	199.00	199.17	199.25	199.30	199.33	199.37	199.42	199.46	199.51
	1	98.49	99.00	99.17	99.25	99.30	99.33	99.36	99.42	99.46	99.50
	2.5	38.51	39.00	39.17	39.25	39.30	39.33	39.37	39.42	39.46	39.50
	5	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.41	19.45	19.50
	10	8.53	9.00	9.16	9.24	9.29	9.33	9.37	9.41	9.45	9.49
3	0.1	167.5	148.5	141.1	137.1	134.6	132.8	130.6	128.3	125.9	123.5
	0.5	55.55	49.80	47.47	46.20	45.39	44.84	44.13	43.39	42.62	41.83
	1	34.12	30.81	29.46	28.71	28.24	27.91	27.49	27.05	26.60	26.12
	2.5	17.44	16.04	15.44	15.10	14.89	14.74	14.54	14.34	14.12	13.90
	5	10.13	9.55	9.28	9.12	9.01	8.94	8.84	8.74	8.64	8.53
	10	5.54	5.46	5.39	5.34	5.31	5.28	5.25	5.22	5.18	5.13
4	0.1	74.14	61.25	56.18	53.44	51.71	50.53	49.00	47.41	45.77	44.05
	0.5	31.33	26.28	24.26	23.16	22.46	21.98	21.35	20.71	20.03	19.33
	1	21.20	18.00	16.69	15.98	15.52	15.21	14.80	14.37	13.93	13.46
	2.5	12.22	10.65	9.98	9.60	9.36	9.20	8.98	8.75	8.51	8.26
	5	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.91	5.77	5.63
	10	4.54	4.32	4.19	4.11	4.05	4.01	3.95	3.90	3.83	3.76
5	0.1	47.04	36.61	33.20	31.09	29.75	28.84	27.64	26.42	25.14	23.78
	0.5	22.79	18.31	16.53	15.56	14.94	14.51	13.96	13.38	12.78	12.14
	1	16.26	13.27	12.06	11.39	10.97	10.67	10.29	9.89	9.47	9.02
	2.5	10.01	8.43	7.76	7.39	7.15	6.98	6.76	6.52	6.28	6.02
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.68	4.53	4.36
	10	4.06	3.78	3.62	3.52	3.45	3.40	3.34	3.27	3.19	3.10
6	0.1	35.51	27.00	23.70	21.90	20.81	20.03	19.03	17.99	16.89	15.75
	0.5	18.64	14.54	12.92	12.03	11.46	11.07	10.57	10.03	9.47	8.88
	1	13.74	10.92	9.78	9.15	8.75	8.47	8.10	7.72	7.31	6.88
	2.5	8.81	7.26	6.60	6.23	5.99	5.82	5.60	5.37	5.12	4.85
	5	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.00	3.84	3.67
	10	3.78	3.46	3.29	3.18	3.11	3.05	2.98	2.90	2.82	2.72
7	0.1	29.22	21.69	18.77	17.19	16.21	15.52	14.63	13.71	12.73	11.69
	0.5	16.24	12.40	10.88	10.05	9.52	9.16	8.68	8.18	7.65	7.08
	1	12.25	9.55	8.45	7.85	7.46	7.19	6.84	6.47	6.07	5.65
	2.5	8.07	6.54	5.89	5.52	5.29	5.12	4.90	4.67	4.42	4.14
	5	5.59	4.74	4.35	4.12	3.97	3.87	3.73	3.57	3.41	3.23
	10	3.59	3.26	3.07	2.96	2.88	2.83	2.75	2.67	2.58	2.47
8	0.1	25.42	18.49	15.83	14.39	13.49	12.86	12.04	11.19	10.30	9.34
	0.5	14.69	11.04	9.60	8.81	8.30	7.95	7.50	7.01	6.50	5.95

(continued)

APPENDIX: Statistical Tables

TABLE 3 Continued

df_1 df_2		1	2	3	4	5	6	8	12	24	∞
9	1%	11.26	8.65	7.59	7.01	6.63	6.37	6.03	5.67	5.28	4.86
	2.5%	7.57	6.06	5.42	5.05	4.82	4.65	4.43	4.20	3.95	3.67
	5%	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.28	3.12	2.93
	10%	3.46	3.11	2.92	2.81	2.73	2.67	2.59	2.50	2.40	2.29
	0.1	22.86	16.39	13.90	12.56	11.71	11.13	10.37	9.57	8.72	7.81
	0.5	13.61	10.11	8.72	7.96	7.47	7.13	6.69	6.23	5.73	5.19
	1	10.56	8.02	6.99	6.42	6.06	5.80	5.47	5.11	4.73	4.31
	2.5	7.21	5.71	5.08	4.72	4.48	4.32	4.10	3.87	3.61	3.33
	5	5.12	4.26	3.86	3.63	3.48	3.37	3.23	3.07	2.90	2.71
	10	3.36	3.01	2.81	2.69	2.61	2.55	2.47	2.38	2.28	2.16
10	0.1	21.04	14.91	12.55	11.28	10.48	9.92	9.20	8.45	7.64	6.76
	0.5	12.83	8.08	7.34	6.87	6.54	6.12	5.66	5.17	4.64	
	1	10.04	7.56	6.55	5.99	5.64	5.39	5.06	4.71	4.33	3.91
	2.5	6.94	5.46	4.83	4.47	4.24	4.07	3.85	3.62	3.37	3.08
	5	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.91	2.74	2.54
11	10	3.28	2.92	2.73	2.61	2.52	2.46	2.38	2.28	2.18	2.06
	0.1	19.69	13.81	11.56	10.35	9.58	9.05	8.35	7.63	6.85	6.00
	0.5	12.23	8.91	7.60	6.88	6.42	6.10	5.68	5.24	4.76	4.23
	1	9.65	7.20	6.22	5.67	5.32	5.07	4.74	4.40	4.02	3.60
	2.5	6.72	5.26	4.63	4.28	4.04	4.88	3.66	3.43	3.17	2.88
12	5	4.84	3.98	3.59	3.36	3.20	3.09	2.95	2.79	2.61	2.40
	10	3.23	2.86	2.66	2.54	2.45	2.39	2.30	2.21	2.10	1.97
	0.1	18.64	12.97	10.80	9.63	8.89	8.38	7.71	7.00	6.25	5.42
	0.5	11.75	8.51	7.23	6.52	6.07	5.76	5.35	4.91	4.43	3.90
	1	9.33	6.93	5.95	5.41	5.06	4.82	4.50	4.16	3.78	3.36
13	2.5	6.55	5.10	4.47	4.12	3.89	3.73	3.51	3.28	3.02	2.72
	5	4.75	3.88	3.49	3.26	3.11	3.00	2.85	2.69	2.50	2.30
	10	3.18	2.81	2.61	2.48	2.39	2.33	2.24	2.15	2.04	1.90
	0.1	17.81	12.31	10.21	9.07	8.35	7.86	7.21	6.52	5.78	4.97
	0.5	11.37	8.19	6.93	6.23	5.79	5.48	5.08	4.64	4.17	3.65
14	1	9.07	6.70	5.74	5.20	4.86	4.62	4.30	3.96	3.59	3.16
	2.5	6.41	4.97	4.35	4.00	3.77	3.60	3.39	3.15	2.89	2.60
	5	4.67	3.80	3.41	3.18	3.02	2.92	2.77	2.60	2.42	2.21
	10	3.14	2.76	2.56	2.43	2.35	2.28	2.20	2.10	1.98	1.85
	0.1	17.14	11.78	9.73	8.62	7.92	7.43	6.80	6.13	5.41	4.60
15	0.5	11.06	7.92	6.68	6.00	5.56	5.26	4.86	4.43	3.96	3.44
	1	8.86	6.51	5.56	5.03	4.69	4.46	4.14	3.80	3.43	3.00
	2.5	6.30	4.86	4.24	3.89	3.66	3.50	3.27	3.05	2.79	2.49
	5	4.60	3.74	3.34	3.11	2.96	2.85	2.70	2.53	2.35	2.13
	10	3.10	2.73	2.52	2.39	2.31	2.24	2.15	2.05	1.94	1.80
15	0.1	16.59	11.34	9.30	8.25	7.57	7.09	6.47	5.81	5.10	4.31
	0.5	10.80	7.70	6.48	5.80	5.37	5.07	4.67	4.25	3.79	3.26
	1	8.68	6.36	5.42	4.89	4.56	4.32	4.00	3.67	3.29	2.87
	2.5	8.20	4.77	4.15	3.80	3.58	3.41	3.20	2.96	2.70	2.40

APPENDIX: Statistical Tables

TABLE 3 Continued

df_1 df_2		1	2	3	4	5	6	8	12	24	∞
16	5%	4.54	3.80	3.29	3.06	2.90	2.79	2.64	2.48	2.29	2.07
	10%	3.07	2.70	2.49	2.36	2.27	2.21	2.12	2.02	1.90	1.76
	0.1	16.12	10.97	9.00	7.94	7.27	6.81	6.19	5.55	4.85	4.06
	0.5	10.58	7.51	6.30	5.64	5.21	4.91	4.52	4.10	3.64	3.11
	1	8.53	6.23	5.29	4.77	4.44	4.20	3.89	3.55	3.18	2.75
	2.5	6.12	4.69	4.08	3.73	3.50	3.34	3.12	2.89	2.63	2.32
	5	4.49	3.63	3.24	3.01	2.85	2.74	2.59	2.42	2.24	2.01
17	10	3.05	2.67	2.46	2.33	2.24	2.18	2.09	1.99	1.87	1.72
	0.1	15.72	10.66	8.73	7.68	7.02	6.56	5.96	5.32	4.63	3.85
	0.5	10.38	7.35	6.16	5.50	5.07	4.78	4.39	3.97	3.51	2.98
	1	8.40	6.11	5.18	4.67	4.34	4.10	3.79	3.45	3.08	2.65
	2.5	6.04	4.62	4.01	3.66	3.44	3.28	3.06	2.82	2.56	2.25
	5	4.45	3.59	3.20	2.96	2.81	2.70	2.55	2.38	2.19	1.96
	10	3.03	2.64	2.44	2.31	2.22	2.15	2.06	1.96	1.84	1.69
18	0.1	15.38	10.39	8.49	7.46	6.81	6.35	5.76	5.13	4.45	3.67
	0.5	10.22	7.21	6.03	5.37	4.96	4.66	4.28	3.86	3.40	2.87
	1	8.28	6.01	5.09	4.58	4.25	4.01	3.71	3.37	3.00	2.57
	2.5	5.98	4.56	3.95	3.61	3.38	3.22	3.01	2.77	2.50	2.19
	5	4.41	3.55	3.16	2.93	2.77	2.66	2.51	2.34	2.15	1.92
	10	3.01	2.62	2.42	2.29	2.20	2.13	2.04	1.93	1.81	1.66
	0.1	15.08	10.16	8.28	7.26	6.61	6.18	5.59	4.97	4.29	3.52
19	0.5	10.07	7.09	5.92	5.27	4.85	4.56	4.18	3.76	3.31	2.78
	1	8.18	5.93	5.01	4.50	4.17	3.94	3.63	3.30	2.92	2.49
	2.5	5.92	4.51	3.90	3.56	3.33	3.17	2.96	2.72	2.45	2.13
	5	4.38	3.52	3.13	2.90	2.74	2.63	2.48	2.31	2.11	1.88
	10	2.99	2.61	2.40	2.27	2.18	2.11	2.02	1.91	1.79	1.63
	0.1	14.82	9.95	8.10	7.10	6.46	6.02	5.44	4.82	4.15	3.38
	0.5	9.94	6.99	5.82	5.17	4.76	4.47	4.09	3.68	3.22	2.69
20	1	8.10	5.85	4.94	4.43	4.10	3.87	3.56	3.23	2.86	2.42
	2.5	5.87	4.46	3.86	3.51	3.29	3.13	2.91	2.68	2.41	2.09
	5	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.28	2.08	1.84
	10	2.97	2.59	2.38	2.25	2.16	2.09	2.00	1.89	1.77	1.61
	0.1	14.59	9.77	7.94	6.95	6.32	5.88	5.31	4.70	4.03	3.26
	0.5	9.83	6.89	5.73	5.09	4.68	4.39	4.01	3.60	3.15	2.61
	1	8.02	5.78	4.87	4.37	4.04	3.81	3.51	3.17	2.80	2.36
21	2.5	5.83	4.42	3.82	3.48	3.25	3.09	2.87	2.64	2.37	2.04
	5	4.32	3.47	3.07	2.84	2.68	2.57	2.42	2.25	2.05	1.81
	10	2.96	2.57	2.36	2.23	2.14	2.08	1.98	1.88	1.75	1.59
	0.1	14.38	9.61	7.80	6.81	6.19	5.76	5.19	4.58	3.92	3.15
	0.5	9.73	6.81	5.65	5.02	4.61	4.32	3.94	3.54	3.08	2.55
	1	7.94	5.72	4.82	4.31	3.99	3.76	3.45	3.12	2.75	2.31
	2.5	5.79	4.38	3.78	3.44	3.22	3.05	2.84	2.60	2.33	2.00
22	5	4.30	3.44	3.05	2.82	2.66	2.55	2.40	2.23	2.03	1.78
	10	2.95	2.56	2.35	2.22	2.13	2.06	1.97	1.86	1.73	1.57

(continued)

APPENDIX: Statistical Tables

TABLE 3 Continued

df_1 df_2		1	2	3	4	5	6	8	12	24	∞
23	0.1%	14.19	9.47	7.67	6.69	6.08	5.65	5.09	4.48	3.82	3.05
	0.5%	9.63	6.73	5.58	4.95	4.54	4.26	3.88	3.47	3.02	2.48
	1%	7.88	5.66	4.76	4.26	3.94	3.71	3.41	3.07	2.70	2.26
	2.5%	5.75	4.35	3.75	3.41	3.18	3.02	2.81	2.57	2.30	1.97
	5%	4.28	3.42	3.03	2.80	2.64	2.53	2.38	2.20	2.00	1.76
	10%	2.94	2.55	2.34	2.21	2.11	2.05	1.95	1.84	1.72	1.55
24	0.1	14.03	9.34	7.55	6.59	5.98	5.55	4.00	4.00	3.74	2.97
	0.5	9.55	6.66	5.52	4.89	4.49	4.20	3.83	3.42	2.97	2.43
	1	7.82	5.61	4.72	4.22	3.90	3.67	3.36	3.03	2.66	2.21
	2.5	5.72	4.32	3.72	3.38	3.15	2.99	2.78	2.54	2.27	1.94
	5	4.26	3.40	3.01	2.78	2.62	2.51	2.36	2.18	1.98	1.73
	1.0	2.93	2.54	2.33	2.19	2.10	2.04	1.94	1.83	1.70	1.53
25	0.1	13.88	9.22	7.45	6.49	5.88	5.46	4.91	4.31	3.66	2.89
	0.5	9.48	6.60	5.46	4.84	4.43	4.15	3.78	3.37	2.92	2.38
	1	7.77	5.57	4.68	4.18	3.86	3.63	3.32	2.99	2.62	2.17
	2.5	5.69	4.29	3.69	3.35	3.13	2.97	2.75	2.51	2.24	1.91
	5	4.24	3.38	2.99	2.76	2.60	2.49	2.34	2.16	1.96	1.71
	1.0	2.92	2.53	2.32	2.18	2.09	2.02	1.93	1.82	1.69	1.52
26	0.1	13.74	9.12	7.36	6.41	5.80	5.38	4.83	4.24	3.59	2.82
	0.5	9.41	6.54	5.41	4.79	4.38	4.10	3.73	3.33	2.87	2.33
	1	7.72	5.53	4.64	4.14	3.82	3.59	3.29	2.96	2.58	2.13
	2.5	5.66	4.27	3.67	3.33	3.10	2.94	2.73	2.49	2.22	1.88
	5	4.22	3.37	2.98	2.74	2.59	2.47	2.32	2.15	1.95	1.69
	10	2.91	2.52	2.31	2.17	2.08	2.01	1.92	1.81	1.68	1.50
27	0.1	13.61	9.02	7.27	6.33	5.73	5.31	4.76	4.17	3.52	2.75
	0.5	9.34	6.49	5.36	4.74	4.34	4.06	3.69	3.28	2.83	2.29
	1	7.68	5.49	4.60	4.11	3.78	3.56	3.26	2.93	2.55	2.10
	2.5	5.63	4.24	3.65	3.31	3.08	2.92	2.71	2.47	2.19	1.85
	5	4.21	3.35	2.96	2.73	2.57	2.46	2.30	2.13	1.93	1.67
	10	2.90	2.51	2.30	2.17	2.07	2.00	1.91	1.80	1.67	1.49
28	0.1	13.50	8.93	7.19	6.25	5.66	5.24	4.69	4.11	3.46	2.70
	0.5	9.28	6.44	5.32	4.70	4.30	4.02	3.65	3.25	2.79	2.25
	1	7.64	5.45	4.57	4.07	3.75	3.53	3.23	2.90	2.52	2.06
	2.5	5.61	4.22	3.63	3.29	3.06	2.90	2.69	2.45	2.17	1.83
	5	4.20	3.34	2.95	2.71	2.56	2.44	2.29	2.12	1.91	1.65
	10	2.89	2.50	2.29	2.16	2.06	2.00	1.90	1.79	1.66	1.48
29	0.1	13.39	8.85	7.12	6.19	5.59	5.18	4.64	4.05	3.41	2.64
	0.5	9.23	6.40	5.28	4.66	4.26	3.98	3.61	3.21	2.76	2.21
	1	7.60	5.42	4.54	4.04	3.73	3.50	3.20	2.87	2.49	2.03
	2.5	5.59	4.20	3.61	3.27	3.04	2.88	2.67	2.43	2.15	1.81
	5	4.18	3.33	2.93	2.70	2.54	2.43	2.28	2.10	1.90	1.64
	10	2.89	2.50	2.28	2.15	2.06	1.99	1.89	1.78	1.65	1.47
30	0.1	13.29	8.77	7.05	6.12	5.53	5.12	4.58	4.00	3.36	2.59
	0.5	9.18	6.35	5.24	4.62	4.23	3.95	3.58	3.18	2.73	2.18

APPENDIX: Statistical Tables

TABLE 3 Continued

df ₁ df ₂		1	2	3	4	5	6	8	12	24	∞
40	1%	7.56	5.39	4.51	4.02	3.70	3.47	3.17	2.84	2.47	2.01
	2.5%	5.57	4.18	3.59	3.25	3.03	2.87	2.65	2.41	2.14	1.79
	5%	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.09	1.89	1.62
	10%	2.88	2.49	2.28	2.14	2.05	1.98	1.88	1.77	1.64	1.46
	0.1	12.61	8.25	6.60	5.70	5.13	4.73	4.21	3.64	3.01	2.23
	0.5	8.83	6.07	4.98	4.37	3.99	3.71	3.35	2.95	2.50	1.93
	1	7.31	5.18	4.31	3.83	3.51	3.29	2.99	2.66	2.29	1.80
	2.5	5.42	4.05	3.46	3.13	2.90	2.74	2.53	2.29	2.01	1.64
	5	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.00	1.79	1.51
	10	2.84	2.44	2.23	2.09	2.00	1.93	1.83	1.71	1.57	1.38
60	0.1	11.97	7.76	6.17	5.31	4.76	4.37	3.87	3.31	2.69	1.90
	0.5	8.49	5.80	4.73	4.14	3.76	3.49	3.13	2.74	2.29	1.69
	1	7.08	4.98	4.13	3.65	3.34	3.12	2.82	2.50	2.12	1.60
	2.5	5.29	3.93	3.34	3.01	2.79	2.63	2.41	2.17	1.88	1.48
	5	4.00	3.15	2.76	2.52	2.37	2.25	2.10	1.92	1.70	1.39
	10	2.70	2.30	2.10	2.04	1.05	1.37	1.77	1.66	1.51	1.20
	0.1	11.38	7.31	5.79	4.95	4.42	4.04	3.55	3.02	2.40	1.56
	0.5	8.18	5.54	4.50	3.92	3.55	3.28	2.93	2.54	2.09	1.43
	1	6.85	4.79	3.95	3.48	3.17	2.96	2.66	2.34	1.95	1.38
	2.5	5.15	3.80	3.23	2.89	2.67	2.52	2.30	2.05	1.76	1.31
120	5	3.92	3.07	2.68	2.45	2.29	2.17	2.02	1.83	1.61	1.25
	10	2.75	2.35	2.13	1.99	1.90	1.82	1.72	1.60	1.45	1.19
	0.1	10.83	6.91	5.42	4.62	4.10	3.74	3.27	2.74	2.13	1.00
	0.5	7.88	5.30	4.28	3.72	3.35	3.09	2.74	2.36	1.90	1.00
	1	6.64	4.60	3.78	3.32	3.02	2.80	2.51	2.18	1.79	1.00
	2.5	5.02	3.69	3.12	2.79	2.57	2.41	2.19	1.94	1.64	1.00
	5	3.84	2.99	2.60	2.37	2.21	2.09	1.94	1.75	1.52	1.00
	10	2.71	2.30	2.08	1.94	1.85	1.77	1.67	1.55	1.35	1.00
∞											

Source: Adapted from Table 18 in *Biometrika Tables for Statisticians*, vol. 1, 3d ed., edited by E. S. Pearson and H. O. Hartley (New York: Cambridge University Press, 1958).

APPENDIX: Statistical Tables

TABLE 4 Critical Values of Chi Square (χ^2)

df	0.250	0.100	0.050	0.025	0.010	0.005	0.001
1	1.32330	2.70554	3.84146	5.02389	6.63490	7.87944	10.828
2	2.77259	4.60517	5.99147	7.37776	9.21034	10.5966	13.816
3	4.10835	6.25139	7.81473	9.34840	11.3449	12.8381	16.266
4	5.38527	7.77944	9.48773	11.1433	13.2767	14.8602	18.467
5	6.62568	9.23635	11.0705	12.8325	15.0863	16.7496	20.515
6	7.84080	10.6446	12.5916	14.4494	16.8119	18.5476	22.458
7	9.03715	12.0170	14.0671	16.0128	18.4753	20.2777	24.322
8	10.2188	13.3616	15.5073	17.5346	20.0902	21.9550	26.125
9	11.3887	14.6837	16.9190	19.0228	21.6660	23.5893	27.877
10	12.5489	15.9871	18.3070	20.4831	23.2093	25.1882	29.588
11	13.7007	17.2750	19.6751	21.9200	24.7250	26.7569	31.264
12	14.8454	18.5494	21.0261	23.3367	26.2170	28.2995	32.909
13	15.9839	19.8119	22.3621	24.7356	27.6883	29.8194	34.528
14	17.1770	21.0642	23.6848	26.1190	29.1413	31.3193	36.123
15	18.2451	22.3072	24.9958	27.4884	30.5779	32.8013	37.697
16	19.3688	23.5418	26.2962	28.8454	31.9999	34.2672	39.252
17	20.4887	24.7690	27.5871	30.1910	33.4087	35.7185	40.790
18	21.6049	25.9894	28.8693	31.5264	34.8053	37.1564	42.312
19	22.7178	27.2036	30.1435	32.8523	36.1908	38.5822	43.820
20	23.8277	28.4120	31.4104	34.1696	37.5662	39.9968	45.315
21	24.9348	29.6151	32.6705	35.4789	38.9321	41.4010	46.797
22	26.0393	30.8133	33.9244	36.7807	40.2894	42.7956	48.268
23	27.1413	32.0069	35.1725	38.0757	41.6384	44.1813	49.728
24	28.2412	33.1963	36.4151	39.3641	42.9798	45.5585	51.179
25	29.3389	34.3816	37.6525	40.6465	44.3141	46.9278	52.620
26	30.4345	35.5631	38.8852	41.9232	45.6417	48.2899	54.052
27	31.5284	36.7412	40.1133	43.1944	46.9630	49.6449	55.476
28	32.6205	37.9159	41.3372	44.4607	48.2782	50.9933	56.892
29	33.7109	39.0875	42.5569	45.7222	49.5879	52.3356	58.302
30	34.7998	40.2560	43.7729	46.9792	50.8922	53.6720	59.703
40	45.6160	51.8050	65.7585	59.3417	63.6907	66.7659	73.402
50	56.3336	63.1671	67.5048	71.4202	76.1539	79.4900	86.661
60	66.9814	74.3970	79.0819	83.2976	88.3794	91.9517	99.607
70	77.5766	85.5271	90.5312	95.0231	100.425	104.215	112.317
80	88.1303	96.5782	101.879	106.629	112.329	116.321	124.839
90	98.6499	107.565	113.145	118.136	124.116	128.299	137.208
100	109.141	118.498	124.342	129.561	135.807	140.169	149.449

Source: Adapted from Table 8 in *Biometrika Tables for Statisticians*, vol. 1, 3d ed., edited by E. S. Pearson and H. O. Hartley (New York: Cambridge University Press, 1958).

APPENDIX: Statistical Tables

**TABLE 5 Critical Values for Squared Multiple Correlation (R^2)
in Forward Stepwise Selection: $\alpha = .05$**

k	F	$N-k-1$															
		10	12	14	16	18	20	25	30	35	40	50	60	80	100	150	200
2	2	43	38	33	30	27	24	20	16	14	13	10	8	6	5	3	2
2	3	40	36	31	27	24	22	18	15	13	11	9	7	5	4	2	2
2	4	38	33	29	26	23	21	17	14	12	10	8	7	5	4	3	2
3	2	49	43	39	35	32	29	24	21	18	16	12	10	8	7	4	2
3	3	45	40	36	32	29	26	22	19	17	15	11	9	7	6	4	3
3	4	42	36	33	29	27	25	20	17	15	13	11	9	7	5	4	3
4	2	54	48	44	39	35	33	27	23	20	18	15	12	10	8	5	4
4	3	49	43	39	36	33	30	25	22	19	17	14	11	8	7	5	4
4	4	45	39	35	32	29	27	22	19	17	15	12	10	8	6	5	3
5	2	58	52	47	43	39	36	31	26	23	21	17	14	11	9	6	5
5	3	52	46	42	38	35	32	27	24	21	19	16	13	9	8	5	4
5	4	46	41	38	35	32	29	24	21	18	16	13	11	9	7	5	4
6	2	60	54	50	46	41	39	33	29	25	23	19	16	12	10	7	5
6	3	54	48	44	40	37	34	29	25	22	20	17	14	10	8	6	5
6	4	48	43	39	36	33	30	26	23	20	17	14	12	9	7	5	4
7	2	61	56	51	48	44	41	35	30	27	24	20	17	13	11	7	5
7	3	59	50	46	42	39	36	31	26	23	21	18	15	11	9	7	5
7	4	50	45	41	38	35	32	27	24	21	18	15	13	10	8	6	4
8	2	62	58	53	49	46	43	37	31	28	26	21	18	14	11	8	6
8	3	57	52	47	43	40	37	32	28	24	22	19	16	12	10	7	5
8	4	51	46	42	39	36	33	28	25	22	19	16	14	11	9	7	5
9	2	63	59	54	51	47	44	38	33	30	27	22	19	15	12	9	6
9	3	58	53	49	44	41	38	33	29	25	23	20	16	12	10	7	6
9	4	52	46	43	40	37	34	29	25	23	20	17	14	11	10	7	6
10	2	64	60	55	52	49	46	39	34	31	28	23	20	16	13	10	7
10	3	59	54	50	45	42	39	34	30	26	24	20	17	13	11	8	6
10	4	52	47	44	41	38	35	30	26	24	21	18	15	12	10	8	6
12	2	66	62	57	54	51	48	42	37	33	30	25	22	17	14	10	8
12	3	60	55	52	47	44	41	36	31	28	25	22	19	14	12	9	7
12	4	53	48	45	41	39	36	31	27	25	22	19	16	13	11	9	7
14	2	68	64	60	56	53	50	44	39	35	32	27	24	18	15	11	8
14	3	61	57	53	49	46	43	37	32	29	27	23	20	15	13	10	8
14	4	43	49	46	42	40	37	32	29	26	23	20	17	13	11	9	7
16	2	69	66	61	58	55	53	46	41	37	34	29	25	20	17	12	9
16	3	61	58	54	50	47	44	38	34	31	28	24	21	17	14	11	8
16	4	53	50	46	43	40	38	33	30	27	24	21	18	14	12	10	8
18	2	70	67	65	60	57	55	49	44	40	36	31	27	21	18	13	9
18	3	62	59	55	51	49	46	40	35	32	30	26	23	18	15	12	9
18	4	54	50	46	44	41	38	34	31	28	25	22	19	15	13	11	8
20	2	72	68	64	62	59	56	50	46	42	38	33	28	22	19	14	10
20	3	62	60	56	52	50	47	42	37	34	31	27	24	19	16	12	9
20	4	54	50	46	44	41	37	35	32	29	26	23	20	16	14	11	8

(continued)

APPENDIX: Statistical Tables

TABLE 5 Continued $\alpha = .01$

k	F	$N-k-1$															
		10	12	14	16	18	20	25	30	35	40	50	60	80	100	150	200
2	2	59	53	48	43	40	36	30	26	23	20	17	14	11	9	7	5
2	3	58	52	46	42	38	35	30	25	22	19	16	13	10	8	6	4
2	4	57	49	44	39	36	32	26	22	19	16	13	11	8	7	5	4
3	2	67	60	55	50	46	42	35	30	27	24	20	17	13	11	7	5
3	3	63	58	52	47	43	40	34	29	25	22	19	16	12	10	7	5
3	4	61	54	48	44	40	37	31	26	23	20	16	14	11	9	6	5
4	2	70	64	58	53	49	46	39	34	30	27	23	19	15	12	8	6
4	3	67	62	56	51	47	44	37	32	28	25	21	18	14	11	8	6
4	4	64	58	52	47	43	40	34	29	26	23	19	16	13	11	7	6
5	2	73	67	61	57	52	49	42	37	32	29	25	21	16	13	9	7
5	3	70	65	59	54	50	46	39	34	30	27	23	19	15	12	9	7
5	4	65	60	55	50	46	43	36	31	28	25	20	17	14	12	8	6
6	2	74	69	63	59	55	51	44	39	34	31	26	23	18	14	10	8
6	3	72	67	61	56	51	48	41	36	32	28	24	20	16	13	10	7
6	4	66	61	56	52	48	45	38	33	29	26	22	19	15	13	9	7
7	2	76	70	65	60	56	53	46	40	36	33	28	25	19	15	11	9
7	3	73	68	62	57	53	50	42	37	33	30	25	21	17	14	10	8
7	4	67	62	58	54	49	46	40	35	31	28	23	20	16	14	10	8
8	2	77	72	66	62	58	55	48	42	38	34	29	26	20	16	12	9
8	3	74	69	63	58	54	51	44	39	34	31	26	22	18	15	11	9
8	4	67	63	59	55	50	47	41	36	32	29	24	21	17	15	11	9
9	2	78	73	67	63	60	56	49	43	39	36	31	27	21	17	12	10
9	3	74	69	64	59	56	52	45	40	35	32	27	23	19	16	12	9
9	4	68	63	60	56	51	48	42	37	33	30	25	22	18	16	12	9
10	2	79	74	68	65	61	58	51	45	40	37	32	28	22	18	13	10
10	3	74	69	65	60	57	53	47	41	37	33	28	24	20	17	13	10
10	4	68	64	61	56	52	49	43	38	34	31	26	23	19	17	13	9
12	2	80	75	70	66	63	60	53	48	43	39	34	30	24	20	14	11
12	3	74	70	66	62	58	55	48	43	39	35	30	26	21	18	14	10
12	4	69	65	61	57	53	50	44	40	35	32	27	24	20	18	13	10
14	2	81	76	71	68	65	62	55	50	45	41	36	32	25	21	15	11
14	3	74	70	67	63	60	56	50	45	41	37	31	27	22	19	15	11
14	4	69	65	61	57	54	52	45	41	36	33	28	25	21	19	14	10
16	2	82	77	72	69	66	63	57	52	47	43	38	34	27	22	16	12
16	3	74	70	67	64	61	58	52	47	42	39	33	29	23	20	15	11
16	4	70	66	62	58	55	52	46	42	37	34	29	26	22	20	14	11
18	2	82	78	73	70	67	65	59	54	49	45	39	35	28	23	17	12
18	3	74	70	67	65	62	59	53	48	44	41	35	30	24	21	16	12
18	4	70	65	62	58	55	53	47	43	38	35	30	27	23	20	15	11
20	2	82	78	74	71	68	66	60	55	50	46	41	36	29	24	18	13
20	3	74	70	67	65	62	60	55	60	46	42	36	32	26	22	17	12
20	4	70	66	62	58	55	53	47	43	39	36	31	28	24	21	16	11

Note: Decimals are omitted; k = number of candidate predictors; N = sample size; F = criterion F -to-enter.

Source: Adapted from Tables 1 and 2 in "Tests of significance in forward selection regression," by Wilkinson and Dallal, *Technometrics*, 1981, 23(4), 377-380. Vol. 23 No.4, November 1981 "Reprinted with permission from *Technometrics*. Copyright 1981 by the American Statistical Association. All rights reserved."

APPENDIX: Statistical Tables

TABLE 6 Critical Values for $F_{\text{MAX}} (S^2_{\text{MAX}}/S^2_{\text{MIN}})$ Distribution for $\alpha = .05$ and $.01$

$\alpha = .05$											
k df	2	3	4	5	6	7	8	9	10	11	12
4	9.60	15.5	20.6	25.2	29.5	33.6	37.5	41.1	44.6	48.0	51.4
5	7.15	10.8	13.7	16.3	18.7	20.8	22.9	24.7	26.5	28.2	29.9
6	5.82	8.38	10.4	12.1	13.7	15.0	16.3	17.5	18.6	19.7	20.7
7	4.99	6.94	8.44	9.70	10.8	11.8	12.7	13.5	14.3	15.1	15.8
8	4.43	6.00	7.18	8.12	9.03	9.78	10.5	11.1	11.7	12.2	12.7
9	4.03	5.34	6.31	7.11	7.80	8.41	8.95	9.45	9.91	10.3	10.7
10	3.72	4.85	5.67	6.34	6.92	7.42	7.87	8.28	8.66	9.01	9.34
12	3.28	4.16	4.79	5.30	5.72	6.09	6.42	6.72	7.00	7.25	7.48
15	2.86	3.54	4.01	4.37	4.68	4.95	5.19	5.40	5.59	5.77	5.93
20	2.46	2.95	3.29	3.54	3.76	3.94	4.10	4.24	4.37	4.49	4.59
30	2.07	2.40	2.61	2.78	2.91	3.02	3.12	3.21	3.29	3.36	3.39
60	1.67	1.85	1.96	2.04	2.11	2.17	2.22	2.26	2.30	2.33	2.36
∞	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
$\alpha = .01$											
k df	2	3	4	5	6	7	8	9	10	11	12
4	23.2	37	49	59	69	79	89	97	106	113	120
5	14.9	22	28	33	38	42	46	50	54	57	60
6	11.1	15.5	19.1	22	25	27	30	32	34	36	37
7	8.89	12.1	14.5	16.5	18.4	20	22	23	24	26	27
8	7.50	9.9	11.7	13.2	14.5	15.8	16.9	17.9	18.9	19.8	21
9	6.54	8.5	9.9	11.1	12.1	13.1	13.9	14.7	15.3	16.0	16.6
10	5.85	7.4	8.6	9.6	10.4	11.1	11.8	12.4	12.9	13.4	13.9
12	4.91	6.1	6.9	7.6	8.2	8.7	9.1	9.5	9.9	10.2	10.6
15	4.07	4.9	5.5	6.0	6.4	6.7	7.1	7.3	7.5	7.8	8.0
20	3.32	3.8	4.3	4.6	4.9	5.1	5.3	5.5	5.6	5.8	5.9
30	2.63	3.0	3.3	3.4	3.6	3.7	3.8	3.9	4.0	4.1	4.2
60	1.96	2.2	2.3	2.4	2.4	2.5	2.5	2.6	2.6	2.7	2.7
∞	1.00	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

Note: S^2_{MAX} is the largest and S^2_{MIN} the smallest in a set of k independent mean squares, each based on degrees of freedom (df).
Source: Adapted from Table 31 in *Biometrika Tables for Statisticians*, vol. 1, 3d ed., edited by E. S. Pearson and H. O. Hartley (New York: Cambridge University Press, 1958).

APPENDIX

Research Designs for Complete Examples

1 Women's Health and Drug Study

Data used in most of the large sample examples were collected with the aid of a grant from the National Institute on Drug Abuse (#DA 00847) to L. S. Fidell and J. E. Prather in 1974–1976. Methods of collecting the data and references to the measures included in the study are described here approximately as they have been previously reported (Hoffman & Fidell, 1979).

Method

A structured interview, containing a variety of health, demographic, and attitudinal measures, was given to a randomly selected group of 465 female, 20- to 59-year-old, English-speaking residents of the San Fernando Valley, a suburb of Los Angeles, in February 1975. A second interview, focusing primarily on health variables but also containing the Bem Sex Role Inventory (BSRI; Bem, 1974) and the Eysenck Personality Inventory (EPI; Eysenck & Eysenck, 1963), was conducted with 369 (79.4%) of the original respondents in February 1976.

The 1975 target sample of 703 names was approximately a .003 probability sample of appropriately aged female residents of the San Fernando Valley, and was randomly drawn from lists prepared by listers during the weeks immediately preceding the sample selection. Lists were prepared for census blocks that had been randomly drawn (proportional to population) from 217 census tracts, which were themselves randomly drawn after they were stratified by income and assigned probabilities proportional to their populations. Respondents were contacted after first receiving a letter soliciting their cooperation. Substitutions were not allowed. A minimum of four callbacks was required before the attempt to obtain an interview was terminated. The completion rate for the target sample was 66.1%, with a 26% refusal rate and a 7.9% “unobtainable” rate.

The demographic characteristics of the 465 respondents who cooperated in 1975 confirmed the essentially white, middle- and working-class composition of the San Fernando Valley, and agreed, for the most part, with the profile of characteristics of women in the valley that was calculated from 1970 Census Bureau data. The final sample was 91.2% white, with a median family income (before taxes) of \$17,000 per year and an average Duncan scale (Featherman, 1973) socioeconomic level (SEL) rating of 51. Respondents were also well educated (13.2 years of school completed, on average), and predominantly Protestant (38%), with 26% Catholic, 20% Jewish, and the remainder “none” or “other.” A total of 52.9% worked (either full-time—33.5%—or part-time—19.4%). Seventy-eight percent were living with

husbands at the time of the first interview, with 9% divorced, 6% single, 3% separated, 3% widowed, and fewer than 1% “living together.” Altogether, 82.4 of the women had children; the average number of children was 2.7, with 2.1 children, on the average, still living in the same house as the respondent.

Of the original 465 respondents, 369 (79.4%) were re-interviewed a year later. Of the 96 respondents who were not re-interviewed, 51 refused, 36 had moved and could not be relocated, 8 were known to be in the Los Angeles area but were not contacted after a minimum of 5 attempts, and 1 was deceased. Those who were and were not re-interviewed were similar (by analyses of variance) on health and attitudinal variables. They differed, however, on some demographic measures. Those who were re-interviewed tended to be higher-SEL, higher-income white women who were better-educated, were older, and had experienced significantly fewer life change units (Rahe, 1974) in 1975.

The 1975 interview schedule was composed of items assessing a number of demographic, health, and attitudinal characteristics (see Table 1). Insofar as possible, previously tested and validated items and measures were used, although time constraints prohibited including all items from some measures. Coding on most items was prearranged so that responses given large numbers reflected increasingly unfavorable attitudes, dissatisfaction, poorer health, lower income, increasing stress, increasing use of drugs, and so forth.

The 1976 interview schedule repeated many of the health items, with a shorter set of items assessing changes in marital status and satisfaction, changes in work status and satisfaction, and so forth. The BSRI and EPI were also included, as previously mentioned. The interview schedules for both 1975 and 1976 took 75 minutes on average to administer and were conducted in respondent’s homes by experienced and trained interviewers.

To obtain median values for the masculine and feminine scores of the BSRI for a comparable sample of men, the BSRI was mailed to the 369 respondents who cooperated in 1976, with instructions to ask a man near to them (husband, friend, brother, etc.) to fill out and return it. The completed BSRI was received from 162 (46%) men, of whom 82% were husbands, 8.6% friends, 3.7% fiancés, 1.9% brothers, 1.2% sons, 1.2% ex-husbands, 0.6% brothers-in-law, and 0.6% fathers. Analyses of variance were used to compare the demographic characteristics of the men who returned the BSRI with those who did not (insofar as such characteristics could be determined by responses of the women to questions in the 1975 interview). The two groups differed in that, as with the re-interviewed women, the men who responded presented an advantaged socioeconomic picture relative to those who did not. Respondents had higher SEL2 ratings, were better educated, and enjoyed higher income. The unweighted averages of the men’s and women’s median masculine scores and median feminine scores were used to split the sample of women into those who were feminine, masculine, androgynous, and undifferentiated.

2 Sexual Attraction Study

Data used in the large sample multiway frequency analysis example were collected in 1984 as part of a survey assessing issues surrounding the nature of sexual attraction to clients among clinical psychologists. Data-collection methods and demographic characteristics that follow are approximately as they appear in an *American Psychologist* paper (Pope, Keith-Spiegel, & Tabachnick, 1986).

APPENDIX: Research Designs for Complete Examples

TABLE 1 Description of Some of the Variables Available from 1975–1976 Interviews

Variable	Abbreviation	Brief Description	Source
Demographic variables			
Socioeconomic level	SEL, SEL2	Measure of deference accorded employment categories (SEL2 from second interview)	Featherman (1973), update of Duncan Scale
Education	EDUC EDCODE	Number of years completed Categorical variable assessing whether or not education proceeded beyond high school	
Income	INCOME INCODE	Total family income before taxes Categorical variable assessing family income	
Age	AGE	Chronological age in 5-year categories	
Marital status	MARITAL	A categorical variable assessing current marital status	
	MSTATUS	A dichotomous variable assessing whether or not currently married	
Parenthood	CHILDREN	A categorical variable assessing whether or not one has children	
Ethnic group membership	RACE	A categorical variable assessing ethnic affiliation	
Employment status	EMPLMNT	A categorical variable assessing whether or not one is currently employed	
	WORKSTAT	A categorical variable assessing current employment status and, if not, attitude toward unemployed status	
Religious affiliation	RELIGION	A categorical variable assessing religious affiliation	
Attitudinal variables			
Attitudes toward housework	ATTHOUSE	Frequency of experiencing various favorable and unfavorable attitudes toward homemaking	Derived from Johnson (1955)
Attitudes toward paid work	ATTWORK	Frequency of experiencing various favorable and unfavorable attitudes toward paid work	Johnson (1955)
Attitudes toward role of women	ATTROLE	Measure of conservative or liberal attitudes toward role of women	Spence and Helmreich (1972)
Locus of control	CONTROL	Measure of control ideology; internal or external	Rotter (1966)

(continued)

TABLE 1 Continued

Variable	Abbreviation	Brief Description	Source
Attitudinal variables (<i>cont.</i>)			
Attitudes toward marital status	ATTMAR	Satisfaction with current marital status	From Burgess & Locke (1960); Locke & Wallace (1959); and Rollins & Feldman (1970)
Personality variables			
Self-esteem	ESTEEM	Measures of self-esteem and confidence in various situations	Rosenberg (1965)
Neuroticism-stability index	NEUROTIC	A scale derived from factor analysis to measure neuroticism vs. stability	Eysenck & Eysenck (1963)
Introversion-extraversion index	INTEXT	A scale derived from factor analysis to measure introversion vs. extraversion	Eysenck & Eysenck (1963)
Androgyny measure	ANDRM	A categorical variable based on femininity and masculinity	Derived from Bem (1974)
Health variables			
Mental health	MENHEAL	Frequency count of mental health problems (feeling somewhat apart, can't get along, etc.)	Langer (1962)
Physical health	PHYHEAL	Frequency count of problems with various body systems (circulation, digestion, etc.), general description of health	
Number of visits	TIMEDRS	Frequency count of visits to physical and mental health professionals	
Use of psycho-tropic drugs	DRUGUSE	A frequency, recency measure of involvement with prescription and nonprescription major and minor tranquilizers, sedatives-hypnotics, antidepressants, and stimulants	Balter & Levine (1971)
Use of psycho-tropic and over-the-counter drugs	PSYDRUG	DRUGUSE plus a frequency, recency measure of over-the-counter mood-modifying drugs	
Attitudes toward medication	ATTDRUG	Items concerning attitudes toward use of medication	
Life change units	STRESS	Weighted items reflecting number and importance of change in life situation	Rahe (1974)

Method

A cover letter, a brief 17-item questionnaire (15 structured questions and 2 open-ended questions), and a return envelope were sent to 1000 psychologists (500 men and 500 women) randomly selected from the 4356 members of Division 42 (Psychologists in Private Practice) as listed in the 1983 Membership Register of the American Psychological Association.

The questionnaire requested respondents to provide information about their gender, age group, and years of experience in the field. Information was elicited about the respondent's incidence of sexual attraction to male and female clients; clients' reactions to this experience of attraction; beliefs about the clients' awareness of and reciprocation of the attraction; the impact of the attraction on the therapy process; how such feelings were managed; the incidence of sexual fantasies about clients; why, if relevant, respondents chose to refrain from acting out their attraction through actual sexual intimacies with clients; what features determined which clients would be perceived as sexually attractive; incidence of actual sexual activity with clients; and the extent to which the respondents' graduate training and internship experiences had dealt with issues related to sexual attraction to clients.

Questionnaires were returned by 585 respondents. Of these 59.7% were men. Sixty-eight percent of the male respondents returned their questionnaires as compared with 49% of the female respondents. Approximately half of the respondents were 45 years of age and under. The sample's median age was approximately 46 years as compared with the median age of 40 years reported in a 1983 survey of mental health service providers (VandenBos & Stapp, 1983).

Respondents averaged 16.99 ($SD = 8.43$) years of professional experience with no significant differences between male and female psychologists. Younger therapists averaged 11.36 ($SD = 8.43$) years of experience and older therapists averaged 21.79 ($SD = 8.13$) years of experience. Only 77 of the 585 therapists reported never being attracted to any client.

3 Learning Disabilities Data Bank

Data (profile analysis of subscales of the WISC) were taken from a data bank developed at the California Center for Educational Therapy (CCET) in the San Fernando Valley.

All children who were referred to the CCET were given an extensive battery of psychodiagnostic tests to measure current intellectual functioning, perceptual development, psycholinguistic abilities, visual and auditory functioning, and achievement in a number of academic subjects. In addition, an extensive Parent Information Outline queries parents about demographic variables, family health history, as well as child's developmental history, strengths, weaknesses, preferences, and the like. The entire data bank consists of 112 variables from the testing battery plus 155 variables from the Parent Information Outline.

Data collection began in July 1972 and continued to 1993. Some samples include children tested before February 1984 who were administered the Wechsler Intelligence Scale for Children, who were diagnosed as learning-disabled, whose parents agreed to be included in the data bank, and whose parents answered a question about the child's preference for playmates' age. Available

answers to this question were: (1) older, (2) younger, (3) same age, and (4) no preference. Of the 261 children tested between 1972 and 1984, 177 were eligible for inclusion in the sample.

For the entire sample of 261 cases, average age is 10.58 years with a range of 5 to 61 years. About 75% of the entire sample is male. At the time of testing, 63% of the entire sample attended public school; 33% were enrolled in various types of private schools. Of the 94% of parents who revealed their educational level, mothers had completed an average of 13.6 years of schooling and fathers had completed an average of 14.9 years.

4 Reaction Time to Identify Figures

Data for this study were collected by Damos (1989). Twenty right-handed males were required to respond “standard” or “mirror” on a keypad as quickly as possible to a stimulus (the letter G or a symbol) and its mirror image. Each trial consisted of 30 presentations of the stimulus and 30 presentations of its mirror image, five presentations in each of six orientations: upright (0), 60, 120, 180, 240, and 300 degrees of rotation. In all, the experiment required ten sessions over two consecutive weeks. Each session had four blocks of nine trials distributed over morning and afternoon periods. Thus, each subject made 21,600 responses during the study. Half the subjects were given stimulus G in the first week and a symbol in the second week; the others participated only for one week and were given the symbol stimulus. Order of presentation of all stimuli was random on all trials.

The two DVs were average correct reaction time and error rate. Responses to 60° of absolute rotation were averaged with those of 300°, and responses to 120° were averaged with those of 240°. Averages were calculated separately for standard and mirror-image trials. Linear regression on the averages for each subject in each session yielded a slope and intercept.

Data selected for analysis were slopes and intercepts from the first four sessions: morning and afternoon sessions for days 1 and 2 of the first week, so that practice effects could be observed. Only the trials on which the subject responded “standard” were used. Thus, each of the 20 subjects provided eight scores and a slope and an intercept for each of the four sessions.

5 Field Studies of Noise-Induced Sleep Disturbance

These data are from field studies of the effects of nighttime noise on sleep disturbance reported by Fidell et al. (1995). Measurements were taken in the vicinity of two airports and one control site. Airports were Castle Air Force Base in Merced, CA, neighborhoods in the Los Angeles area that were not exposed to nighttime aircraft noise but were exposed to high levels of road traffic noise, and LAX. Indoor and outdoor noise exposure was measured for periods of approximately one month in 45 homes of 82 test participants for a total of 1887 subject nights. Responses included time to fall asleep, number of recalled awakenings, behavioral awakenings, sleep quality, sleep time, annoyance, tiredness in the morning, reported time awake, and awakenings by aircraft noise.

A number of personal characteristics also were recorded, including age, gender, spontaneous awakening rate, duration of residence, use of alcohol and medications, and tiredness upon retiring. Event-based awakening was predicted in a logistic regression analysis by event noise level, ambient noise level, rate of spontaneous awakening, age, time since retiring, duration of residence, and tiredness on retiring. However, the slope of the relationship between awakening and noise exposure was rather shallow. Results were found to be in good agreement with those of other field studies, but quite different from laboratory studies.

6 Clinical Trial for Primary Biliary Cirrhosis

Data in this example were collected during a double-blind randomized clinical trial conducted at the Mayo Clinic January 1974 and May 1984. Of the 424 patients who met eligibility requirements for the study, 312 (with their physicians) agreed to participate in the randomized trial. A number of clinical, biochemical, serologic, and histologic measurements were made for the 312 patients, in addition to the date of random assignment to either a placebo group or to a group treated with the drug D-penicillamine. Data from the clinical trial were analyzed in 1986 for presentation in the clinical literature.

Data from the 312 cases used in the survival analysis included 125 who had died (the 11 patients whose deaths were not attributable to PBC were not distinguished) and 19 who had undergone liver transplant. The data and some analyses are described by Fleming and Harrington (1991, Chapters 1 and 4) and by Markus, et al. (1989). The latter paper focuses on the efficacy of liver transplantation, but also discusses the Mayo model which determined the variables used in the survival analysis.

Fleming and Harrington describe PBC as “. . . a rare but fatal chronic liver disease of unknown cause, with a prevalence of about 50 cases per million population. The primary pathologic event appears to be the destruction of interlobular bile ducts, which may be mediated by immunologic mechanisms” (p. 2).

7 Impact of Seat Belt Law

Rock (1992) applied an ARIMA approach to evaluating the impact of the Illinois seat belt use law on accidents, deaths, and various degrees of injury. Data were collected monthly starting January 1980 and continued through December 1990, with the seat belt law taking effect in 1985. Accident statistics are available from the IDOT. Rock considered other statistical techniques in addition to ARIMA, which he concluded was preferable (less biased).

DVs, evaluated separately, were monthly fatalities, accidents, and A-, B-, and C-level injuries. A-level injuries are the most serious, defined as incapacitating; C-level are the least severe, possibly not visible. Different ARIMA models were applied to the five DVs, with a statistically significant impact of the law found only for A-level injuries. Rock concluded that overall the net impact of the law was a reduction in injury severity.

APPENDIX

A Skimpy Introduction to Matrix Algebra

The purpose of this appendix is to provide readers with sufficient background to follow, and duplicate as desired the calculations. The purpose is not to provide a thorough review of matrix algebra or even to facilitate an in-depth understanding of it. The reader who is interested in more than calculational rules has several excellent discussions available, particularly those in Tatsuoka (1971), Carroll, Green, and Chattervedi (1997), and Rummel (1970).

Most of the algebraic manipulations with which the reader is familiar—addition, subtraction, multiplication, and division—have counterparts in matrix algebra. In fact, the algebra that most of us learned is a special case of matrix algebra involving only a single number, a scalar, instead of an ordered array of numbers, a matrix. Some generalizations from scalar algebra to matrix algebra seem “natural” (i.e., matrix addition and subtraction) while others (multiplication and division) are convoluted. Nonetheless, matrix algebra provides an extremely powerful and compact method for manipulating sets of numbers to arrive at desirable statistical products.

The matrix calculations illustrated here are calculations performed on square matrices. Square matrices have the same number of rows as columns. Sums-of-squares and cross-products matrices, variance-covariance matrices, and correlation matrices are all square. In addition, these three very commonly encountered matrices are symmetrical, having the same value in row 1, column 2, as in column 1, row 2, and so forth. Symmetrical matrices are mirror images of themselves about the main diagonal (the diagonal going from top left to bottom right in the matrix).

There is a more complete matrix algebra that includes nonsquare matrices as well. However, once one proceeds from the data matrix, which has as many rows as research units (subjects) and as many columns as variables, to the sum-of-squares and cross-products matrix. For purposes of numerical illustration, two very simple matrices, square, but not symmetrical (to eliminate any uncertainty regarding which elements are involved in calculations), will be defined as follows:

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 \\ 7 & 5 & 0 \\ 1 & 0 & 8 \end{bmatrix}$$
$$\mathbf{B} = \begin{bmatrix} r & s & t \\ u & v & w \\ x & y & z \end{bmatrix} = \begin{bmatrix} 6 & 1 & 0 \\ 2 & 8 & 7 \\ 3 & 4 & 5 \end{bmatrix}$$

1 The Trace of a Matrix

The trace of a matrix is the sum of the numbers on the diagonal that runs from the upper left to lower right. For matrix **A**, the trace is 16 ($3 + 5 + 8$); for matrix **B** it is 19. If the matrix is a sum-of-squares and cross-products matrix, then the trace is the sum of squares. If it is a variance-covariance matrix, the trace is the sum of variances. If it is a correlation matrix, the trace is the number of variables (each having contributed a value of 1 to the trace).

2 Addition or Subtraction of a Constant to a Matrix

If one has a matrix, **A**, and wants to add or subtract a constant, k , to the elements of the matrix, one simply adds (or subtracts) the constant to every element in the matrix.

$$\mathbf{A} + k = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} + k = \begin{bmatrix} a+k & b+k & c+k \\ d+k & e+k & f+k \\ g+k & h+k & i+k \end{bmatrix} \quad (1)$$

If $k = -3$, then

$$\mathbf{A} + k = \begin{bmatrix} 0 & -1 & 1 \\ 4 & 2 & -3 \\ -2 & -3 & 5 \end{bmatrix}$$

3 Multiplication or Division of a Matrix by a Constant

Multiplication or division of a matrix by a constant is a straightforward process.

$$\begin{aligned} k\mathbf{A} &= k \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \\ k\mathbf{A} &= \begin{bmatrix} ka & kb & kc \\ kd & ke & kf \\ kg & kh & ki \end{bmatrix} \end{aligned} \quad (2)$$

and

$$\frac{1}{k}\mathbf{A} = \begin{bmatrix} \frac{a}{k} & \frac{b}{k} & \frac{c}{k} \\ \frac{d}{k} & \frac{e}{k} & \frac{f}{k} \\ \frac{g}{k} & \frac{h}{k} & \frac{i}{k} \end{bmatrix} \quad (3)$$

Numerically, if $k = 2$, then

$$kA = \begin{bmatrix} 6 & 4 & 8 \\ 14 & 10 & 0 \\ 2 & 0 & 16 \end{bmatrix}$$

4 Addition and Subtraction of Two Matrices

These procedures are straightforward, as well as useful. If matrices **A** and **B** are as defined at the beginning of this appendix, one simply performs the addition or subtraction of corresponding elements.

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} + \begin{bmatrix} r & s & t \\ u & v & w \\ x & y & z \end{bmatrix} = \begin{bmatrix} a+r & b+s & c+t \\ d+u & e+v & f+w \\ g+x & h+y & i+z \end{bmatrix} \quad (4)$$

And

$$\mathbf{A} - \mathbf{B} = \begin{bmatrix} a-r & b-s & c-t \\ d-u & e-v & f-w \\ g-x & h-y & i-z \end{bmatrix} \quad (5)$$

For the numerical example:

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 3 & 2 & 4 \\ 7 & 5 & 0 \\ 1 & 0 & 8 \end{bmatrix} + \begin{bmatrix} 6 & 1 & 0 \\ 2 & 8 & 7 \\ 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 9 & 3 & 4 \\ 9 & 13 & 7 \\ 4 & 4 & 13 \end{bmatrix}$$

Calculation of a difference between two matrices is required when, for instance, one desires a residuals matrix, the matrix obtained by subtracting a reproduced matrix from an obtained matrix. Or if the matrix that is subtracted happens to consist of columns with appropriate means of variables inserted in every slot, then the difference between it and a matrix of raw scores produces a deviation matrix.

5 Multiplication, Transposes, and Square Roots of Matrices

Matrix multiplication is both unreasonably complicated and undeniably useful. Note that the ij th element of the resulting matrix is a function of row i of the first matrix and column j of the second.

$$\mathbf{AB} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} r & s & t \\ u & v & w \\ x & y & z \end{bmatrix} = \begin{bmatrix} ar+bu+cx & as+bv+cy & at+bw+cz \\ rd+eu+fx & ds+ev+fy & dt+ew+fz \\ gr+hu+ix & gs+hv+iy & gt+hw+iz \end{bmatrix} \quad (6)$$

Numerically,

$$\begin{aligned}
 \mathbf{AB} &= \begin{bmatrix} 3 & 2 & 4 \\ 7 & 5 & 0 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} 6 & 1 & 0 \\ 2 & 8 & 7 \\ 3 & 4 & 5 \end{bmatrix} \\
 &= \begin{bmatrix} 3 \cdot 6 + 2 \cdot 2 + 4 \cdot 3 & 3 \cdot 1 + 2 \cdot 8 + 4 \cdot 4 & 3 \cdot 0 + 2 \cdot 7 + 4 \cdot 5 \\ 7 \cdot 6 + 5 \cdot 2 + 0 \cdot 3 & 7 \cdot 1 + 5 \cdot 8 + 0 \cdot 4 & 7 \cdot 0 + 5 \cdot 7 + 0 \cdot 5 \\ 1 \cdot 6 + 0 \cdot 2 + 8 \cdot 3 & 1 \cdot 1 + 0 \cdot 8 + 8 \cdot 4 & 1 \cdot 0 + 0 \cdot 7 + 8 \cdot 5 \end{bmatrix} \\
 &= \begin{bmatrix} 34 & 35 & 34 \\ 52 & 47 & 35 \\ 30 & 33 & 40 \end{bmatrix}
 \end{aligned}$$

Regrettably, $\mathbf{AB} \neq \mathbf{BA}$ in matrix algebra. Thus

$$\mathbf{BA} = \begin{bmatrix} 6 & 1 & 0 \\ 2 & 8 & 7 \\ 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 3 & 2 & 4 \\ 7 & 5 & 0 \\ 1 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 25 & 17 & 24 \\ 69 & 44 & 64 \\ 42 & 26 & 52 \end{bmatrix}$$

If another concept of matrix algebra is introduced, some useful statistical properties of matrix algebra can be shown. The transpose of a matrix is indicated by a prime (') and stands for a rearrangement of the elements of the matrix such that the first row becomes the first column, the second row the second column, and so forth. Thus

$$\mathbf{A}' = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} = \begin{bmatrix} 3 & 7 & 1 \\ 2 & 5 & 0 \\ 4 & 0 & 8 \end{bmatrix} \quad (7)$$

When transposition is used in conjunction with multiplication, then some advantages of matrix multiplication become clear, namely,

$$\begin{aligned}
 \mathbf{AA}' &= \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} \\
 &= \begin{bmatrix} a^2 + b^2 + c^2 & ad + be + cf & ag + bh + ci \\ ad + be + cf & d^2 + e^2 + f^2 & dg + eh + fi \\ ag + bh + ci & dg + eh + fi & g^2 + h^2 + i^2 \end{bmatrix} \quad (8)
 \end{aligned}$$

The elements in the main diagonal are the sums of squares while those off the diagonal are cross products.

Had $\mathbf{A}$ been multiplied by itself, rather than by a transpose of itself, a different result would have been achieved.

$$\mathbf{AA} = \begin{bmatrix} a^2 + bd + cg & ab + be + ch & ac + bf + ci \\ da + ed + fg & db + e^2 + fh & dc + ef + fi \\ ga + hd + ig & gb + he + ih & gc + hf + i^2 \end{bmatrix}$$

If $\mathbf{AA} = \mathbf{C}$, then $\mathbf{C}^{1/2} = \mathbf{A}$. That is, there is a parallel in matrix algebra to squaring and taking the square root of a scalar, but it is a complicated business because of the complexity of matrix multiplication. If, however, one has a matrix $\mathbf{C}$ from which a square root is desired, one searches for a matrix, $\mathbf{A}$, which, when multiplied by itself, produces $\mathbf{C}$. If, for example,

$$\mathbf{C} = \begin{bmatrix} 27 & 16 & 44 \\ 56 & 39 & 28 \\ 11 & 2 & 68 \end{bmatrix}$$

then

$$\mathbf{C}^{1/2} = \begin{bmatrix} 3 & 2 & 4 \\ 7 & 5 & 0 \\ 1 & 0 & 8 \end{bmatrix}$$

6 Matrix “Division” (Inverses and Determinants)

If you liked matrix multiplication, you’ll love matrix inversion. Logically, the process is analogous to performing division for single numbers by finding the reciprocal of the number and multiplying by the reciprocal: if $a^{-1} = 1/a$, then $(a)(a^{-1}) = a/a = 1$. That is, the reciprocal of a scalar is a number that, when multiplied by the number itself, equals 1. Both the concepts and the notation are similar in matrix algebra, but they are complicated by the fact that a matrix is an array of numbers.

To determine if the reciprocal of a matrix has been found, one needs the matrix equivalent of the 1 as employed in the preceding paragraph. The identity matrix, $\mathbf{I}$, a matrix with 1s in the main diagonal and zeros elsewhere, is such a matrix. Thus

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{9}$$

Matrix division, then, becomes a process of finding $\mathbf{A}^{-1}$ such that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{AA}^{-1} = \mathbf{I} \tag{10}$$

One way of finding $\mathbf{A}^{-1}$ requires a two-stage process, the first of which consists of finding the determinant of $\mathbf{A}$, noted $|\mathbf{A}|$. The determinant of a matrix is sometimes said to represent the generalized variance of the matrix, as most readily seen in a 2×2 matrix. Thus we define a new matrix as follows:

$$\mathbf{D} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

where

$$|\mathbf{D}| = ad - bc \quad (11)$$

If $\mathbf{D}$ is a variance-covariance matrix where a and d are variances while b and c are covariances, then $ad - bc$ represents variance minus covariance. It is this property of determinants that makes them useful for hypothesis testing.

Calculation of determinants becomes rapidly more complicated as the matrix gets larger. For example, in our 3 by 3 matrix,

$$|\mathbf{A}| = a(ei - fh) + b(fg - di) + c(dh - eg) \quad (12)$$

Should the determinant of $\mathbf{A}$ equal 0, then the matrix cannot be inverted because the next operation in inversion would involve division by zero. Multicollinear or singular matrices have zero determinants that prohibit inversion.

A full inversion of $\mathbf{A}$ is

$$\begin{aligned} \mathbf{A}^{-1} &= \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}^{-1} \\ &= \frac{1}{|\mathbf{A}|} \begin{bmatrix} ei - fh & ch - bi & bf - ce \\ fg - di & ai - cg & cd - af \\ dh - eg & bg - ah & ae - bd \end{bmatrix} \end{aligned} \quad (13)$$

Please recall that because $\mathbf{A}$ is not a variance-covariance matrix, a negative determinant is possible, even somewhat likely. Thus, in the numerical example,

$$|\mathbf{A}| = 3(5 \cdot 8 - 0 \cdot 0) + 2(0 \cdot 1 - 7 \cdot 8) + 4(7 \cdot 0 - 5 \cdot 1) = -12$$

and

$$= \frac{1}{-12} \begin{bmatrix} 5 \cdot 8 - 0 \cdot 0 & 4 \cdot 0 - 2 \cdot 8 & 2 \cdot 0 - 4 \cdot 5 \\ 0 \cdot 1 - 7 \cdot 8 & 3 \cdot 8 - 4 \cdot 1 & 4 \cdot 7 - 3 \cdot 0 \\ 7 \cdot 0 - 5 \cdot 1 & 2 \cdot 1 - 3 \cdot 0 & 3 \cdot 5 - 2 \cdot 7 \end{bmatrix}$$

$$\begin{aligned}
 &= \begin{bmatrix} \frac{40}{-12} & \frac{-16}{-12} & \frac{-20}{-12} \\ \frac{-56}{-12} & \frac{20}{-12} & \frac{28}{-12} \\ \frac{-5}{-12} & \frac{2}{-12} & \frac{1}{-12} \end{bmatrix} \\
 &= \begin{bmatrix} -3.33 & 1.33 & 1.67 \\ 4.67 & -1.67 & -2.33 \\ 0.42 & -0.17 & -0.08 \end{bmatrix}
 \end{aligned}$$

Confirm that, within rounding error, Equation 10 in this appendix is true. Once the inverse of $\mathbf{A}$ is found, “division” by it is accomplished whenever required by using the inverse and performing matrix multiplication.

7 Eigenvalues and Eigenvectors: Procedures for Consolidating Variance From a Matrix

We promised you a demonstration of computation of eigenvalues and eigenvectors for a matrix, so here it is. However, you may well find that this discussion satisfies your appetite for only a couple of hours. During that time, round up Tatsuoka (1971), get the cat off your favorite chair, and prepare for an intelligible, if somewhat lengthy, description of the same subject.

Most of the multivariate procedures rely on eigenvalues and their corresponding eigenvectors (also called characteristic roots and vectors) in one way or another because they consolidate the variance in a matrix (the *eigenvalue*) while providing the linear combination of variables (the *eigenvector*) to do it. The coefficients applied to variables to form linear combinations of variables in all the multivariate procedures are rescaled elements from eigenvectors. The variance that the solution “accounts for” is associated with the eigenvalue, and is sometimes called so directly.

Calculation of eigenvalues and eigenvectors is best left up to a computer with any realistically sized matrix. For illustrative purposes, a 2×2 matrix will be used here. The logic of the process is also somewhat difficult, involving several of the more abstract notions and relations in matrix algebra, including the equivalence between matrices, systems of linear equations with several unknowns, and roots of polynomial equations.

Solution of an eigenproblem involves solution of the following equation:

$$(\mathbf{D} - \lambda \mathbf{I})\mathbf{V} = 0 \quad (14)$$

where λ is the eigenvalue and $\mathbf{V}$ the eigenvector to be sought. Expanded, this equation becomes

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

or

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

or, by applying Equation 5 in this Appendix,

$$\begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \quad (15)$$

If one considers the matrix $\mathbf{D}$, whose eigenvalues are sought, a variance-covariance matrix, one can see that a solution is desired to “capture” the variance in $\mathbf{D}$ while rescaling the elements in $\mathbf{D}$ by v_1 and v_2 to do so.

It is obvious from Equation 15 in this Appendix that a solution is always available when v_1 and v_2 are 0. A nontrivial solution may also be available when the determinant of the leftmost matrix in Equation 15 in this Appendix is 0.¹ That is, if (following Equation 11 in this Appendix)

$$(a - \lambda)(d - \lambda) - bc = 0 \quad (16)$$

then there may exist values of λ and values of v_1 and v_2 that satisfy the equation and are not 0. However, expansion of Equation 16 in this Appendix gives a polynomial equation, in λ , of degree 2:

$$\lambda^2 - (a + d)\lambda + ad - bc = 0 \quad (17)$$

Solving for the eigenvalues, λ , requires solving for the roots of this polynomial. If the matrix has certain properties (see footnote 1), there will be as many positive roots to the equation as there are rows (or columns) in the matrix.

If Equation 17 in this Appendix is rewritten as $x\lambda^2 + y\lambda + z = 0$, the roots may be found by applying the following equation:

$$\lambda = \frac{-y \pm \sqrt{y^2 - 4xz}}{2x} \quad (18)$$

For a numerical example, consider the following matrix.

$$\mathbf{D} = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Applying Equation 17 in this Appendix, we obtain

$$\lambda^2 - (5 + 2)\lambda + 5 \cdot 2 - 1 \cdot 4 = 0$$

or

$$\lambda^2 - 7\lambda + 6 = 0$$

¹Read Tatsuoka (1971); a matrix is said to be positive definite when all $\lambda_i > 0$, positive semidefinite when all $\lambda_i \geq 0$, and ill-conditioned when some $\lambda_i < 0$.

The roots to this polynomial may be found by Equation 18 in this Appendix as follows:

$$\lambda = \frac{-(-7) + \sqrt{(-7)^2 - 4 \cdot 1 \cdot 6}}{2 \cdot 1} = 6$$

and

$$\lambda = \frac{-(-7) - \sqrt{(-7)^2 - 4 \cdot 1 \cdot 6}}{2 \cdot 1} = 1$$

(The roots could also be found by factoring to get $[\lambda - 6][\lambda - 1]$.)

Once the roots are found, they may be used in Equation 15 in this Appendix to find v_1 and v_2 , the eigenvector. There will be one set of eigenvectors for the first root and a second set for the second root. Both solutions require solving sets of two simultaneous equations in two unknowns, to wit, for the first root, 6, and applying Equation 15 in this Appendix.

$$\begin{bmatrix} 5 - 6 & 1 \\ 4 & 2 - 6 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

or

$$\begin{bmatrix} -1 & 1 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

so that

$$-1v_1 + 1v_2 = 0$$

and

$$4v_1 - 4v_2 = 0$$

When $v_1 = 1$ and $v_2 = 1$, a solution is found.

For the second root, 1, the equations become

$$\begin{bmatrix} 5 - 1 & 1 \\ 4 & 2 - 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

or

$$\begin{bmatrix} 4 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

so that

$$4v_1 + 1v_2 = 0$$

and

$$4v_1 + 1v_2 = 0$$

When $v_1 = -1$ and $v_2 = 4$, a solution is found. Thus the first eigenvalue is 6, with $[1, 1]$ as a corresponding eigenvector, while the second eigenvalue is 1, with $[-1, 4]$ as a corresponding eigenvector.

Because the matrix was 2×2 , the polynomial for eigenvalues was quadratic and there were two equations in two unknowns to solve for eigenvectors. Imagine the joys of a matrix 15×15 , a polynomial with terms to the 15th power for the first half of the solution and 15 equations in 15 unknowns for the second half. A little more appreciation for your computer, please, next time you use it!

Introduction

1 Multivariate Statistics: Why?

Multivariate statistics are increasingly popular techniques used for analyzing complicated data sets. They provide analysis when there are many independent variables (IVs) and/or many dependent variables (DVs), all correlated with one another to varying degrees. Because of the difficulty in addressing complicated research questions with univariate analyses and because of the availability of canned software for performing multivariate analyses, multivariate statistics have become widely used. Indeed, a standard univariate statistics course only begins to prepare a student to read research literature or a researcher to produce it.

But how much harder are the multivariate techniques? Compared with the multivariate methods, univariate statistical methods are so straightforward and neatly structured that it is hard to believe they once took so much effort to master. Yet many researchers apply and correctly interpret results of intricate analysis of variance before the grand structure is apparent to them. The same can be true of multivariate statistical methods. Although we are delighted if you gain insights into the full multivariate general linear model, we have accomplished our goal if you feel comfortable selecting and setting up multivariate analyses and interpreting the computer output.

Multivariate methods are more complex than univariate by at least an order of magnitude. However, for the most part, the greater complexity requires few conceptual leaps. Familiar concepts such as sampling distributions and homogeneity of variance simply become more elaborate.

Multivariate models have not gained popularity by accident—or even by sinister design. Their growing popularity parallels the greater complexity of contemporary research. In psychology, for example, we are less and less enamored of the simple, clean, laboratory study, in which pliant, first-year college students each provides us with a single behavioral measure on cue.

1.1 The Domain of Multivariate Statistics: Numbers of IVs and DVs

Multivariate statistical methods are an extension of univariate and bivariate statistics. Multivariate statistics are the *complete* or general case, whereas univariate and bivariate statistics are special cases of the multivariate model. If your design has many variables, multivariate techniques often let you perform a single analysis instead of a series of univariate or bivariate analyses.

Variables are roughly dichotomized into two major types—independent and dependent. Independent variables (IVs) are the differing conditions (treatment vs. placebo) to which you expose your subjects, or the characteristics (tall or short) that the subjects themselves bring into the

research situation. IVs are usually considered predictor variables because they predict the DVs—the response or outcome variables. Note that IV and DV are defined within a research context; a DV in one research setting may be an IV in another.

Additional terms for IVs and DVs are predictor-criterion, stimulus-response, task-performance, or simply input-output. We use IV and DV throughout this chapter to identify variables that belong on one side of an equation or the other, without causal implication. That is, the terms are used for convenience rather than to indicate that one of the variables caused or determined the size of the other.

The term *univariate statistics* refers to analyses in which there is a single DV. There may be, however, more than one IV. For example, the amount of social behavior of graduate students (the DV) is studied as a function of course load (one IV) and type of training in social skills to which students are exposed (another IV). Analysis of variance is a commonly used univariate statistic.

Bivariate statistics frequently refers to analysis of two variables, where neither is an experimental IV and the desire is simply to study the relationship between the variables (e.g., the relationship between income and amount of education). Bivariate statistics, of course, can be applied in an experimental setting, but usually they are not. Prototypical examples of bivariate statistics are the Pearson product-moment correlation coefficient and chi-square analysis.

With multivariate statistics, you simultaneously analyze multiple dependent and multiple independent variables. This capability is important in both nonexperimental (correlational or survey) and experimental research.

1.2 Experimental and Nonexperimental Research

A critical distinction between experimental and nonexperimental research is whether the researcher manipulates the levels of the IVs. In an experiment, the researcher has control over the levels (or conditions) of at least one IV to which a subject is exposed by determining what the levels are, how they are implemented, and how and when cases are assigned and exposed to them. Further, the experimenter randomly assigns subjects to levels of the IV and controls all other influential factors by holding them constant, counterbalancing, or randomizing their influence. Scores on the DV are expected to be the same, within random variation, except for the influence of the IV (Campbell & Stanley, 1966). If there are systematic differences in the DV associated with levels of the IV, these differences are attributed to the IV.

For example, if groups of undergraduates are randomly assigned to the same material but different types of teaching techniques, and afterward some groups of undergraduates perform better than others, the difference in performance is said, with some degree of confidence, to be caused by the difference in teaching technique. In this type of research, the terms *independent* and *dependent* have obvious meaning: the value of the DV depends on the manipulated level of the IV. The IV is manipulated by the experimenter and the score on the DV depends on the level of the IV.

In nonexperimental (correlational or survey) research, the levels of the IV(s) are not manipulated by the researcher. The researcher can define the IV, but has no control over the assignment of subjects to levels of it. For example, groups of people may be categorized into geographic area of residence (Northeast, Midwest, etc.), but only the definition of the variable is under researcher control. Except for the military or prison, place of residence is rarely subject to manipulation by a researcher. Nevertheless, a naturally occurring difference like this is often considered an IV and is

used to predict some other nonexperimental (dependent) variable such as income. In this type of research, the distinction between IVs and DVs is usually arbitrary and many researchers prefer to call IVs *predictors* and DVs *criterion variables*.

In nonexperimental research, it is very difficult to attribute causality to an IV. If there is a systematic difference in a DV associated with levels of an IV, the two variables are said (with some degree of confidence) to be related, but the cause of the relationship is unclear. For example, income as a DV might be related to geographic area, but no causal association is implied.

Nonexperimental research takes many forms, but a common example is the survey. Typically, many people are surveyed, and each respondent provides answers to many questions, producing a large number of variables. These variables are usually interrelated in highly complex ways, but univariate and bivariate statistics are not sensitive to this complexity. Bivariate correlations between all pairs of variables, for example, could not reveal that the 20 to 25 variables measured really represent only two or three “supervariables.”

If a research goal is to distinguish among subgroups in a sample (e.g., between Catholics and Protestants) on the basis of a variety of attitudinal variables, we could use several univariate *t* tests (or analyses of variance) to examine group differences on each variable separately. But if the variables are related, which is highly likely, the results of many *t* tests are misleading and statistically suspect.

With the use of multivariate statistical techniques, complex interrelationships among variables are revealed and assessed in statistical inference. Further, it is possible to keep the overall Type I error rate at, say, 5%, no matter how many variables are tested.

Although most multivariate techniques were developed for use in nonexperimental research, they are also useful in experimental research, in which there may be multiple IVs and multiple DVs. With multiple IVs, the research is usually designed so that the IVs are independent of each other and a straightforward correction for numerous statistical tests is available. With multiple DVs, a problem of inflated error rate arises if each DV is tested separately. Further, at least some of the DVs are likely to be correlated with each other, so separate tests of each DV reanalyze some of the same variance. Therefore, multivariate tests are used.

Experimental research designs with multiple DVs were unusual at one time. Now, however, with attempts to make experimental designs more realistic, and with the availability of computer programs, experiments often have several DVs. It is dangerous to run an experiment with only one DV and risk missing the impact of the IV because the most sensitive DV is not measured. Multivariate statistics help the experimenter design more efficient and more realistic experiments by allowing measurement of multiple DVs without violation of acceptable levels of Type I error.

One of the few considerations not relevant to choice of *statistical* technique is whether the data are experimental or correlational. The statistical methods “work” whether the researcher manipulated the levels of the IV or not. But attribution of causality to results is crucially affected by the experimental–nonexperimental distinction.

1.3 Computers and Multivariate Statistics

One answer to the question “Why multivariate statistics?” is that the techniques are now accessible by computer. Only the most dedicated number cruncher would consider doing real-life-sized problems in multivariate statistics without a computer. Fortunately, excellent multivariate programs are available in a number of computer packages.

Two packages are demonstrated in this chapter. Examples are based on programs in IBM SPSS and SAS.

If you have access to both packages, you are indeed fortunate. Programs within the packages do not completely overlap, and some problems are better handled through one package than the other. For example, doing several versions of the same basic analysis on the same set of data is particularly easy with IBM SPSS, whereas SAS has the most extensive capabilities for saving derived scores from data screening or from intermediate analyses.

Recent versions of the multivariate programs are implemented in Windows. All multivariate techniques may be implemented through syntax, and syntax itself is generated through menus. Then you may add or change syntax as desired for your analysis. For example, you may “paste” menu choices into a syntax window in IBM SPSS, edit the resulting text, and then run the program. Also, syntax generated by IBM SPSS menus is saved in the “journal” file (statistics.jnl), which may also be accessed and copied into a syntax window. Syntax generated by SAS menus is recorded in a “log” file. The contents may then be copied to an interactive window, edited, and run. Do not overlook the help files in these programs. Indeed, SAS and IBM SPSS now provide the entire set of user manuals on CD, often with more current information than is available in printed manuals.

With commercial computer packages, you need to know which version of the package you are using. Programs are continually being changed, and not all changes are immediately implemented at each facility. Therefore, many versions of the various programs are simultaneously in use at different institutions; even at one institution, more than one version of a package is sometimes available.

Program updates are often corrections of errors discovered in earlier versions. Occasionally, though, there are major revisions in one or more programs or a new program is added to the package. Sometimes defaults change with updates, so that the output looks different although syntax is the same. Check to find out which version of each package you are using. Then be sure that the manual you are using is consistent with the version in use at your facility. Also check updates for error correction in previous releases that may be relevant to some of your previous runs.

Information on availability and versions of software, macros, books, and the like changes almost daily. We recommend the Internet as a source of “keeping up.”

1.4 Garbage In, Roses Out?

The trick in multivariate statistics is not in computation. This is easily done as discussed above. The trick is to select reliable and valid measurements, choose the appropriate program, use it correctly, and know how to interpret the output. Output from commercial computer programs, with their beautifully formatted tables, graphs, and matrices, can make garbage look like roses.

Second, when you use multivariate statistics, you rarely get as close to the raw data as you do when you apply univariate statistics to a relatively few cases. Errors and anomalies in the data that would be obvious if the data were processed by hand are less easy to spot when processing is entirely by computer. But the computer packages have programs to graph and describe your data in the simplest univariate terms and to display bivariate relationships among your variables. These programs provide preliminary analyses that are absolutely necessary if the results of multivariate programs are to be believed.

There are also certain costs associated with the benefits of using multivariate procedures. Benefits of increased flexibility in research design, for instance, are sometimes paralleled by increased ambiguity in interpretation of results. In addition, multivariate results can be quite sensitive to which analytic strategy is chosen (cf. Section 2.4) and do not always provide better protection against statistical errors than their univariate counterparts. Add to this the fact that occasionally you still cannot get a firm statistical answer to your research questions, and you may wonder if the increase in complexity and difficulty is warranted.

Frankly, we think it is. Slippery as some of the concepts and procedures are, these statistics provide insights into relationships among variables that may more closely resemble the complexity of the “real” world. And sometimes you get at least partial answers to questions that could not be asked at all in the univariate framework. For a complete analysis, making sense of your data usually requires a judicious mix of multivariate and univariate statistics.

The addition of multivariate statistical methods to your repertoire makes data analysis a lot more fun. If you liked univariate statistics, you will love multivariate statistics!¹

2 Some Useful Definitions

In order to describe multivariate statistics easily, it is useful to review some common terms in research design and basic statistics. Distinctions were made between IVs and DVs and between experimental and nonexperimental research in preceding sections.

2.1 Continuous, Discrete, and Dichotomous Data

In applying statistical techniques of any sort, it is important to consider the type of measurement and the nature of the correspondence between the numbers and the events that they represent. The distinction made here is among continuous, discrete, and dichotomous variables; you may prefer to

¹Don’t even think about it.

substitute the terms *interval* or *quantitative* for *continuous* and *nominal*, *categorical* or *qualitative* for *dichotomous* and *discrete*.

Continuous variables are measured on a scale that changes values smoothly rather than in steps. Continuous variables take on any values within the range of the scale, and the size of the number reflects the amount of the variable. Precision is limited by the measuring instrument, not by the nature of the scale itself. Some examples of continuous variables are time as measured on an old-fashioned analog clock face, annual income, age, temperature, distance, and grade point average (GPA).

Discrete variables take on a finite and usually small number of values, and there is no smooth transition from one value or category to the next. Examples include time as displayed by a digital clock, continents, categories of religious affiliation, and type of community (rural or urban).

Sometimes discrete variables are used in multivariate analyses as if continuous if there are numerous categories and the categories represent a quantitative attribute. For instance, a variable that represents age categories (where, say, 1 stands for 0 to 4 years, 2 stands for 5 to 9 years, 3 stands for 10 to 14 years, and so on up through the normal age span) can be used because there are a lot of categories and the numbers designate a quantitative attribute (increasing age). But the same numbers used to designate categories of religious affiliation are not in appropriate form for analysis with many of the techniques² because religions do not fall along a quantitative continuum.

Discrete variables composed of qualitatively different categories are sometimes analyzed after being changed into a number of dichotomous or two-level variables (e.g., Catholic vs. non-Catholic, Protestant vs. non-Protestant, Jewish vs. non-Jewish, and so on until the degrees of freedom are used). Recategorization of a discrete variable into a series of dichotomous ones is called *dummy variable coding*. The conversion of a discrete variable into a series of dichotomous ones is done to limit the relationship between the dichotomous variables and others to linear relationships. A discrete variable with more than two categories can have a relationship of any shape with another variable, and the relationship is changed arbitrarily if the assignment of numbers to categories is changed. Dichotomous variables, however, with only two points, can have only linear relationships with other variables; they are, therefore, appropriately analyzed by methods using correlation in which only linear relationships are analyzed.

The distinction between continuous and discrete variables is not always clear. If you add enough digits to the digital clock, for instance, it becomes for all practical purposes a continuous measuring device, whereas time as measured by the analog device can also be read in discrete categories such as hours or half hours. In fact, any continuous measurement may be rendered discrete (or dichotomous) with some loss of information, by specifying cutoffs on the continuous scale.

The property of variables that is crucial to the application of multivariate procedures is not the type of measurement so much as the shape of distribution. Non-normally distributed continuous variables and dichotomous variables with very uneven splits between the categories present problems to several of the multivariate analyses.

Another type of measurement that is used sometimes produces a rank order (ordinal) scale. This scale assigns a number to each subject to indicate the subject's position vis-à-vis other subjects along some dimension. For instance, ranks are assigned to contestants (first place, second place, third place, etc.) to provide an indication of who is the best—but not by how much. A problem with

²Some multivariate techniques (e.g., logistic regression, SEM) are appropriate for all types of variables.

ordinal measures is that their distributions are rectangular (one frequency per number) instead of normal, unless tied ranks are permitted and they pile up in the middle of the distribution.

In practice, we often treat variables as if they are continuous when the underlying scale is thought to be continuous, but the measured scale actually is ordinal, the number of categories is large—say, seven or more, and the data meet other assumptions of the analysis. For instance, the number of correct items on an objective test is technically not continuous because fractional values are not possible, but it is thought to measure some underlying continuous variable such as course mastery. Another example of a variable with ambiguous measurement is one measured on a Likert-type scale, in which consumers rate their attitudes toward a product as “strongly like,” “moderately like,” “mildly like,” “neither like nor dislike,” “mildly dislike,” “moderately dislike,” or “strongly dislike.” As mentioned previously, even dichotomous variables may be treated as if continuous under some conditions. Thus, we often use the term “*continuous*” throughout the remainder of this chapter, whether the measured scale itself is continuous or the variable is to be treated as if continuous. We use the term “*discrete*” for variables with a few categories, whether the categories differ in type or quantity.

2.2 Samples and Populations

Samples are measured to make generalizations about populations. Ideally, samples are selected, usually by some random process, so that they represent the population of interest. In real life, however, populations are frequently best defined in terms of samples, rather than vice versa; the population is the group from which you were able to randomly sample.

Sampling has somewhat different connotations in nonexperimental and experimental research. In nonexperimental research, you investigate relationships among variables in some predefined population. Typically, you take elaborate precautions to ensure that you have achieved a representative sample of that population; you define your population, and then do your best to randomly sample from it.³

In experimental research, you attempt to create different populations by treating subgroups from an originally homogeneous group differently. The sampling objective here is to ensure that all subjects come from the same population before you treat them differently. Random sampling consists of randomly assigning subjects to treatment groups (levels of the IV) to ensure that, before differential treatment, all subsamples come from the same population. Statistical tests provide evidence as to whether, after treatment, all samples still come from the same population. Generalizations about treatment effectiveness are made to the type of subjects who participated in the experiment.

2.3 Descriptive and Inferential Statistics

Descriptive statistics describe samples of subjects in terms of variables or combinations of variables. Inferential statistical techniques test hypotheses about differences in populations on the basis of measurements made on samples of subjects. If reliable differences are found, descriptive statistics are then used to provide estimations of central tendency, and the like, in the population. Descriptive statistics used in this way are called *parameter estimates*.

³Strategies for random sampling are discussed in many sources, including Levy and Lemenshow (1999), Rea and Parker (1997), and de Vaus (2002).

Use of inferential and descriptive statistics is rarely an either–or proposition. We are usually interested in both describing and making inferences about a data set. We describe the data, find reliable differences or relationships, and estimate population values for the reliable findings. However, there are more restrictions on inference than there are on description. Many assumptions of multivariate statistical methods are necessary only for inference. If simple description of the sample is the major goal, many assumptions are relaxed.

2.4 Orthogonality: Standard and Sequential Analyses

Orthogonality is a perfect nonassociation between variables. If two variables are orthogonal, knowing the value of one variable gives no clue as to the value of the other; the correlation between them is zero.

Orthogonality is often desirable in statistical applications. For instance, factorial designs for experiments are orthogonal when two or more IVs are completely crossed with equal sample sizes in each combination of levels. Except for use of a common error term, tests of hypotheses about main effects and interactions are independent of each other; the outcome of each test gives no hint as to the outcome of the others. In orthogonal experimental designs with random assignment of subjects, manipulation of the levels of the IV, and good controls, changes in value of the DV can be unambiguously attributed to various main effects and interactions.

Similarly, in multivariate analyses, there are advantages if sets of IVs or DVs are orthogonal. If all pairs of IVs in a set are orthogonal, each IV adds, in a simple fashion, to prediction of the DV. Consider income as a DV with education and occupational prestige as IVs. If education and occupational prestige are orthogonal, and if 35% of the variability in income may be predicted from education and a different 45% is predicted from occupational prestige, then 80% of the variance in income is predicted from education and occupational prestige together.

Orthogonality can easily be illustrated in Venn diagrams, as shown in Figure 1. Venn diagrams represent shared variance (or correlation) as overlapping areas between two (or more) circles. The total variance for income is one circle. The section with horizontal stripes represents the part of income predictable from education, and the section with vertical stripes represents the part predictable from occupational prestige; the circle for education overlaps the circle for income 35% and the circle for occupational prestige overlaps 45%. Together, they account for 80% of the variability in income because education and occupational prestige are orthogonal and do not themselves overlap. There are similar advantages if a set of DVs is orthogonal. The overall effect of an IV can be partitioned into effects on each DV in an additive fashion.

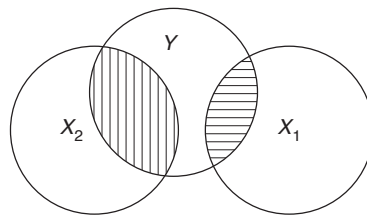


FIGURE 1 Venn diagram for Y (income), X_1 (education), and X_2 (occupational prestige).

Usually, however, the variables are correlated with each other (nonorthogonal). IVs in nonexperimental designs are often correlated naturally; in experimental designs, IVs become correlated when unequal numbers of subjects are measured in different cells of the design. DVs are usually correlated because individual differences among subjects tend to be consistent over many attributes.

When variables are correlated, they have shared or overlapping variance. In the example of Figure 2, education and occupational prestige correlate with each other. Although the independent contribution made by education is still 35% and that by occupational prestige is 45%, their joint contribution to prediction of income is not 80%, but rather something smaller due to the overlapping area shown by the arrow in Figure 2(a). A major decision for the multivariate analyst is how to handle the variance that is predictable from more than one variable. Many multivariate techniques have at least two strategies for handling it, but some have more.

In standard analysis, the overlapping variance contributes to the size of summary statistics of the overall relationship but is not assigned to either variable. Overlapping variance is disregarded in assessing the contribution of each variable to the solution. Figure 2(a) is a Venn diagram of a standard analysis in which overlapping variance is shown as overlapping areas in circles; the unique contributions of X_1 and X_2 to prediction of Y are shown as horizontal and vertical areas, respectively, and the total relationship between Y and the combination of X_1 and X_2 is those two areas plus the area with the arrow. If X_1 is education and X_2 is occupational prestige, then in standard analysis, X_1 is “credited with” the area marked by the horizontal lines and X_2 by the area marked by vertical lines. Neither of the IVs is assigned the area designated with the arrow. When X_1 and X_2 substantially overlap each other, very little horizontal or vertical area may be left for either of them, despite the fact that they are both related to Y . They have essentially knocked each other out of the solution.

Sequential analyses differ, in that the researcher assigns priority for entry of variables into equations, and the first one to enter is assigned both unique variance and any overlapping variance it has with other variables. Lower-priority variables are then assigned on entry their unique and any remaining overlapping variance. Figure 2(b) shows a sequential analysis for the same case as Figure 2(a), where X_1 (education) is given priority over X_2 (occupational prestige). The total variance explained is the same as in Figure 2(a), but the relative contributions of X_1 and X_2 have changed;

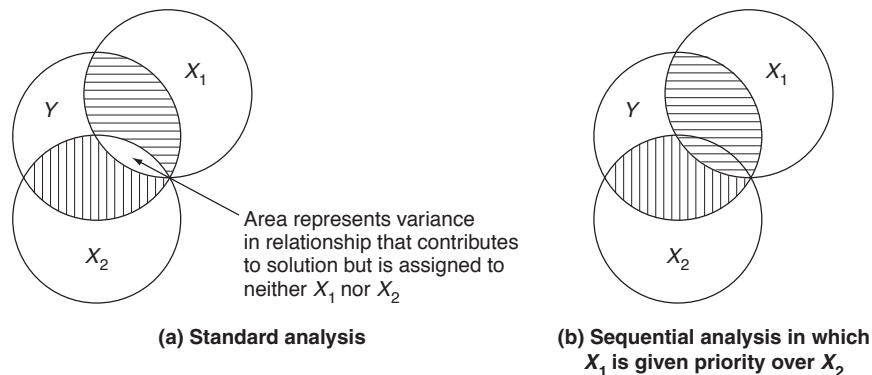


FIGURE 2 Standard (a) and sequential (b) analyses of the relationship between Y , X_1 , and X_2 . Horizontal shading depicts variance assigned to X_1 . Vertical shading depicts variance assigned to X_2 .

education now shows a stronger relationship with income than in the standard analysis, whereas the relation between occupational prestige and income remains the same.

The choice of strategy for dealing with overlapping variance is not trivial. If variables are correlated, the overall relationship remains the same, but the apparent importance of variables to the solution changes depending on whether a standard or a sequential strategy is used. If the multivariate procedures have a reputation for unreliability, it is because solutions change, sometimes dramatically, when different strategies for entry of variables are chosen. However, the strategies also ask different questions of the data, and it is incumbent on the researcher to determine exactly which question to ask.

3 Linear Combinations of Variables

Multivariate analyses combine variables to do useful work, such as predict scores or predict group membership. The combination that is formed depends on the relationships among the variables and the goals of analysis, but in most cases, the combination is linear. A linear combination is one in which each variable is assigned a weight (e.g., W_1), and then the products of weights and the variable scores are summed to predict a score on a combined variable. In Equation 1, Y' (the predicted DV) is predicted by a linear combination of X_1 and X_2 (the IVs).

$$Y' = W_1X_1 + W_2X_2 \quad (1)$$

If, for example, Y' is predicted income, X_1 is education, and X_2 is occupational prestige, the best prediction of income is obtained by weighting education (X_1) by W_1 and occupational prestige (X_2) by W_2 before summing. No other values of W_1 and W_2 produce as good a prediction of income.

Notice that Equation 1 includes neither X_1 nor X_2 raised to powers (exponents) nor a product of X_1 and X_2 . This seems to severely restrict multivariate solutions until one realizes that X_1 could itself be a product of two different variables or a single variable raised to a power. For example, X_1 might be education squared. A multivariate solution does not produce exponents or cross-products of IVs to improve a solution, but the researcher can include X s that are cross-products of IVs or are IVs raised to powers. Inclusion of variables raised to powers or cross-products of variables has both theoretical and practical implications for the solution. Berry (1993) provides a useful discussion of many of the issues.

The size of the W values (or some function of them) often reveals a great deal about the relationship between DVs and IVs. If, for instance, the W value for some IV is zero, the IV is not needed in the best DV–IV relationship. Or if some IV has a large W value, then the IV tends to be important to the relationship. Although complications prevent interpretation of the multivariate solution from the sizes of the W values alone, they are nonetheless important in most multivariate procedures.

The combination of variables can be considered a supervariable, not directly measured but worthy of interpretation. The supervariable may represent an underlying dimension that predicts something or optimizes some relationship. Therefore, the attempt to understand the meaning of the combination of IVs is worthwhile in many multivariate analyses.

In the search for the best weights to apply in combining variables, computers do not try out all possible sets of weights. Various algorithms have been developed to compute the weights. Most

algorithms involve manipulation of a correlation matrix, a variance–covariance matrix, or a sum-of-squares and cross-products matrix. Section 6 describes these matrices in very simple terms and shows their development from a very small data set. Appendix ‘A Skimpy Introduction to Matrix Algebra’ describes some terms and manipulations appropriate to matrices.

4 Number and Nature of Variables to Include

Attention to the number of variables included in analysis is important. A general rule is to get the best solution with the fewest variables. As more and more variables are included, the solution usually improves, but only slightly. Sometimes the improvement does not compensate for the cost in degrees of freedom of including more variables, so the power of the analyses diminishes.

A second problem is *overfitting*. With overfitting, the solution is very good; so good, in fact, that it is unlikely to generalize to a population. Overfitting occurs when too many variables are included in an analysis relative to the sample size. With smaller samples, very few variables can be analyzed. Generally, a researcher should include only a limited number of uncorrelated variables in each analysis,⁴ fewer with smaller samples.

Additional considerations for inclusion of variables in a multivariate analysis include cost, availability, meaning, and theoretical relationships among the variables. Except in analysis of structure, one usually wants a small number of valid, cheaply obtained, easily available, uncorrelated variables that assess all the theoretically important dimensions of a research area. Another important consideration is reliability. How stable is the position of a given score in a distribution of scores when measured at different times or in different ways? Unreliable variables degrade an analysis, whereas reliable ones enhance it. A few reliable variables give a more meaningful solution than a large number of less reliable variables. Indeed, if variables are sufficiently unreliable, the entire solution may reflect only measurement error. Further considerations for variable selection are mentioned as they apply to each analysis.

5 Statistical Power

A critical issue in designing any study is whether there is adequate power. Power, as you may recall, represents the probability that effects that actually exist have a chance of producing statistical significance in your eventual data analysis. For example, do you have a large enough sample size to show a significant relationship between GRE and GPA if the actual relationship is fairly large? What if the relationship is fairly small? Is your sample large enough to reveal significant effects of treatment on your DV(s)?

⁴The exceptions are analysis of structure, such as factor analysis, in which numerous correlated variables are measured.

Issues of power are best considered in the planning state of a study when the researcher determines the required sample size. The researcher estimates the size of the anticipated effect (e.g., an expected mean difference), the variability expected in assessment of the effect, the desired alpha level (ordinarily .05), and the desired power (often .80). These four estimates are required to determine the necessary sample size. Failure to consider power in the planning stage often results in failure to find a significant effect (and an unpublishable study). The interested reader may wish to consult Cohen (1965, 1988), Rossi (1990), or Sedlmeier and Gigerenzer (1989) for more detail.

There is a great deal of software available to help you estimate the power available with various sample sizes for various statistical techniques, and to help you determine necessary sample size given a desired level of power (e.g., an 80% probability of achieving a significant result if an effect exists) and expected sizes of relationships. One of these programs that estimates power for several techniques is NCSS PASS (Hintze, 2011). Many other programs are reviewed (and sometimes available as shareware) on the Internet.

6 Data Appropriate for Multivariate Statistics

An appropriate data set for multivariate statistical methods consists of values on a number of variables for each of several subjects or cases. For continuous variables, the values are scores on variables. For example, if the continuous variable is the GRE (Graduate Record Examination), the values for the various subjects are scores such as 500, 420, 650, and so on. For discrete variables, values are number codes for group membership or treatment. For example, if there are three teaching techniques, students who receive one technique are arbitrarily assigned a “1,” those receiving another technique are assigned a “2,” and so on.

6.1 The Data Matrix

The data matrix is an organization of scores in which rows (lines) represent subjects and columns represent variables. An example of a data matrix with six subjects⁵ and four variables is given in Table 1. For example, X_1 might be type of teaching technique, X_2 score on the GRE, X_3 GPA, and X_4 gender, with women coded 1 and men coded 2.

TABLE 1 A Data Matrix of Hypothetical Scores

Student	X_1	X_2	X_3	X_4
1	1	500	3.20	1
2	1	420	2.50	2
3	2	650	3.90	1
4	2	550	3.50	2
5	3	480	3.30	1
6	3	600	3.25	2

⁵Normally, of course, there are many more than six subjects.

Data are entered into a data file with long-term storage accessible by computer in order to apply computer techniques to them. Each subject starts with a new row (line). Information identifying the subject is typically entered first, followed by the value of each variable for that subject.

Scores for each variable are entered in the same order for each subject. If there are more data for each subject than can be accommodated on a single line, the data are continued on additional lines, but all of the data for each subject are kept together. All of the computer package manuals provide information on setting up a data matrix.

In this example, there are values for every variable for each subject. This is not always the case with research in the real world. With large numbers of subjects and variables, scores are frequently missing on some variables for some subjects. For instance, respondents may refuse to answer some kinds of questions, or some students may be absent the day when a particular test is given, and so forth. This creates missing values in the data matrix. To deal with missing values, first build a data file in which some symbol is used to indicate that a value on a variable is missing in data for a subject. The various programs have standard symbols, such as a dot (.), for this purpose. You can also use other symbols, but it is often just as convenient to use one of the default symbols.

6.2 The Correlation Matrix

Most readers are familiar with **R**, a correlation matrix. **R** is a square, symmetrical matrix. Each row (and each column) represents a different variable, and the value at the intersection of each row and column is the correlation between the two variables. For instance, the value at the intersection of the second row, third column, is the correlation between the second and the third variables. The same correlation also appears at the intersection of the third row, second column. Thus, correlation matrices are said to be symmetrical about the main diagonal, which means they are mirror images of themselves above and below the diagonal from top left to bottom right. Hence, it is common practice to show only the bottom half or the top half of an *R* matrix. The entries in the main diagonal are often omitted as well, since they are all ones—correlations of variables with themselves.⁶

Table 2 shows the correlation matrix for X_2 , X_3 , and X_4 of Table 1. The value .85 is the correlation between X_2 and X_3 and it appears twice in the matrix (as do other values). Other correlations are as indicated in the table.

Many programs allow the researcher a choice between analysis of a correlation matrix and analysis of a variance–covariance matrix. If the correlation matrix is analyzed, a unit-free result is produced. That is, the solution reflects the relationships among the variables but not in the metric in which they are measured. If the metric of the scores is somewhat arbitrary, analysis of **R** is appropriate.

TABLE 2 Correlation Matrix for Part of Hypothetical Data for Table 1

	X_2	X_3	X_4
X_2	1.00	.85	-.13
R = X_3	.85	1.00	-.46
X_4	-.13	-.46	1.00

⁶Alternatively, other information such as standard deviations is inserted.

6.3 The Variance–Covariance Matrix

If scores are measured along a meaningful scale, it is sometimes appropriate to analyze a variance–covariance matrix. A variance–covariance matrix, Σ , is also square and symmetrical, but the elements in the main diagonal are the variances of each variable, and the off-diagonal elements are covariances between pairs of different variables.

Variances, as you recall, are averaged squared deviations of each score from the mean of the scores. Since the deviations are averaged, the number of scores included in computation of a variance is not relevant, but the metric in which the scores are measured is relevant. Scores measured in large numbers tend to have large numbers as variances, and scores measured in small numbers tend to have small variances.

Covariances are averaged cross-products (product of the deviation between one variable and its mean and the deviation between a second variable and its mean). Covariances are similar to correlations except that they, like variances, retain information concerning the scales in which the variables are measured. The variance–covariance matrix for the continuous data in Table 1 appears in Table 3.

6.4 The Sum-of-Squares and Cross-Products Matrix

The matrix, $\mathbf{S}$, is a precursor to the variance–covariance matrix in which deviations are not yet averaged. Thus, the size of the entries depends on the number of cases as well as on the metric in which the elements were measured. The sum-of-squares and cross-products matrix for X_2 , X_3 , and X_4 in Table 1 appears in Table 4.

The entry in the major diagonal of the matrix $\mathbf{S}$ is the sum of squared deviations of scores from the mean for that variable, hence, “sum of squares,” or SS. That is, for each variable, the value in the major diagonal is

$$SS(X_j) = \sum_{i=1}^N (X_{ij} - \bar{X}_j)^2 \quad (2)$$

where $i = 1, 2, \dots, N$
 N = the number of subjects
 j = the variable identifier
 X_{ij} = the score on variable j by subject i
 $\bar{X}_j$ = the mean of all scores on the j th variable

TABLE 3 Variance–Covariance Matrix for Part of Hypothetical Data of Table 1

	X_2	X_3	X_4
X_2	7026.66	32.80	−6.00
X_3	32.80	.21	−.12
X_4	−6.00	−.12	.30

TABLE 4 Sum-of-Squares and Cross-Products Matrix for Part of Hypothetical Data of Table 1

	X_2	X_3	X_4
X_2	35133.33	164.00	−30.00
X_3	164.00	1.05	−0.58
X_4	−30.00	−0.58	1.50

Introduction

For example, for X_4 , the mean is 1.5. The sum of squared deviations around the mean and the diagonal value for the variable is

$$\begin{aligned}\sum_{i=1}^6 (X_{i4} - \bar{X}_4)^2 &= (1 - 1.5)^2 + (2 - 1.5)^2 + (1 - 1.5)^2 + (2 - 1.5)^2 + (1 - 1.5)^2 + (2 - 1.5)^2 \\ &= 1.50\end{aligned}$$

The off-diagonal elements of the sum-of-squares and cross-products matrix are the cross-products—the sum of products (SP)—of the variables. For each pair of variables, represented by row and column labels in Table 4, the entry is the sum of the product of the deviation of one variable around its mean times the deviation of the other variable around its mean.

$$SP(X_j X_k) = \sum_{i=1}^N (X_{ij} - \bar{X}_j)(X_{ik} - \bar{X}_k) \quad (3)$$

where j identifies the first variable, k identifies the second variable, and all other terms are as defined in Equation 1. (Note that if $j = k$, Equation 3 becomes identical to Equation 2.)

For example, the cross-product term for variables X_2 and X_3 is

$$\begin{aligned}\sum_{i=1}^N (X_{i2} - \bar{X}_2)(X_{i3} - \bar{X}_3) &= (500 - 533.33)(3.20 - 3.275) + (420 - 533.33)(2.50 - 3.275) \\ &\quad + \dots + (600 - 533.33)(3.25 - 3.275) = 164.00\end{aligned}$$

Most computations start with **S** and proceed to Σ or **R**. The progression from a sum-of-squares and cross-products matrix to a variance–covariance matrix is simple.

$$\Sigma = \frac{1}{N - 1} \mathbf{S} \quad (4)$$

The variance–covariance matrix is produced by dividing every element in the sum-of-squares and cross-products matrix by $N - 1$, where N is the number of cases.

The correlation matrix is derived from an **S** matrix by dividing each sum-of-squares by itself (to produce the 1s in the main diagonal of **R**) and each cross-product of the **S** matrix by the square root of the product of the sum-of-squared deviations around the mean for each of the variables in the pair. That is, each cross-product is divided by

$$\text{Denominator}(X_j X_k) = \sqrt{\sum (X_{ij} - \bar{X}_j)^2 \sum (X_{ik} - \bar{X}_k)^2} \quad (5)$$

where terms are defined as in Equation 3.

For some multivariate operations, it is not necessary to feed the data matrix to a computer program. Instead, an **S** or an **R** matrix is entered, with each row (representing a variable) starting a

new line. Often, considerable computing time and expense are saved by entering one or the other of these matrices rather than raw data.

6.5 Residuals

Often a goal of analysis or test of its efficiency is its ability to reproduce the values of a DV or the correlation matrix of a set of variables. For example, we might want to predict scores on the GRE (X_2) of Table 1 from knowledge of GPA (X_3) and gender (X_4). After applying the proper statistical operations—a multiple regression in this case—a predicted GRE score for each student is computed by applying the proper weights for GPA and gender to the GPA, and gender scores for each student. But because we already obtained GRE scores for the sample of students, we are able to compare the predicted score with the obtained GRE score. The difference between the predicted and obtained values is known as the *residual* and is a measure of error of prediction.

In most analyses, the residuals for the entire sample sum to zero. That is, sometimes the prediction is too large and sometimes it is too small, but the average of all the errors is zero. The squared value of the residuals, however, provides a measure of how good the prediction is. When the predictions are close to the obtained values, the squared errors are small. The way that the residuals are distributed is of further interest in evaluating the degree to which the data meet the assumptions of multivariate analyses.

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A Guide to Statistical Techniques

1 Research Questions and Associated Techniques

This chapter organizes the statistical techniques by major research question. A decision tree at the end of this chapter leads you to an appropriate analysis for your data. On the basis of your major research question and a few characteristics of your data set, you determine which statistical technique(s) is appropriate. The first and the most important criterion for choosing a technique is the major research question to be answered by the statistical analysis. Here, the research questions are categorized into degree of relationship among variables, significance of group differences, prediction of group membership, structure, and questions that focus on the time course of events. This chapter emphasizes differences in research questions answered by the different techniques described in nontechnical terms.

1.1 Degree of Relationship Among Variables

If the major purpose of analysis is to assess the associations among two or more variables, some form of correlation/regression or chi-square is appropriate. The choice among five different statistical techniques is made by determining the number of independent and dependent variables, the nature of the variables (continuous or discrete), and whether any of the IVs are best conceptualized as covariates.¹

1.1.1 Bivariate r

Bivariate correlation and regression assess the degree of relationship between two continuous variables, such as belly dancing skill and years of musical training. Bivariate correlation measures the association between two variables with no distinction necessary between IV and DV. Bivariate regression, on the other hand, predicts a score on one variable from knowledge of the score on another variable (e.g., predicts skill in belly dancing as measured by a single index, such as knowledge of steps, from a single predictor, such as years of musical training).

The predicted variable is considered the DV, whereas the predictor is considered the IV. Bivariate correlation and regression are not multivariate techniques, but they are integrated into the general linear model.

¹If the effects of some IVs are assessed after the effects of other IVs are statistically removed, the latter are called *covariates*.

1.1.2 Multiple R

Multiple correlation assesses the degree to which one continuous variable (the DV) is related to a set of other (usually) continuous variables (the IVs) that have been combined to create a new, composite variable. Multiple correlation is a bivariate correlation between the original DV and the composite variable created from the IVs. For example, how large is the association between the belly dancing skill and the number of IVs, such as years of musical training, body flexibility, and age?

Multiple regression is used to predict the score on the DV from scores on several IVs. In the preceding example, belly dancing skill measured by knowledge of steps is the DV (as it is for bivariate regression), and we have added body flexibility and age to years of musical training as IVs. Other examples are prediction of success in an educational program from scores on a number of aptitude tests, prediction of the sizes of earthquakes from a variety of geological and electromagnetic variables, or stock market behavior from a variety of political and economic variables.

As for bivariate correlation and regression, multiple correlation emphasizes the degree of relationship between the DV and the IVs, whereas multiple regression emphasizes the prediction of the DV from the IVs. In multiple correlation and regression, the IVs may or may not be correlated with each other. With some ambiguity, the techniques also allow assessment of the relative contribution of each of the IVs toward predicting the DV.

1.1.3 Sequential R

In sequential (sometimes called hierarchical) multiple regression, IVs are given priorities by the researcher before their contributions to the prediction of the DV are assessed. For example, the researcher might first assess the effects of age and flexibility on belly dancing skill before looking at the contribution that years of musical training makes to that skill. Differences among dancers in age and flexibility are statistically “removed” before assessment of the effects of years of musical training.

In the example of an educational program, success of outcome might first be predicted from variables such as age and IQ. Then scores on various aptitude tests are added to see if prediction of outcome is enhanced after adjustment for age and IQ.

In general, then, the effects of IVs that enter first are assessed and removed before the effects of IVs that enter later are assessed. For each IV in a sequential multiple regression, higher-priority IVs act as covariates for lower-priority IVs. The degree of relationship between the DV and the IVs is reassessed at each step of the sequence. That is, multiple correlation is recomputed as each new IV (or set of IVs) is added. Sequential multiple regression, then, is also useful for developing a reduced set of IVs (if that is desired) by determining when IVs no longer add to predictability.

1.1.4 Canonical R

In canonical correlation, there are several continuous DVs as well as several continuous IVs, and the goal is to assess the relationship between the two sets of variables. For example, we might study the relationship between a number of indices of belly dancing skill (the DVs, such as knowledge of steps, ability to play finger cymbals, and responsiveness to the music) and the IVs (such as flexibility, musical training, and age). Thus, canonical correlation adds DVs (e.g., further indices of belly dancing skill) to the single index of skill used in bivariate and multiple correlations, so that there are multiple DVs as well as multiple IVs in canonical correlation.

Or we might ask whether there is a relationship among achievements in arithmetic, reading, and spelling as measured in elementary school and a set of variables reflecting early childhood development (e.g., age at first speech, walking, and toilet training). Such research questions are answered by canonical correlation.

1.1.5 Multiway Frequency Analysis

A goal of multiway frequency analysis is to assess relationships among discrete variables where none is considered a DV. For example, you might be interested in the relationships among gender, occupational category, and preferred type of reading material. Or the research question might involve relationships among gender, categories of religious affiliation, and attitude toward abortion.

When one of the variables is considered a DV with the rest serving as IVs, multiway frequency analysis is called *logit analysis*, as described in Section 1.3.3.

1.1.6 Multilevel Modeling

In many research applications, cases are nested in (normally occurring) groups, which may, in turn, be nested in other groups. The quintessential example is students nested in classrooms, which are, in turn, nested in schools. (Another common example involves repeated measures where, e.g., scores are nested within students who are, in turn, nested in classrooms, which, in turn, are nested in schools.) However, students in the same classroom are likely to have scores that correlate more highly than those of students in general. This creates problems with an analysis that pools all students into one very large group, ignoring classroom and school designations. Multilevel modeling is a somewhat complicated but increasingly popular strategy for analyzing data in these situations.

1.2 Significance of Group Differences

When subjects are randomly assigned to groups (treatments), the major research question usually is the extent to which statistically significant mean differences on DVs are associated with group membership. Once significant differences are found, the researcher often assesses the degree of relationship (effect size or strength of association) between IVs and DVs.

The choice among techniques hinges on the number of IVs and DVs and whether some variables are conceptualized as covariates. Further distinctions are made as to whether all DVs are measured on the same scale and how within-subjects IVs are to be treated.

1.2.1 One-Way ANOVA and *t* Test

The two statistics, one-way ANOVA and *t* test, are strictly univariate in nature and are adequately covered in most standard statistical texts.

1.2.2 One-Way ANCOVA

One-way analysis of covariance is designed to assess group differences on a single DV after the effects of one or more covariates are statistically removed. Covariates are chosen because of their known association with the DV; otherwise, there is no point in using them. For example, age and

degree of reading disability are usually related to the outcome of a program of educational therapy (the DV). If groups are formed by randomly assigning children to different types of educational therapies (the IV), it is useful to remove differences in age and degree of reading disability before examining the relationship between outcome and type of therapy. Prior differences among children in age and reading disability are used as covariates. The ANCOVA question is: Are there mean differences in outcome associated with type of educational therapy after adjusting for differences in age and degree of reading disability?

ANCOVA gives a more powerful look at the IV–DV relationship by minimizing error variance. The stronger the relationship between the DV and the covariate(s), the greater the power of ANCOVA over ANOVA.

ANCOVA is also used to adjust for differences among groups, when groups are naturally occurring and random assignment to them is not possible. For example, one might ask if attitude toward abortion (the DV) varies as a function of religious affiliation. However, it is not possible to randomly assign people to religious affiliation. In this situation, there could easily be other systematic differences among groups, such as level of education, that are also related to attitude toward abortion. Apparent differences among religious groups might well be due to differences in education rather than differences in religious affiliation. To get a “purer” measure of the relationship between attitude and religious affiliation, attitude scores are first adjusted for educational differences, that is, education is used as a covariate.

When there are more than two groups, planned or post hoc comparisons are available in ANCOVA just as in ANOVA. With ANCOVA, selected and/or pooled group means are adjusted for differences on covariates before differences in means on the DV are assessed.

1.2.3 Factorial ANOVA

Factorial ANOVA is the subject of numerous statistics texts (e.g., Brown, Michels, & Winer, 1991; Keppel & Wickens, 2004; Myers & Well, 2002; Tabachnick & Fidell, 2007) and is introduced in most elementary texts.

1.2.4 Factorial ANCOVA

Factorial ANCOVA differs from one-way ANCOVA only in that there is more than one IV. The desirability and the use of covariates are the same. For instance, in the educational therapy example of Section 1.2.2, another interesting IV might be gender of the child. The effects of gender, the type of educational therapy, and their interaction on the outcome are assessed after adjusting for age and prior degree of reading disability. The interaction of gender with type of therapy asks if boys and girls differ as to which type of educational therapy is more effective after adjustment for covariates.

1.2.5 Hotelling's T^2

Hotelling's T^2 is used when the IV has only two groups and there are several DVs. For example, there might be two DVs, such as score on an academic achievement test and attention span in the classroom, and two levels of type of educational therapy, emphasis on perceptual training versus emphasis on academic training. It is not legitimate to use separate t tests for each DV to look for differences between groups because that inflates Type I error due to unnecessary multiple significance

tests with (likely) correlated DVs. Instead, Hotelling's T^2 is used to see if groups differ on the two DVs combined. The researcher asks if there are non-chance differences in the centroids (average on the combined DVs) for the two groups.

Hotelling's T^2 is a special case of multivariate analysis of variance, just as the t test is a special case of univariate analysis of variance, when the IV has only two groups.

1.2.6 One-Way MANOVA

Multivariate analysis of variance evaluates differences among centroids (composite means) for a set of DVs when there are two or more levels of an IV (groups). MANOVA is useful for the educational therapy example in the preceding section with two groups and also when there are more than two groups (e.g., if a nontreatment control group is added).

With more than two groups, planned and post hoc comparisons are available. For example, if a main effect of treatment is found in MANOVA, it might be interesting to ask post hoc if there are differences in the centroids of the two groups given different types of educational therapies, ignoring the control group, and, possibly, if the centroid of the control group differs from the centroid of the two educational therapy groups combined.

Any number of DVs may be used; the procedure deals with correlations among them, and the entire analysis is accomplished within the preset level for Type I error. Once statistically significant differences are found, techniques are available to assess which DVs are influenced by which IV. For example, assignment to treatment group might affect the academic DV but not attention span.

MANOVA is also available when there are within-subject IVs. For example, children might be measured on both DVs three times: 3, 6, and 9 months after therapy begins. Profile analysis is an alternative to one-way between-subjects MANOVA when the DVs are all measured on the same scale. Discriminant analysis is an alternative to one-way between-subjects designs, as described in Section 1.3.1.

1.2.7 One-Way MANCOVA

In addition to dealing with multiple DVs, multivariate analysis of variance can be applied to problems when there are one or more covariates. In this case, MANOVA becomes multivariate analysis of covariance—MANCOVA. In the educational therapy example of Section 1.2.6, it might be worthwhile to adjust the DV scores for pretreatment differences in academic achievement and attention span. Here the covariates are pretests of the DVs, a classic use of covariance analysis. After adjustment for pretreatment scores, differences in posttest scores (DV) can be more clearly attributed to treatment (the two types of educational therapies plus control group that make up the IV).

In the one-way ANCOVA example of religious groups in Section 1.2.2, it might be interesting to test political liberalism versus conservatism and attitude toward ecology, as well as attitude toward abortion, to create three DVs. Here again, differences in attitudes might be associated with both differences in religion and differences in education (which, in turn, varies with religious affiliation). In the context of MANCOVA, education is the covariate, religious affiliation the IV, and attitudes the DVs. Differences in attitudes among groups with different religious affiliations are assessed after adjustment for differences in education.

If the IV has more than two levels, planned and post hoc comparisons are useful, with adjustment for covariates. MANCOVA is available for both the main analysis and the comparisons.

1.2.8 Factorial MANOVA

Factorial MANOVA is the extension of MANOVA to designs with more than one IV and multiple DVs. For example, gender (a between-subjects IV) might be added to the type of educational therapy (another between-subjects IV) with both academic achievement and attention span used as DVs. In this case, the analysis is a two-way between-subjects factorial MANOVA that provides tests of the main effects of gender and type of educational therapy and their interaction on the centroids of the DVs.

Duration of therapy (3, 6, and 9 months) might be added to the design as a within-subjects IV with type of educational therapy a between-subjects IV to examine the effects of duration, the type of educational therapy, and their interaction on the DVs. In this case, the analysis is a factorial MANOVA with one between- and one within-subjects IV.

Comparisons can be made among margins or cells in the design, and the influence of various effects on combined or individual DVs can be assessed. For instance, the researcher might plan (or decide post hoc) to look for linear trends in scores associated with duration of therapy for each type of therapy separately (the cells) or across all types of therapies (the margins). The search for linear trend could be conducted among the combined DVs or separately for each DV with appropriate adjustments for Type I error rate.

Virtually any complex ANOVA design with multiple DVs can be analyzed through MANOVA, given access to appropriate computer programs.

1.2.9 Factorial MANCOVA

It is sometimes desirable to incorporate one or more covariates into a factorial MANOVA design to produce factorial MANCOVA. For example, pretest scores on academic achievement and attention span could serve as covariates for the two-way between-subjects design with gender and type of educational therapy serving as IVs and posttest scores on academic achievement and attention span serving as DVs. The two-way between-subjects MANCOVA provides tests of gender, type of educational therapy, and their interaction on adjusted, combined centroids for the DVs.

Here again, procedures are available for comparisons among groups or cells and for evaluating the influences of IVs and their interactions on the various DVs.

1.2.10 Profile Analysis of Repeated Measures

A special form of MANOVA is available when all of the DVs are measured on the same scale (or on scales with the same psychometric properties) and you want to know if groups differ on the scales. For example, you might use the subscales of the Profile of Mood States as DVs to assess whether mood profiles differ between a group of belly dancers and a group of ballet dancers.

There are two ways to conceptualize this design. The first is as a one-way between-subjects design in which the IV is the type of dancer and the DVs are the Mood States subscales; one-way

MANOVA provides a test of the main effect of type of dancer on the combined DVs. The second way is as a profile study with one grouping variable (type of dancer) and the several subscales; profile analysis provides tests of the main effects of type of dancer and of subscales as well as their interaction (frequently the effect of greatest interest to the researcher).

If there is a grouping variable and a repeated measure such as trials in which the same DV is measured several times, there are three ways to conceptualize the design. The first is as a one-way between-subjects design with several DVs (the score on each trial); MANOVA provides a test of the main effect of the grouping variable. The second is as a two-way between- and within-subjects design; ANOVA provides tests of groups, trials, and their interaction, but with some very restrictive assumptions that are likely to be violated. Third is as a profile study, in which profile analysis provides tests of the main effects of groups and trials and their interaction, but without the restrictive assumptions. This is sometimes called *the multivariate approach to repeated-measures ANOVA*.

Finally, you might have a between- and within-subjects design (groups and trials), in which several DVs are measured on each trial. For example, you might assess groups of belly and ballet dancers on the Mood States subscales at various points in their training. This application of profile analysis is frequently referred to as *doubly multivariate*.

1.3 Prediction of Group Membership

In research where groups are identified, the emphasis is frequently on predicting group membership from a set of variables. Discriminant analysis, logit analysis, and logistic regression are designed to accomplish this prediction. Discriminant analysis tends to be used when all IVs are continuous and nicely distributed, logit analysis when IVs are all discrete, and logistic regression when IVs are a mix of continuous and discrete and/or poorly distributed.

1.3.1 One-Way Discriminant Analysis

In one-way discriminant analysis, the goal is to predict membership in groups (the DV) from a set of IVs. For example, the researcher might want to predict category of religious affiliation from attitude toward abortion, liberalism versus conservatism, and attitude toward ecological issues. The analysis tells us if group membership is predicted at a rate that is significantly better than chance. Or the researcher might try to discriminate belly dancers from ballet dancers from scores on Mood States subscales.

These are the same questions as those addressed by MANOVA, but turned around. Group membership serves as the IV in MANOVA and the DV in discriminant analysis. If groups differ significantly on a set of variables in MANOVA, the set of variables significantly predicts group membership in discriminant analysis. One-way between-subjects designs can be fruitfully analyzed through either procedure and are often best analyzed with a combination of both procedures.

As in MANOVA, there are techniques for assessing the contribution of various IVs to the prediction of group membership. For example, the major source of discrimination among religious groups might be abortion attitude, with little predictability contributed by political and ecological attitudes.

In addition, discriminant analysis offers classification procedures to evaluate how well individual cases are classified into their appropriate groups on the basis of their scores on the IVs.

1.3.2 Sequential One-Way Discriminant Analysis

Sometimes IVs are assigned priorities by the researcher, so their effectiveness as predictors of group membership is evaluated in the established order in sequential discriminant analysis. For example, when attitudinal variables are predictors of religious affiliation, variables might be prioritized according to their expected contribution to prediction, with abortion attitude given the highest priority, political liberalism versus conservatism second priority, and ecological attitude the lowest priority. Sequential discriminant analysis first assesses the degree to which religious affiliation is predicted from abortion attitude at a better-than-chance rate. Gain in prediction is then assessed with the addition of political attitude, and then with the addition of ecological attitude.

Sequential analysis provides two types of useful information. First, it is helpful in eliminating predictors that do not contribute more than predictors already in the analysis. For example, if political and ecological attitudes do not add appreciably to abortion attitude in predicting religious affiliation, they can be dropped from further analysis. Second, sequential discriminant analysis is a covariance analysis. At each step of the hierarchy, higher-priority predictors are covariates for lower-priority predictors. Thus, the analysis permits you to assess the contribution of a predictor with the influence of other predictors removed.

Sequential discriminant analysis is also useful for evaluating sets of predictors. For example, if a set of continuous demographic variables is given higher priority than an attitudinal set in prediction of group membership, one can see if attitudes significantly add to prediction after adjustment for demographic differences. However, it is usually more efficient to answer such questions through sequential logistic regression, particularly when some of the predictor variables are continuous and others discrete (see Section 1.3.5).

1.3.3 Multiway Frequency Analysis (Logit)

The logit form of multiway frequency analysis may be used to predict group membership when all of the predictors are discrete. For example, you might want to predict whether someone is a belly dancer (the DV) from knowledge of gender, occupational category, and preferred type of reading material (science fiction, romance, history, and statistics).

This technique allows evaluation of the odds that a case is in one group (e.g., belly dancer) based on membership in various categories of predictors (e.g., female professors who read science fiction).

1.3.4 Logistic Regression

Logistic regression allows prediction of group membership when predictors are continuous, discrete, or a combination of the two. Thus, it is an alternative to both discriminant analysis and logit analysis. For example, prediction of whether someone is a belly dancer may be based on gender, occupational category, preferred type of reading material, and age.

Logistic regression allows one to evaluate the odds (or probability) of membership in one of the groups (e.g., belly dancer) based on the combination of values of the predictor variables (e.g., 35-year-old female professors who read science fiction).

1.3.5 Sequential Logistic Regression

As in sequential discriminant analysis, sometimes predictors are assigned priorities and then assessed in terms of their contribution to prediction of group membership given their priority. For example, one can assess how well the preferred type of reading material predicts whether someone

is a belly dancer after adjusting for differences associated with age, gender, and occupational category.

1.3.6 Factorial Discriminant Analysis

If groups are formed on the basis of more than one attribute, prediction of group membership from a set of IVs can be performed through factorial discriminant analysis. For example, respondents might be classified on the basis of both gender and religious affiliation. One could use attitudes toward abortion, politics, and ecology to predict gender (ignoring religion) or religion (ignoring gender), or both gender and religion. But this is the same problem as addressed by factorial MANOVA. For a number of reasons, programs designed for discriminant analysis do not readily extend to factorial arrangements of groups. Unless some special conditions are met, it is usually better to rephrase the research question so that factorial MANOVA can be used.

1.3.7 Sequential Factorial Discriminant Analysis

Difficulties inherent in factorial discriminant analysis extend to sequential arrangements of predictors. Usually, however, questions of interest can readily be rephrased in terms of factorial MANCOVA.

1.4 Structure

Another set of questions is concerned with the latent structure underlying a set of variables. Depending on whether the search for structure is empirical or theoretical, the choice is principal components, factor analysis, or structural equation modeling. Principal components is an empirical approach, whereas factor analysis and structural equation modeling tend to be theoretical approaches.

1.4.1 Principal Components

If scores on numerous variables are available from a group of subjects, the researcher might ask if and how the variables group together. Can the variables be combined into a smaller number of supervariables on which the subjects differ? For example, suppose people are asked to rate the effectiveness of numerous behaviors for coping with stress (e.g., talking to a friend, going to a movie, jogging, and making lists of ways to solve the problem). The numerous behaviors may be empirically related to just a few basic coping mechanisms, such as increasing or decreasing social contact, engaging in physical activity, and instrumental manipulation of stress producers.

Principal components analysis uses the correlations among the variables to develop a small set of components that empirically summarizes the correlations among the variables. It provides a description of the relationship rather than a theoretical analysis.

1.4.2 Factor Analysis

When there is a theory about underlying structure or when the researcher wants to understand underlying structure, factor analysis is often used. In this case, the researcher believes that responses to many different questions are driven by just a few underlying structures called *factors*. In the example of mechanisms for coping with stress, one might hypothesize ahead of time that there are two major factors: general approach to problems (escape vs. direct confrontation) and the use of social supports (withdrawing from people vs. seeking them out).

Factor analysis is useful in developing and assessing theories. What is the structure of personality? Are there some basic dimensions of personality on which people differ? By collecting scores from many people on numerous variables that may reflect different aspects of personality, researchers address questions about underlying structure through factor analysis.

1.4.3 Structural Equation Modeling

Structural equation modeling combines factor analysis, canonical correlation, and multiple regression. Like factor analysis, some of the variables can be latent, whereas others are directly observed. Like canonical correlation, there can be many IVs and many DVs. And similar to multiple regression, the goal may be to study the relationships among many variables.

For example, one may want to predict birth outcome (the DVs) from several demographic, personality, and attitudinal measures (the IVs). The DVs are a mix of several observed variables such as birth weight, a latent assessment of mother's acceptance of the child based on several measured attitudes, and a latent assessment of infant responsiveness; the IVs are several demographic variables such as socioeconomic status, race, and income, several latent IVs based on personality measures, and prebirth attitudes toward parenting.

The technique evaluates whether the model provides a reasonable fit to the data and the contribution of each of the IVs to the DVs. Comparisons among alternative models, as well as evaluation of differences between groups, are also possible.

1.5 Time Course of Events

Two techniques focus on the time course of events. Survival/failure analysis asks how long it takes for something to happen. Time-series analysis looks at the change in a DV over the course of time.

1.5.1 Survival/Failure Analysis

Survival/failure analysis is a family of techniques dealing with the time it takes for something to happen: a cure, a failure, an employee leaving, a relapse, a death, and so on. For example, what is the life expectancy of someone diagnosed with breast cancer? Is the life expectancy longer with chemotherapy? Or, in the context of failure analysis, what is the expected time before a hard disk fails? Do DVDs last longer than CDs?

Two major varieties of survival/failure analysis are life tables, which describe the course of survival of one or more groups of cases (e.g., DVDs and CDs), and determination of whether survival time is influenced by some variables in a set. The latter technique encompasses a set of regression techniques in which the DV is the survival time.

1.5.2 Time-Series Analysis

Time-series analysis is used when the DV is measured over a very large number of time periods—at least 50; time is the major IV. Time-series analysis is used to forecast future events (stock markets' indices, crime statistics, etc.) based on a long series of past events. Time-series analysis is also used to evaluate the effect of an intervention, such as implementation of a water-conservation program by observing water usage for many periods before and after the intervention.

2 Some Further Comparisons

When assessing the degree of relationship among variables, bivariate r is appropriate when only two variables (one DV and one IV) are involved, while multiple R is appropriate when there are several variables on the IV side (one DV and several IVs). The multivariate analysis adjusts for correlations that are likely present among the IVs. Canonical correlation is available to study the relationship between several DVs and several IVs, adjusting for correlations among all of them. These techniques are usually applied to continuous (and dichotomous) variables. When all variables are discrete, multiway frequency analysis (vastly expanded chi square) is the choice.

Numerous analytic strategies are available to study mean differences among groups, depending on whether there is a single DV or multiple DVs, and whether there are covariates. The familiar ANOVA (and ANCOVA) is used with a single DV, while MANOVA (and MANCOVA) is used when there are multiple DVs. Essentially, MANOVA uses weights to combine multiple DVs into a new DV and then performs ANOVA.

A third important issue when studying mean differences among groups is whether there are repeated measures (the familiar within-subjects ANOVA). You may recall the restrictive and often-violated assumption of sphericity with this type of ANOVA. The two multivariate extensions of repeated-measures ANOVA (profile analysis of repeated measures and doubly multivariate profile analysis) circumvent this assumption by combining the DVs; MANOVA combines different DVs, while profile analysis combines the same DV measured repeatedly. Another variation of profile analysis (here called profile analysis of repeated measures) is a multivariate extension of the familiar “mixed” (between-within-subjects) ANOVA. None of the multivariate extensions is usually as powerful as its univariate “parent.”

The DV in both discriminant analysis and logistic regression is a discrete variable. In discriminant analysis, the IVs are usually continuous variables. A complication arises with discriminant analysis when the DV has more than two groups because there can be as many ways to distinguish the groups from each other as there are degrees of freedom for the DV. For example, if there are three levels of the DV, there are two degrees of freedom and therefore two potential ways to combine the IVs to separate the levels of the DV. The first combination might, for instance, separate members of the first group from the second and third groups (but not them from each other); the second combination might, then, separate members of group two from group three. Those of you familiar with comparisons in ANOVA probably recognize this as a familiar process for working with more than two groups; the difference is that in ANOVA *you* create the comparison coefficients used in the analysis while in discriminant analysis, the analysis tells *you* how the groups are best discriminated from each other (if they are).

Logistic regression analyzes a discrete DV too, but the IVs are often a mix of continuous and discrete variables. For that reason, the goal is to predict the probability that a case will fall into various levels of the DV rather than group membership per se. In this way, the analysis closely resembles the familiar chi-square analysis. In logistic regression, as in all multivariate techniques, the IVs are combined, but in an exponent rather than directly. That makes the analyses conceptually more difficult, but well worth the effort, especially in the medical/biological sciences where the risk ratios, a product of logistic regression, are routinely discussed.

There are several procedures for examining structure (that become increasingly “speculative”). Two very closely aligned techniques are principal components and factor analyses. These techniques are interesting because there is no DV (or, for that matter, IVs). Instead, there is just a bunch of

variables, with the goal of analysis to discover which of them “go” together. The idea is that some latent, underlying structure (e.g., several different factors representing components of personality) is driving similar responses to correlated sets of questions. The trick for the researcher is to divine the “meaning” of the factors that are developed during analysis. The technique of principal components provides an empirical solution while factor analysis provides a more theoretical solution.

Structural equation modeling combines multiple regression with factor analysis. There is a DV in this technique, but the IVs can be both discrete and continuous, both latent and observed. That is, the researcher tries to predict the values on an observed DV (continuous or discrete) using both observed variables (continuous and discrete) and the latent ones (factors derived from many observed variables during the analysis). Structural equation modeling is undergoing rapid development at present, with expansion to MANOVA-like analyses, longitudinal analysis, sophisticated procedures for handling missing data, and the like.

Multilevel modeling assesses the significance of variables where the cases are nested into different levels (e.g., students nested in classes nested in schools; patients nested in wards nested in hospitals). There is a DV at the lowest (student) level, but some IVs pertain to students, some to classes, and some to schools. The analysis takes into account the (likely) higher correlations among scores of students nested in the same class and of classes nested in the same school. Relationships (regressions) developed at one level (e.g., predicting student scores on the SAT from parental educational level) become the DVs for the next level, and so on.

Finally, we present two techniques for analyzing the time course of events, survival analysis and time-series analysis. One underlying IV for both of these is time; there may be other IVs as well. In survival analysis, the goal is often to determine whether a treated group survives longer than an untreated group, given the current standard of care. (In manufacturing, it is called failure analyses, and the goal, for instance, is to see if a part manufactured from a new alloy fails later than the part manufactured from the current alloy.) One advantage of this technique, at least in medicine, is its ability to analyze data for cases that have disappeared for one reason or another (moved away, gone to another clinic for treatment, or died of another cause) before the end of the study; these are called censored cases.

Time-series analysis tracks the pattern of the DV over multiple measurements (at least 50) and may or may not have an IV. If there is an IV, the goal is to determine if the pattern seen in the DV over time is the same for the group in one level of the IV as for the group in the other level. The IV can be naturally occurring or manipulated.

Generally, statistics are like tools—you pick the wrench you need to do the job.

3 A Decision Tree

A decision tree starting with major research questions appears in Table 1. For each question, choice among techniques depends on number of IVs and DVs (sometimes an arbitrary distinction) and whether some variables are usefully viewed as covariates. The table also briefly describes analytic goals associated with some techniques.

The paths in Table 1 are only recommendations concerning an analytic strategy. Researchers frequently discover that they need two or more of these procedures or, even more frequently, a judicious mix of univariate and multivariate procedures to fully answer their research questions. We recommend a flexible approach to data analysis in which both univariate and multivariate procedures are used to clarify the results.

TABLE 1 Choosing Among Statistical Techniques

Major Research Question	Number (Kind) of Dependent Variables	Number (Kind) of Independent Variables	Covariates	Analytic Strategy	Goal of Analysis
Degree of relationship among variables	One (continuous)	One (continuous)		Bivariate r	Create a linear combination of IVs to optimally predict DV.
		Multiple (continuous)	None	Multiple R	
			Some	Sequential multiple R	
	Multiple (continuous)	Multiple (continuous)		Canonical R	Maximally correlate a linear combination of DVs with a linear combination of IVs.
	One (may be repeated)	Multiple (continuous and discrete; cases and IVs are nested)		Multilevel modeling	Create linear combinations of IVs at one level to serve as DVs at another level.
Significance of group differences	One (continuous)	One (discrete)	None	One-way ANOVA or t test	Determine significance of mean group differences.
			Some	One-way ANCOVA	
		Multiple (discrete)	None	Factorial ANOVA	
			Some	Factorial ANCOVA	
	Multiple (continuous)	One (discrete)	None	One-way MANOVA or Hotelling's T^2	Create a linear combination of DVs to maximize mean group differences.
			Some	One-way MANCOVA	
		Multiple (discrete)	None	Factorial MANOVA	
			Some	Factorial MANCOVA	
	One (continuous)	Multiple (one discrete within S)		Profile analysis of repeated measures	Create linear combinations of DVs to maximize mean group differences and differences between levels of within-subjects IVs.
	Multiple (continuous/commensurate)	One (discrete)		Profile analysis	
	Multiple (continuous)	Multiple (one discrete within S)		Doubly multivariate profile analysis	

(continued)

TABLE 1 Continued

Major Research Question	Number (Kind) of Dependent Variables	Number (Kind) of Independent Variables	Covariates	Analytic Strategy	Goal of Analysis
Prediction of group membership	One (discrete)	Multiple (continuous)	None	One-way discriminant function	Create a linear combination of IVs to maximize group differences.
			Some	Sequential one-way discriminant function	
		Multiple (discrete)		Multiway frequency analysis (logit)	Create a log-linear combination of IVs to optimally predict DV.
		Multiple (continuous and/or discrete)	None	Logistic regression	Create a linear combination of the log of the odds of being in one group.
			Some	Sequential logistic regression	
	Multiple (discrete)	Multiple (continuous)	None	Factorial discriminant function	Create a linear combination of IVs to maximize group differences (DVs).
Structure			Some	Sequential factorial discriminant function	
	Multiple (continuous observed)	Multiple (latent)		Factor analysis (theoretical)	Create linear combinations of observed variables to represent latent variables.
	Multiple (latent)	Multiple (continuous observed)		Principal components (empirical)	
	Multiple (continuous observed and/or latent)	Multiple (continuous observed and/or latent)		Structural equation modeling	Create linear combinations of observed and latent IVs to predict linear combinations of observed and latent DVs.

TABLE 1 Continued

Major Research Question	Number (Kind) of Dependent Variables	Number (Kind) of Independent Variables	Covariates	Analytic Strategy	Goal of Analysis
Time course of events	One (time)	None	None	Survival analysis (life tables)	Determine how long it takes for something to happen.
		One or more	None or some	Survival analysis (with predictors)	Create a linear combination of IVs and CVs to predict time to an event.
	One (continuous)	Time	None or some	Time-series analysis (forecasting)	Predict future course of DV on basis of past course of DV.
		One or more (including time)	None or some	Time-series analysis (intervention)	Determine whether course of DV changes with intervention.

4 Preliminary Check of the Data

Before applying any technique, or sometimes even before choosing a technique, you should determine the fit between your data and some very basic assumptions underlying most of the multivariate statistics.

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Review of Univariate and Bivariate Statistics

This chapter provides a brief review of univariate and bivariate statistics. Although it is probably too “dense” to be a good source from which to learn, it is hoped that it will serve as a useful reminder of material already mastered and will help in establishing a common vocabulary. Section 1 goes over the logic of the statistical hypothesis test, and Sections 2 through 4 skim many topics in analysis of variance. Section 5 summarizes correlation and regression and Section 6 summarizes chi-square (χ^2).

1 Hypothesis Testing

Statistics are used to make rational decisions under conditions of uncertainty. Inferences (decisions) are made about populations based on data from samples that contain incomplete information. Different samples taken from the same population probably differ from one another and from the population. Therefore, inferences regarding the population are always a little risky.

The traditional solution to this problem is statistical decision theory. Two hypothetical states of reality are set up, each represented by a probability distribution. Each distribution represents an alternative hypothesis about the true nature of events. Given the sample results, a best guess is made as to which distribution the sample was taken from using formalized statistical rules to define “best.”

1.1 One-Sample z Test as Prototype

Statistical decision theory is most easily illustrated through a one-sample z test, using the standard normal distribution as the model for two hypothetical states of reality. Suppose there is a sample of 25 IQ scores and a need to decide whether this sample of scores is a random sample of a “normal” population with $\mu = 100$ and $\sigma = 15$, or a random sample from a population with $\mu = 108$ and $\sigma = 15$.

First, note that hypotheses are tested about means, and not individual scores. Therefore, the distributions representing hypothetical states of reality are distributions of means rather than distributions of individual scores. Distributions of means produce “sampling distributions of means” that differ systematically from distributions of individual scores; the mean of a population distribution, μ , is equal to the mean of a sampling distribution, μ , but the standard deviation of a population of individual scores, σ , is not equal to the standard deviation of a sampling distribution, σ_Y . Sampling distributions

$$\sigma_{\bar{Y}} = \frac{\sigma}{\sqrt{N}} \quad (1)$$

have smaller standard deviations than distributions of scores, and the decrease is related to N , the sample size. For the sample, then,

$$\sigma_{\bar{Y}} = \frac{15}{\sqrt{25}} = 3$$

The question being asked, then, is, “Does our mean, taken from a sample of size 25, come from a sampling distribution with $\mu_{\bar{Y}} = 100$ and $\sigma_{\bar{Y}} = 3$ or does it come from a sampling distribution with $\mu_{\bar{Y}} = 108$ and $\sigma_{\bar{Y}} = 3$.” Figure 1(a) shows the first sampling distribution, defined as the null hypothesis, H_0 , that is, the sampling distribution of means calculated from all possible samples of size 25 taken from a population where $\mu = 100$ and $\sigma = 15$.

The sampling distribution for the null hypothesis has a special, fond place in statistical decision theory because it alone is used to define “best guess.” A decision axis for retaining or rejecting H_0 cuts through the distribution so that the probability of rejecting H_0 by mistake is small. “Small” is defined probabilistically as α . An error in rejecting the null hypothesis is referred to as an α or Type I error. There is little choice in picking α . Tradition and journal editors decree that it is .05 or smaller, meaning that the null hypothesis is rejected no more than 5% of the time when it is true.

With a table of areas under the standard normal distribution (the table of z scores or standard normal deviates), the decision axis is placed so that the probability of obtaining a sample mean above that point is 5% or less. Looking up 5% in Table 1, the z corresponding to a 5% cutoff is 1.645 (between 1.64 and 1.65). Notice that the z scale is one of two abscissas in Figure 1(a). If the decision axis is placed where $z = 1.645$, one can translate from the z scale to the $\bar{Y}$ scale to properly position the decision axis. The transformation equation is

$$\bar{Y} = \mu + z\sigma_{\bar{Y}} \quad (2)$$

Equation 2 is a rearrangement of terms from the z test for a single sample:¹

$$z = \frac{\bar{Y} - \mu}{\sigma_{\bar{Y}}} \quad (3)$$

Applying Equation 2 to the example,

$$\bar{Y} = 100 + (1.645)(3) = 104.935$$

The null hypothesis, that the mean IQ of the sampling distribution is 100, is rejected if the mean IQ of the sample is equal to or greater than 104.935 (call it 105).

Frequently, this is as far as the model is taken—the null hypothesis is either retained or rejected. However, if the null hypothesis is rejected, it is rejected in favor of an alternative hypothesis, H_a .

¹The more usual procedure for testing a hypothesis about a single mean is to solve for z on the basis of the sample mean and standard deviation to see if the sample mean is sufficiently far away from the mean of the sampling distribution under the null hypothesis. If z is 1.645 or larger, the null hypothesis is rejected.

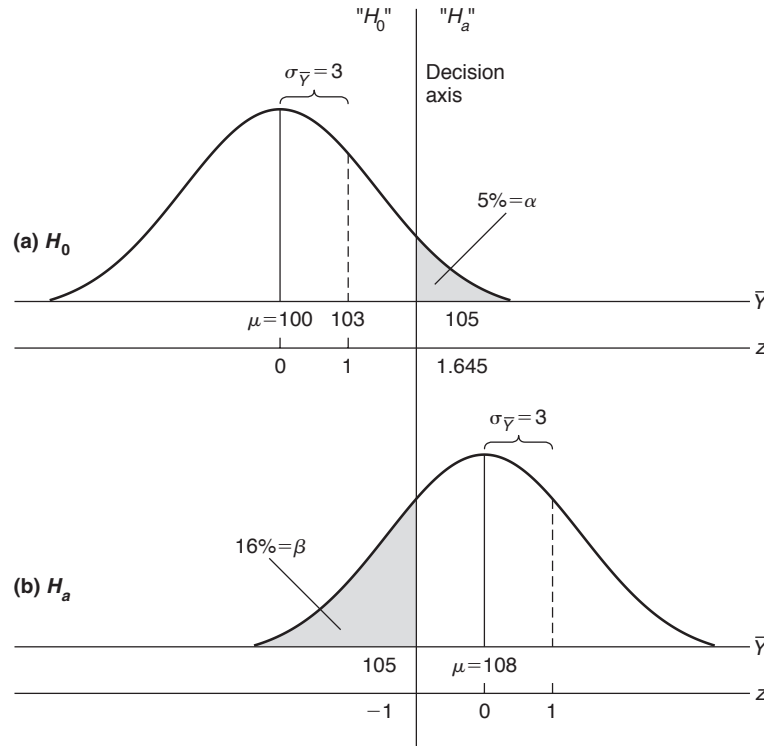


FIGURE 1 Sampling Distribution for Means with $N = 25$ and $\sigma = 15$ under two hypotheses (a) $H_0: \mu = 100$ and (b) $H_a: \mu = 108$.

The alternative hypothesis is not always stated explicitly,² but when it is, one can evaluate the probability of retaining the null hypothesis when it should be rejected, because H_a is true.

This second type of error is called a β or Type II error. Since μ for H_a is 108 in the example, the sampling distribution of means for H_a can be graphed, as shown in Figure 1(b). The decision axis is placed with regard to H_0 . Hence we need to find the probability associated with the place it crosses H_a . The first step is to find z corresponding to an IQ score of 105 in a distribution with $\mu_{\bar{Y}} = 108$ and $\sigma_{\bar{Y}} = 3$. Applying Equation 3, we find that

$$z = \frac{105 - 108}{3} = -1.00$$

By looking up $z = -1.00$, about 16% of the time sample means are ≤ 105 when the population $\mu = 108$ and the alternative hypothesis is true. Therefore, $\beta = .16$.

²Often, the alternative hypothesis is simply that the sample is taken from a population that is not equal to the population represented by the null hypothesis. There is no attempt to specify "not equal to."

H_0 and H_a represent alternative realities, only one of which is true. When the researcher is forced to decide whether to retain or reject H_0 , four things can happen. If the null hypothesis is true, a correct decision is made if the researcher retains H_0 and an error is made if the researcher rejects it. If the probability of making the wrong decision is α , the probability of making the right decision is $1 - \alpha$. If, on the other hand, H_a is true, the probability of making the right decision by rejecting H_0 is $1 - \beta$ and the probability of making the wrong decision is β . This information is summarized in a “confusion matrix” (aptly named, according to beginning statistics students) showing the probabilities of each of these four outcomes:

		Reality	
		H_0	H_a
Statistical decision	“ H_0 ”	$1 - \alpha$	β
	“ H_a ”	α	$1 - \beta$
		1.00	1.00

For the example, the probabilities are

		Reality	
		H_0	H_a
Statistical decision	“ H_0 ”	.95	.16
	“ H_a ”	.05	.84
		1.00	1.00

1.2 Power

The lower right-hand cell of the confusion matrix represents the most desirable outcome and the power of the research. Usually, the researcher believes that H_a is true and hopes that the sample data lead to rejection of H_0 . Power is the probability of rejecting H_0 when H_a is true. In Figure 1(b), power is the portion of the H_a distribution that falls above the decision axis. Many of the choices in designing research are made with an eye toward increasing power because research with low statistical power usually is not worth the effort.

Figure 1 and Equations 1 and 2 suggest some ways to enhance power. One obvious way to increase power is to move the decision axis to the left. However, it cannot be moved far or Type I error rates reach an unacceptable level. Given the choice between .05 and .01 for α error, though, a decision in favor of .05 increases the power. A second strategy is to move the curves farther apart by applying a stronger treatment. Other strategies involve decreasing the standard deviation of the sampling distributions either by decreasing variability in scores (e.g., exerting greater experimental control) or by increasing sample size, N .

This model for statistical decisions and these strategies for increasing power generalize to other sampling distributions and to tests of hypotheses other than a single sample mean against a hypothesized population mean.

There is occasionally the danger of *too much power*. The null hypothesis is probably never exactly true and any sample is likely to be slightly different from the population value. With a large

enough sample, rejection of H_0 is virtually certain. Hence, a “minimal meaningful difference” and acceptable effect size should guide the selection of sample size (Kirk, 1995). The sample size should be large enough to be likely to reveal a minimal meaningful difference. Rejection of the null hypothesis may be trivial if the sample is large enough to reveal any difference whatever. This issue is considered further in Section 4.

1.3 Extensions of the Model

The z test for the difference between a sample mean and a population mean readily extends to a z test of the difference between two sample means. A sampling distribution is generated for the difference between means under the null hypothesis that $\mu_1 = \mu_2$ and is used to position the decision axis. The power of an alternative hypothesis is calculated with reference to the decision axis, as earlier.

When population variances are unknown, it is desirable to evaluate the probabilities using Student’s t rather than z , even for large samples. Numerical examples of use of t to test differences between two means are available in most univariate statistics books and are not presented here. The logic of the process, however, is identical to that described in Section 1.1.

1.4 Controversy Surrounding Significance Testing

While the statistical significance test is pervasive in the social sciences, its use is not without controversy. The latest round of arguments against the use of statistical significance testing began with an article by Carver in 1978, and updated in 1993. In these articles, Carver argues that the significance test, used by itself, does not answer most research questions. These articles, and many others in a rather large literature, are summarized by McLean and Ernest (1998). The significance test, they assert, tells whether the result was likely obtained by chance, but does not convey information about the practical importance of the difference (effect size), the quality of the research design, the reliability and validity of the measures, the fidelity of the treatment, or whether the results are replicable. Thus, a significance test is properly only one among many criteria by which a finding is assessed.

Because of the controversy, the Task Force on Statistical Inference was convened by the American Psychological Association in 1996 and produced a final report in 1999 (Wilkinson & Task Force on Statistical Inference, 1999), where the authors stress the importance of the factors listed above, along with the importance of data screening prior to analysis. Like those who oppose the use of statistical significance testing, they urge reporting effect size, and particularly confidence intervals around effect size estimates. Another approach (Cumming & Finch, 2005) involves plots of means with error bars as a way of accomplishing statistical inference by eye. They propose “7 rules of eye to guide the inferential use of figures with error bars” (p. 170).

2 Analysis of Variance

Analysis of variance is used to compare two or more means to see if there are any statistically significant differences among them. Distributions of scores for three hypothetical samples are provided in Figure 2. Analysis of variance evaluates the differences among means relative to the dispersion in the sampling distributions. The null hypothesis is that $\mu_1 = \mu_2 = \dots = \mu_k$ as estimated from $\bar{Y}_1 = \bar{Y}_2 = \dots = \bar{Y}_k$, where k is equal to the number of means being compared.

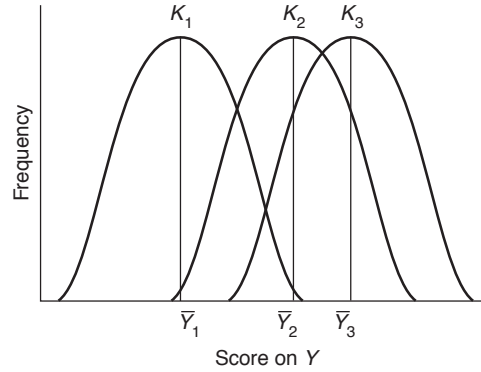


FIGURE 2 Idealized frequency distribution of three samples and their means.

Analysis of variance (ANOVA) is really a set of analytic procedures based on a comparison of two estimates of variance. One estimate comes from differences among scores within each group; this estimate is considered random or error variance. The second estimate comes from differences in group means and is considered a reflection of group differences or treatment effects plus error. If these two estimates of variance do not differ appreciably, one concludes that all of the group means come from the same sampling distribution of means and that the slight differences among them are due to random error. If, on the other hand, the group means differ more than expected, it is concluded that they were drawn from different sampling distributions of means and the null hypothesis that the means are the same is rejected.

Differences among variances are evaluated as ratios, where the variance associated with differences among sample means is in the numerator, and the variance associated with error is in the denominator. The ratio between these two variances forms an F distribution. F distributions change shape depending on degrees of freedom in both numerator and denominator of the F ratio. Thus, tables of critical F , for testing the null hypothesis, depend on two degree-of-freedom parameters (cf. Appendix 'Statistical Tables', Table 3).

The many varieties of analysis of variance are conveniently summarized in terms of the partition of *sums of squares*, that is, sums of squared differences between scores and their means. A sum of squares (SS) is simply the numerator of a variance,

$$S^2 = \frac{\sum(Y - \bar{Y})^2}{N - 1} \quad (4)$$

$$SS = \sum(Y - \bar{Y})^2 \quad (5)$$

The square root of variance is standard deviation, S , which is the measure of variability that is in the metric of the original scores.

$$S = \sqrt{S^2} \quad (6)$$

2.1 One-Way Between-Subjects ANOVA

DV scores appropriate to one-way between-subjects ANOVA with equal n are presented in a table, with k columns representing groups (levels of the IV) and n scores within each group.³ Table 1 shows how subjects are assigned to groups within this design.

Each column has a mean, $\bar{Y}_j$ where $j = 1, 2, \dots, k$ levels of treatment. Each score is designated $\bar{Y}_{ij}$ where $i = 1, 2, \dots, n$ scores within each treatment. Each case provides a single score on the DV. The symbol GM represents the grand mean of all scores over all groups.

The difference between each score and the grand mean ($Y_{ij} - \text{GM}$) is considered the sum of two component differences, the difference between the score and its own group mean and the difference between that mean and the overall mean.

$$Y_{ij} - \text{GM} = (Y_{ij} - \bar{Y}_j) + (\bar{Y}_j - \text{GM}) \quad (7)$$

This result is achieved by first subtracting and then adding the group mean to the equation. Each term is then squared and summed separately to produce the sum of squares for error and the sum of squares for treatment, respectively. The basic partition holds because, conveniently, the cross-product terms produced by squaring and summing cancel each other out. Across all scores, the partition is

$$\sum_i \sum_j (Y_{ij} - \text{GM})^2 = \sum_i \sum_j (Y_{ij} - \bar{Y}_j)^2 + n \sum_j (\bar{Y}_j - \text{GM})^2 \quad (8)$$

Each of these terms is a sum of squares (SS)—a sum of squared differences between scores (with means sometimes treated as scores) and their associated means. That is, each term is a special case of Equation 5.

The term on the left-hand side of the equation is the total sum of squared differences between scores and the grand mean, ignoring groups with which scores are associated, designated SS_{total} . The first term on the right-hand side is the sum of squared deviations between each score and its group mean. When summed over all groups, it becomes the sum of squares within groups, SS_{wg} .

TABLE 1 Assignment of Subjects in a One-Way Between-Subjects ANOVA

Treatment		
K_1	K_2	K_3
S_1	S_4	S_7
S_2	S_5	S_8
S_3	S_6	S_9

³Throughout the chapter, n is used for sample size within a single group or cell, and N is used for total sample size.

The last term is the sum of squared deviations between each group mean and the grand mean, the sum of squares between groups, SS_{bg} . Equation 8 is also symbolized as

$$SS_{total} = SS_{wg} + SS_{bg} \quad (9)$$

Degrees of freedom in ANOVA partition the same way as sums of squares:

$$df_{total} = df_{wg} + df_{bg} \quad (10)$$

Total degrees of freedom are the number of scores minus 1. The 1 df is lost when the grand mean is estimated. Therefore,

$$df_{total} = N - 1 \quad (11)$$

Within-groups degrees of freedom are the number of scores minus k , lost when the means for each of the k groups are estimated. Therefore,

$$df_{wg} = N - k \quad (12)$$

Between-groups degrees of freedom are k “scores” (each group mean treated as a score) minus 1, lost when the grand mean is estimated, so that

$$df_{bg} = k - 1 \quad (13)$$

Verifying the equality proposed in Equation 10, we get

$$N - 1 = N - k + k - 1$$

As in the partition of sums of squares, the term associated with group means is subtracted out of the equation and then added back in.

Another common notation for the partition of Equation 7 is

$$SS_{total} = SS_k + SS_{S(K)} \quad (14)$$

as shown in Table 2(a). In this notation, the total sum of squares is partitioned into a sum of squares due to the k groups, SS_K , and a sum of squares due to subjects within the groups, $SS_{S(K)}$. (Notice that the order of terms on the right-hand side of the equation is the reverse of that in Equation 9.)

The division of a sum of squares by degrees of freedom produces variance, called mean square (MS), in ANOVA. Variance, then, is an “average” sum of squares. ANOVA produces three variances: one associated with total variability among scores, MS_{total} ; one associated with variability within

TABLE 2 Partition of Sums of Squares and Degrees of Freedom for Several ANOVA Designs

(a) One-way between-subjects ANOVA

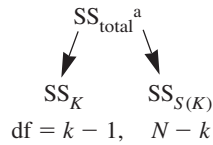
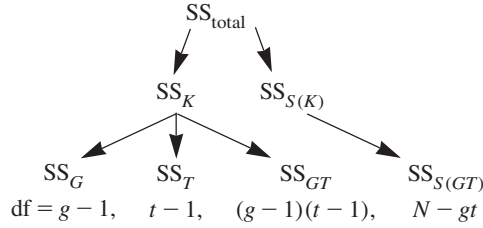
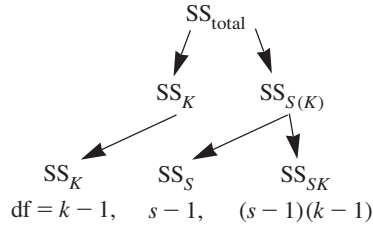


TABLE 2 Continued

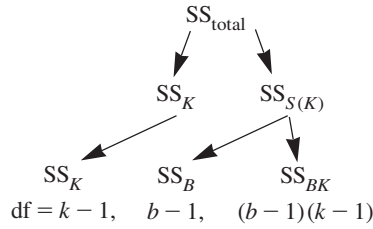
(b) Factorial between-subjects ANOVA



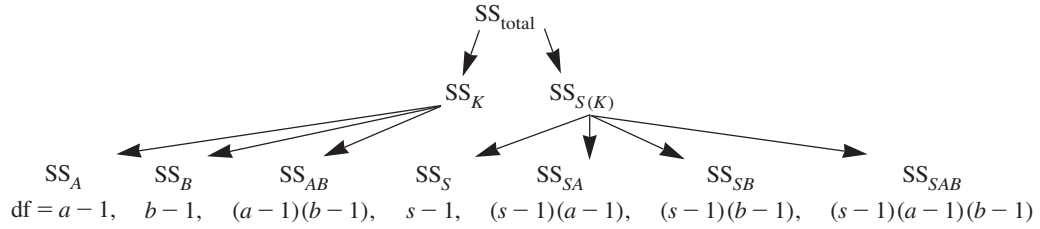
(c) One-way within-subjects ANOVA



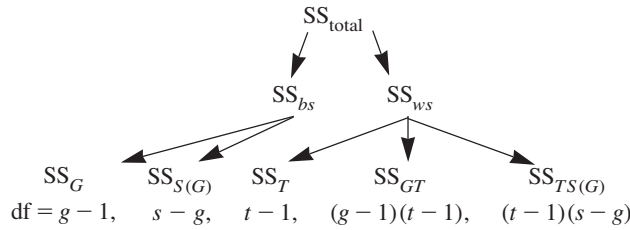
(d) One-way matched-randomized ANOVA



(e) Factorial within-subjects ANOVA



(f) Mixed within-between-subjects ANOVA

^a For all SS_{total} , $df = N - 1$.

groups, MS_{bg} or MS_K ; and one associated with variability between groups, MS_{wg} or $MS_{S/K}$. MS_K and $MS_{S/K}$ provide the variances for an F ratio to test the null hypothesis that $\mu_1 = \mu_2 = \cdots = \mu_k$.

$$F = \frac{MS_K}{MS_{S(K)}} \quad df = (k - 1), N - k \quad (15)$$

Once F is computed, it is tested against critical F obtained from a table, such as Table 3, with numerator $df = k - 1$ and denominator $df = N - k$ at a desired alpha level. If obtained F exceeds critical F , the null hypothesis is rejected in favor of the hypothesis that there is a difference among the means in the k groups.

Anything that increases the obtained F increases the power. Power is increased by decreasing the error variability or increasing the sample size in the denominator ($MS_{S(K)}$) or by increasing differences among means in the numerator (MS_K).

2.2 Factorial Between-Subjects ANOVA

If groups are formed along more than one dimension, differences among means are attributable to more than one source. Consider an example with six groups, three of women and three of men, in which the DV is scores on a final examination in statistics. One source of variation in means is due to gender, SS_G . If the three groups within each gender are exposed to three different methods of teaching statistics, a second source of differences among means is teaching method, SS_T . The final source of known differences among means is the interaction between gender and teaching methods, SS_{GT} . The interaction tests whether effectiveness of teaching methods varies with gender.

Allocation of subjects in this design is shown in Table 3. Sums of squares and degrees of freedom are partitioned as in Table 2(b). Error is estimated by variation in scores within each of the six cells, $SS_{S(GT)}$. Three null hypotheses are tested using the F distribution.

The first test asks if means for men and women are likely to have come from the same sampling distribution of means. Scores are averaged across teaching methods to eliminate that source of variability. Gender differences are tested in the F ratio:

$$F = \frac{MS_G}{MS_{S(GT)}} \quad df = (g - 1), N - gt \quad (16)$$

TABLE 3 Assignment of Subjects in a Factorial Between-Subjects Design

		Teaching Techniques		
		T_1	T_2	T_3
Gender	G_1	S_1	S_5	S_9
		S_2	S_6	S_{10}
	G_2	S_3	S_7	S_{11}
		S_4	S_8	S_{12}

Rejection of the null hypothesis supports an interpretation of differences between women and men in performance on the final exam.

The second test asks if means from the three teaching methods are likely to have come from the same sampling distribution of means, averaged across women and men. This is tested as

$$F = \frac{MS_T}{MS_{S(GT)}} \quad df = (t - 1), N - gt \quad (17)$$

Rejection of the null hypothesis supports an interpretation of differences in effectiveness of the three teaching methods.

The third test asks if the cell means, the means for women and the means for men within each teaching method, are likely to have come from the same sampling distribution of differences between means.

$$F = \frac{MS_{GT}}{MS_{S(GT)}} \quad df = (g - 1)(t - 1), N - gt \quad (18)$$

Rejection of the null hypothesis supports an interpretation that men and women differ regarding the most effective teaching methods.

In each case, the estimate of normally occurring variability in test scores, error, is $MS_{S(GT)}$ or within-cell variance. In each case, critical F is read from a table with appropriate degrees of freedom and desired alpha, and if obtained F (Equation 16, 17, or 18) is greater than critical F , the null hypothesis is rejected. When there is an equal number of scores in each cell, the three tests are independent (except for use of a common error term): the test of each main effect (gender and teaching method) is not related to the test of the other main effect or the test of the interaction.

In a one-way between-subjects design (Section 2.1), $(k - 1)$ degrees of freedom are used to test the null hypothesis of differences among groups. If k is equal to the number of cells in a two-way design, tests of G , T , and GT use up the $(k - 1)$ degrees of freedom. With proper partitioning, then, of a two-way factorial design, you get three tests for the price of one.

With higher-order between-subjects factorial designs, variation due to differences among groups is partitioned into main effects for each IV, two-way interactions between each pair of IVs, three-way interactions among each trio of IVs, and so on. In any between-subjects factorial design, error sum of squares is the sum of squared differences within each cell of the design.

2.3 Within-Subjects ANOVA

In some designs, the means that are tested are derived from the same subjects measured on different occasions, as shown in Table 4, rather than from different groups of subjects.⁴ In these designs, computation of sum of squares and mean square for the effect of the IV is the same as for the between-subject designs. However, the error term is further partitioned into individual differences due to subjects, SS_S , and interaction of individual differences with treatment, SS_{SK} . Because subjects are measured repeatedly, their effect as a source of variability in scores is estimated and

⁴This design is also called repeated measures, one score per cell, randomized blocks, matched-randomized changeover, or crossover.

TABLE 4 Assignment of Subjects in a One-Way Within-Subjects Design

		Treatment		
		K_1	K_2	K_3
Subjects	S_1	S_1	S_1	S_1
	S_2	S_2	S_2	S_2
	S_3	S_3	S_3	S_3

subtracted from $SS_{S(K)}$, the error term in a corresponding between-subjects design. The interaction of individual differences with treatment, MS_{SK} , is used as the error term:

$$F = \frac{MS_K}{MS_{SK}} \quad df = (k - 1), (k - 1)(s - 1) \quad (19)$$

The partition of sums of squares for a one-way within-subjects design with k levels is shown in Table 2(c), where s is the number of subjects.

MS_{SK} is used as the error term because once SS_S is subtracted, no variation is left within cells of the design—there is, in fact, only one score per cell. The interaction of individual differences with treatment is all that remains to serve as an estimate of error variance. If there are individual differences among subjects in scores, and if individuals react similarly to the IV, the interaction is a good estimate of error. Once individual differences are subtracted, the error term is usually smaller than the error term in a corresponding between-subjects design, so the within-subjects design is more sensitive than the between-subjects design.

But if there are no consistent individual differences in scores,⁵ or if there is an interaction between subjects and treatment, the error term may be larger than that of a between-subjects design. The statistical test is then conservative, it is more difficult to reject the null hypothesis of no difference between means, and the power of the test is reduced. Since Type I error is unaffected, the statistical test is not in disrepute, but, in this case, a within-subjects design is a poor choice of research design.

A within-subjects analysis is also used with a matched-randomized blocks design, as shown in Tables 2(d) and 5. Subjects are first matched on the basis of variables thought to be highly related to the DV. Subjects are then divided into blocks, with as many subjects within each block as there are levels of the IV. Finally, members of each block are randomly assigned to levels of the IV. Although the subjects in each block are actually different people, they are treated statistically as if they were the same person. In the analysis, there is a test of the IV, and a test of blocks (the same as the test of subjects in the within-subjects design), with the interaction of blocks and treatment used as the error term. Since matching is used to produce consistency in performance within blocks and the effect of blocks is subtracted from the error term, this design should also be more sensitive than the between-subjects design. It will not be, however, if the matching fails.

⁵Notice that the degrees of freedom for error, $(k - 1)(s - 1)$, are fewer than in the between-subjects design. Unless the reduction in error variance due to subtraction of SS_S is substantial, the loss of degrees of freedom may overcome the gain due to smaller SS when MS_{SK} is computed.

Factorial within-subjects designs, as shown in Table 6, are an extension of the one-way within-subjects design. The partition of a two-way within-subjects design is in Table 2(e).

In the analysis, the error sum of squares is partitioned into a number of “subjects-by-effects” interactions just as the sum of squares for effects is partitioned into numerous sources. It is common (though not universal) to develop a separate error term for each F test; for instance, the test of the main effect of A is

$$F = \frac{MS_A}{MS_{SA}} \quad df = (a - 1), (a - 1)(s - 1) \quad (20)$$

For the main effect of B , the test is

$$F = \frac{MS_B}{MS_{SB}} \quad df = (b - 1), (b - 1)(s - 1) \quad (21)$$

and for the interaction,

$$F = \frac{MS_{AB}}{MS_{SAB}} \quad df = (a - 1)(b - 1), (a - 1)(b - 1)(s - 1) \quad (22)$$

For higher-order factorial designs, the partition into sources of variability grows prodigiously, with an error term developed for each main effect and interaction tested.

There is controversy in within-subject analyses concerning conservatism of the F tests and whether separate error terms should be used. In addition, if the repeated measurements are on single subjects, there are often carry-over effects that limit generalizability to situations where subjects are tested repeatedly. Finally, when there are more than two levels of the IV, the analysis has the assumption of sphericity. One component of sphericity—homogeneity of covariance—roughly speaking, is the assumption that the subjects “line up” in scores is the same for all pairs of levels of the IV. If some pairs of levels are close in time (e.g., trial 2 and trial 3) and other pairs are distant in time (e.g., trial 1 and trial 10), the assumption is often violated. Such violation is serious because Type I error rate is affected. Sphericity is discussed in greater detail in Tabachnick and Fidell (2007) and Frane (1980).

TABLE 5 Assignment of Subjects in a Matched-Randomized Design^a

		Treatment		
		A_1	A_2	A_3
Blocks	B_1	S_1	S_2	S_3
	B_2	S_4	S_5	S_6
	B_3	S_7	S_8	S_9

^aWhere subjects in the same block have been matched on some relevant variable.

TABLE 6 Assignment of Subjects in a Factorial Within-Subjects Design

		Treatment A		
		A_1	A_2	A_3
Treatment B	B_1	S_1	S_1	S_1
		S_2	S_2	S_2
	B_2	S_1	S_1	S_1
		S_2	S_2	S_2

For these reasons, within-subjects ANOVA is sometimes replaced by profile analysis, where repetitions of DVs are transformed into separate DVs and a multivariate statistical test is used.

2.4 Mixed Between-Within-Subjects ANOVA⁶

Often in factorial designs, one or more IVs are measured between subjects, whereas other IVs are measured within subjects.⁷ The simplest example of the mixed between-within-subjects design involves one between-subjects and one within-subjects IV, as shown in Table 7.⁸

To show the partition, the total SS is divided into a source attributable to the between-subjects part of the design (Groups), and a source attributable to the within-subjects part (Trials), as shown in Table 2(f). Each source is then further partitioned into effects and error components: between-subjects into groups and subjects-within-groups error term; and within-subjects into trials, the group-by-trials interaction, and the trial-by-subjects-within-groups error term.

As more between-subjects IVs are added, between-subjects main effects and interactions expand the between-subjects part of the partition. For all the between-subjects effects, there is a single error term consisting of variance among subjects confined to each combination of the between-subjects IVs. As more within-subjects IVs are added, the within-subjects portion of the design expands. Sources of variability include main effects and interactions of within-subjects IVs and interactions of between- and within-subjects IVs. Separate error terms are developed for each source of variability in the within-subjects segment of the design.⁹

TABLE 7 Assignment of Subjects in a Between-Within-Subjects Design

		Trials		
		T_1	T_2	T_3
Groups	G_1	S_1	S_1	S_1
		S_2	S_2	S_2
	G_2	S_3	S_3	S_3
		S_4	S_4	S_4

⁶This design is also called a split-plot, repeated-measures, or randomized-block factorial design.

⁷Mixed designs can also have “blocks” rather than repeated measures on individual subjects as the within-subjects segment of the design.

⁸When the between-subjects variables are based on naturally occurring differences among subjects (e.g., age and sex), the design is said to be “blocked” on the subject variables. This is a different use of the term *blocks* from that of the preceding section. In a mixed design, both kinds of blocking can occur.

⁹“Subjects” are no longer available as a source of variance for analysis. Because subjects are confined to levels of the between-subjects variable(s), differences between subjects in each group are used to estimate error for testing variance associated with between-subjects variables.

Problems associated with within-subjects designs (e.g., homogeneity of covariance) carry over to mixed designs, and profile analysis is sometimes used to circumvent some of these problems.

2.5 Design Complexity

Discussion of analysis of variance has so far been limited to factorial designs where there are equal numbers of scores in each cell, and levels of each IV are purposely chosen by the researcher. Several deviations from these straightforward designs are possible. A few of the more common types of design complexity are mentioned below, but the reader actually faced with the use of these designs is referred to one of the more complete analysis of variance texts such as by Brown, Michels, and Winer (1991), Keppel and Wickens (2004), Myers and Well (2002), and Tabachnick and Fidell (2007).

2.5.1 Nesting

In between-subjects designs, subjects are said to be nested within levels of the IV. That is, each subject is confined to only one level of each IV or combination of IVs. Nesting also occurs with IVs when levels of one IV are confined to only one level of another IV, rather than factorially crossing over the levels of the other IV.

Take the example where the IV is various levels of teaching methods. Children within the same classroom cannot be randomly assigned to different methods, but whole classrooms can be so assigned. The design is one-way between-subjects where teaching methods is the IV and classrooms serve as subjects. For each classroom, the mean score for all children on the test is obtained, and the means serve as DVs in one-way ANOVA.

If the effect of classroom is also assessed, the design is nested or hierarchical, as shown in Table 8(a). Classrooms are randomly assigned to and nested in teaching methods, and children are nested in classrooms. The error term for the test of classroom is subjects within classrooms and teaching method, and the error term for the test of teaching method is classrooms within teaching technique. Nested models also are analyzed through multilevel modeling.

2.5.2 Latin-Square Designs

The order of presentation of levels of an IV often produces differences in the DV. In within-subjects designs, subjects become practiced or fatigued or experiment wise as they experience more levels of the IV. In between-subjects designs, there are often time-of-day or experimenter effects that change scores on the DV. To get an uncontaminated look at the effects of the IV, it is important to counterbalance the effects of increasing experience, time of day, and the like, so that they are independent of levels of the IV. If the within-subjects IV is something like trials, counterbalancing is not possible because the order of trials cannot be changed. But when the IV is something like background color of slide used to determine if background color affects memory for material on the slide, a Latin-square arrangement is often used to control order effects.

A Latin-square design is shown in Table 8(b). If A_1 is a yellow background, A_2 a blue background, and A_3 a red background, then subjects are presented the slides in the order specified by the Latin square. The first subject is presented the slide with the blue background, then yellow, and

TABLE 8 Some Complex ANOVA Designs

<i>(a) Nested Designs</i>			<i>(b) Latin-Square Designs^a</i>				
Teaching Techniques			Order				
T_1	T_2	T_3		1	2	3	
Classroom 1	Classroom 2	Classroom 3	Subjects	S_1	A_2	A_1	A_3
Classroom 4	Classroom 5	Classroom 6		S_2	A_1	A_3	A_2
Classroom 7	Classroom 8	Classroom 9		S_3	A_3	A_2	A_1

^aWhere the three levels of treatment A are experienced by different subjects in different orders, as indicated.

finally red. The second subject is presented with yellow, then red, then blue, and so on. The yellow slide (A_1) appears once in first position, once in second, and once in third, and so on for the other colors, so that order effects are distributed evenly across the levels of the IV.

The simple design of Table 8(b) produces a test of the IV (A), a test of subjects (if desired), and a test of order. The effect of order (like the effect of subjects) is subtracted out of the error term, leaving it smaller than it is when order effects are not analyzed. The error term itself is composed of fragments of interactions that are not available for analysis because effects are not fully crossed in the design. Thus, the design is more sensitive than a comparable between-subjects design when there are order effects and no interactions and less sensitive when there are no order effects but there are interactions. Refer to Tabachnick and Fidell (2007) or one of the other ANOVA texts for greater detail on this fascinating topic.

2.5.3 Unequal n and Nonorthogonality

In simple one-way between-subjects ANOVA, problems created by unequal group sizes are relatively minor. Computation is slightly more difficult, but that is no real disaster, especially if analysis is by computer. However, as group sizes become more discrepant, the assumption of homogeneity of variance is more important. If the group with the smaller n has the larger variance, the F test is too liberal, leading to increased Type I error rate (inflated alpha level.)

In factorial designs with more than one between-subjects IV, unequal sample sizes in each cell create difficulty in computation and ambiguity of results. With unequal n , a factorial design is nonorthogonal. Hypotheses about main effects and interactions are not independent, and sums of squares are not additive. The various sources of variability contain overlapping variance, and the same variance can be attributed to more than one source. If effects are tested without taking the overlap into account, the probability of a Type I error increases because systematic variance contributes to more than one test. Various strategies are available to deal with the problem, none of them completely satisfactory.

The simplest strategy is to randomly delete cases from cells with greater n until all cells are equal. If unequal n is due to random loss of a few subjects in an experimental design originally set up for equal n , deletion is often a good choice. An alternative strategy with random loss of subjects in an experimental design is an unweighted-means analysis, described in ANOVA textbooks such as Tabachnick and Fidell (2007). The unweighted-means approach has

greater power than random deletion of cases and is the preferred approach as long as computational aids are available.

But in nonexperimental work, unequal n often results from the nature of the population. Differences in sample sizes reflect true differences in numbers of various types of subjects. To artificially equalize n is to distort the differences and lose generalizability. In these situations, decisions are made as to how tests of effects are to be adjusted for overlapping variance.

2.5.4 *Fixed and Random Effects*

In all the ANOVA designs discussed so far, levels of each IV are selected by the researchers on the basis of their interest in testing significance of the IV. This is the usual fixed-effects model. Sometimes, however, there is a desire to generalize to a population of levels of an IV. In order to generalize to the population of levels of the IVs, a number of levels are randomly selected from the population, just as subjects are randomly selected from the population of subjects when the desire is to generalize results to the population of subjects. Consider, for example, an experiment to study effects of word familiarity¹⁰ on recall, where the desire is to generalize results to all levels of word familiarity. A finite set of familiarity levels is randomly selected from the population of familiarity levels. Word familiarity is considered a random-effects IV.

The analysis is set up so that results generalize to levels other than those selected for the experiment—generalize to the population of levels from which the sample was selected. During analysis, alternative error terms for evaluating the statistical significance of random-effects IVs are used. Although computer programs are available for analysis of random-effects IVs, use of them is fairly rare. The interested reader is referred to one of the more sophisticated ANOVA texts, such as Tabachnick and Fidell (2007) or Brown et al. (1991), for a full discussion of the random-effects model.

2.6 Specific Comparisons

When an IV has more than one degree of freedom (more than two levels) or when there is an interaction between two or more IVs, the overall test of the effect is ambiguous. The overall test, with $(k - 1)$ degrees of freedom, is pooled over $(k - 1)$ single-degree-of-freedom subtests. If the overall test is significant, so usually are one or more of the subtests, but there is no way to tell which one(s). To find out which single-degree-of-freedom subtests are significant, comparisons are performed.

In analysis, degrees of freedom are best thought of as a nonrenewable resource. They are analyzed once with conventional alpha levels, but further analyses require very stringent alpha levels. Hence, the best strategy is to plan the analysis very carefully so that the most interesting comparisons are tested with conventional alpha levels. Unexpected findings or less interesting effects are tested later with stringent alpha levels. This is the strategy used by the researcher who has been working in an area for a while and knows precisely what to look for.

Regrettably, research is often more tentative; so the researcher “spends” the degrees of freedom on omnibus (routine) ANOVA testing main effects and interactions at conventional alpha levels

¹⁰Word familiarity is usually operationalized by frequency of usage of words in the English language.

and then snoops the single-degree-of-freedom comparisons of significant effects at stringent alpha levels. Snooping through data after results of ANOVA are known is called *conducting post hoc comparisons*.

We present here the most flexible method of conducting comparisons, with mention of other methods as they are appropriate. The procedure for conducting comparisons is the same for planned and post hoc comparisons up to the point where an obtained F is evaluated against a critical F .

2.6.1 Weighting Coefficients for Comparisons

Comparison of treatment means begins by assigning a weighting factor (w) to each of the cell or marginal means so the weights reflect your null hypotheses. Suppose you have a one-way design with k means and you want to make comparisons. For each comparison, a weight is assigned to each mean. Weights of zero are assigned to means (groups) that are left out of a comparison, although at least two of the means must have nonzero weights. Means that are contrasted with each other are assigned weights with opposite signs (positive or negative) with the constraint that the weights sum to zero, that is,

$$\sum_{j=1}^k w_j = 0$$

For example, consider an IV with four levels, producing $\bar{Y}_1, \bar{Y}_2, \bar{Y}_3$, and $\bar{Y}_4$. If you want to test the hypothesis that $\mu_1 - \mu_3 = 0$, weighting coefficients are 1, 0, -1, 0, producing $1\bar{Y}_1 + 0\bar{Y}_2 + (-1)\bar{Y}_3 + 0\bar{Y}_4$. $\bar{Y}_2$ and $\bar{Y}_4$ are left out while $\bar{Y}_1$ is compared with $\bar{Y}_3$. Or if you want to test the null hypothesis that $(\mu_1 + \mu_2)/2 - \mu_3 = 0$ (to compare the average of means from the first two groups with the mean of the third group leaving out the fourth group), weighting coefficients are 1/2, 1/2, -1, 0 (or any multiple of them, such as 1, 1, -2, 0), respectively. Or if you want to test the null hypothesis that $(\mu_1 + \mu_2)/2 - (\mu_3 + \mu_4)/2 = 0$ (to compare the average mean of the first two groups with the average mean of the last two groups), the weighting coefficients are 1/2, 1/2, -1/2, -1/2 (or 1, 1, -1, -1).

The idea behind the test is that the sum of the weighted means is equal to zero when the null hypothesis is true. The more the sum diverges from zero, the greater the confidence with which the null hypothesis is rejected.

2.6.2 Orthogonality of Weighting Coefficients

In a design with an equal number of cases in each group, any pair of comparisons is orthogonal if the sum of the cross-products of the weights for the two comparisons is equal to zero. For example, in the following three comparisons,

	w_1	w_2	w_3
Comparison 1	1	-1	0
Comparison 2	1/2	1/2	-1
Comparison 3	1	0	-1

the sum of the cross-products of weights for Comparisons 1 2 is

$$(1)(1/2) + (-1)(1/2) + (0)(-1) = 0$$

Therefore, the two comparisons are orthogonal.

Comparison 3, however, is orthogonal to neither of the first two comparisons. For instance, checking it against Comparison 1,

$$(1)(1) + (-1)(0) + (0)(-1) = 1$$

In general, there are as many orthogonal comparisons as there are degrees of freedom. Since $k = 3$ in the example, $df = 2$. There are only two orthogonal comparisons when there are three levels of an IV, and only three orthogonal comparisons when there are four levels of an IV.

There are advantages of using orthogonal comparisons, if they suit the needs of the research. First, there are only as many of them as there are degrees of freedom, so the temptation to “overspend” degrees of freedom is avoided. Second, orthogonal comparisons analyze nonoverlapping variance. If one of them is significant, it has no bearing on the significance of another. Last, because they are independent, if all $(k - 1)$ orthogonal comparisons are performed, the sum of the sum of squares for the comparisons is the same as the sum of squares for the IV in omnibus ANOVA. That is, the sum of squares for the effect has been completely broken down into the $(k - 1)$ orthogonal comparisons that comprise it.

2.6.3 Obtained F for Comparisons

Once the weighting coefficients are chosen, the following equation is used to obtain F for the comparison if sample sizes are equal in each group:

$$F = \frac{n_c (\sum w_j \bar{Y}_j)^2 / \sum w_j^2}{MS_{\text{error}}} \quad (23)$$

where n_c = the number of scores in each of the means to be compared

$(\sum w_j \bar{Y}_j)^2$ = the squared sum of the weighted means

$\sum w_j^2$ = the sum of the squared coefficients

MS_{error} = the mean square for error in the ANOVA

The numerator of Equation 23 is both the sum of squares and the mean square for the comparison because a comparison has only one degree of freedom.

For factorial designs, comparisons are done on either marginal or cell means, corresponding to comparisons on main effects and interactions, respectively. The number of scores per mean and the error term follow from the ANOVA design used. However, if comparisons are made on within-subjects effects, it is customary to develop a separate error term for each comparison, just as separate error terms are developed for omnibus tests of within-subjects IVs.

Once you have obtained F for a comparison, whether by hand calculation or computer, the obtained F is compared with critical F to see if it is statistically reliable.

If obtained F exceeds critical F , the null hypothesis for the comparison is rejected. But which critical F is used depends on whether the comparison is planned or performed post hoc.

2.6.4 Critical F for Planned Comparisons

If you are in the enviable position of having planned your comparisons prior to data collection, and if you have planned no more of them than you have degrees of freedom for effect, critical F is obtained from the tables just as in routine ANOVA. Each comparison is tested against critical F at routine α with one degree of freedom in the numerator and degrees of freedom associated with the MS_{error} in the denominator. If obtained F is larger than critical F , the null hypothesis represented by the weighting coefficients is rejected.

With planned comparisons, omnibus ANOVA is not performed;¹¹ the researcher moves straight to comparisons. Once the degrees of freedom are spent on the planned comparisons, however, it is perfectly acceptable to snoop the data at more stringent α levels (Section 2.6.5), including main effects and interactions from omnibus ANOVA if they are appropriate.

Sometimes, however, the researcher cannot resist the temptation to plan more comparisons than degrees of freedom for effect. When there are too many tests, even if comparisons are planned, the α level across all tests exceeds the α level for any one test and some adjustment of α for each test is needed. It is common practice to use a Bonferroni-type adjustment, where slightly more stringent α levels are used with each test to keep α across all tests at reasonable levels. For instance, when 5 comparisons are planned, if each one of them is tested at $\alpha = .01$ the alpha across all tests is an acceptable .05 (roughly .01 times 5, the number of tests). However, if 5 comparisons are each tested at $\alpha = .05$, the alpha across all tests is approximately .25 (roughly .05 times 5)—unacceptable by most standards.

If you want to keep overall α at, say, .10, and you have 5 tests to perform, you can assign $\alpha = .02$ for each test or you can assign $\alpha = .04$ for two tests with the other three evaluated at $\alpha = .01$ for an overall Type I error rate of roughly .11. The decision about how to apportion α through the tests is also made prior to data collection.

As an aside, it is important to realize that routine ANOVA designs with numerous main effects and interactions suffer from the same problem of inflated Type I error rate across all tests as planned comparisons where there are too many tests. Some adjustment of alpha for separate tests is needed in big ANOVA problems as well, if all effects are evaluated even if the tests are planned.

2.6.5 Critical F for Post Hoc Comparisons

If you are unable to plan your comparisons and choose to start with routine ANOVA instead, you want to follow up significant main effects (with more than two levels) and interactions with post hoc comparisons to find the treatments that are different from one another. Post hoc comparisons are needed to provide adjustment to α level because of two considerations. First, you have already spent

¹¹You might perform routine ANOVA to compute the error term(s).

your degrees of freedom, and your “cheap” α level, conducting routine ANOVA; therefore, you run into rapidly increasing overall error rates if you conduct additional analyses without adjustment. Second, you have already seen the means, and it is much easier to identify comparisons that are likely to be significant. These mean differences, however, may have come from chance fluctuations in the data unless you have some theoretical reason to believe they are real—and if you believe they are real, you should plan to test them ahead of time.

Many procedures for dealing with an inflated Type I error rate are available as described in standard ANOVA texts such as Tabachnick and Fidell (2007). The tests differ in the number and type of comparisons they permit and the amount of adjustment required of α . The tests that permit more numerous comparisons have correspondingly more stringent adjustments to critical F . For instance, the Dunnett test, which compares the mean from a single control group with each of the means of the other groups, in turn, has a less stringent correction than the Tukey test, which allows all pairwise comparisons of means. The name of this game is to choose the most liberal test that permits you to perform the comparisons of interest.

The test described here (Scheffé, 1953) is the most conservative and most flexible of the popular methods. Once critical F is computed with the Scheffé adjustment, there is no limit to the number and complexity of comparisons that can be performed. You can perform all pairwise comparisons and all combinations of treatment means pooled and contrasted with other treatment means, pooled or not, as desired. Some possibilities for pooling are illustrated in Section 2.6.1. Once you pay the “price” in conservatism for this flexibility, you might as well conduct all the comparisons that make sense, given your research design.

The Scheffé method for computing critical F for a comparison on marginal means is

$$F' = (k - 1)F_c \quad (24)$$

where F' is adjusted critical F , $(k - 1)$ is degrees of freedom for the effect, and F_c is tabled F with $(k - 1)$ degrees of freedom in the numerator and degrees of freedom associated with the error term in the denominator.

If obtained F is larger than critical F_s , the null hypothesis represented by the weighting coefficients for the comparison is rejected.

3 Parameter Estimation

If a statistically significant difference among means is found, one is usually interested in reporting the likely population value for each mean. Since sample means are unbiased estimators of population means, the best guess about the size of a population mean (μ) is the mean of the sample randomly selected from that population. In most reports of research, therefore, sample mean values are reported along with statistical results.

Sample means are only estimations of population means. They are unbiased because they are systematically neither large nor small, but they are rarely precisely at the population value—and there is no way to know when they are. Thus, the error in estimation, the familiar confidence interval of introductory statistics, is often reported along with the estimated means. The size of the

confidence interval depends on sample size, the estimation of population variability, and the degree of confidence one wishes to have in estimating μ . Alternatively, cell standard deviations or standard errors are presented along with sample means so that the reader can compute the confidence interval if it is desired.

4 Effect Size¹²

Although significance testing, comparisons, and parameter estimation help illuminate the nature of group differences, they do not assess the degree to which the IV(s) and DV are related. It is important to assess the degree of relationship to avoid publicizing trivial results as though they had practical utility. As discussed in Section 1.2, overly powerful research sometimes produces results that are statistically significant but realistically meaningless.

Effect size reflects the proportion of variance in the DV that is associated with levels of an IV. It assesses the amount of total variance in the DV that is predictable from knowledge of the levels of the IV. If the total variances of the DV and the IV are represented by circles as in a Venn diagram, effect size assesses the degree of overlap of the circles. Statistical significance testing assesses the *reliability* of the association between the IV and the DV. Effect size measures *how much* association there is.

A rough estimate of effect size is available for any ANOVA through η^2 (eta squared).

$$\eta^2 = \frac{SS_{\text{effect}}}{SS_{\text{total}}} \quad (25)$$

When there are two levels of the IV, η^2 is the (squared) point biserial correlation between the continuous variable (the DV) and the dichotomous variable (the two levels of the IV).¹³ After finding a significant main effect or interaction, η^2 shows the proportion of variance in the DV (SS_{total}) attributable to the effect (SS_{effect}). In a balanced, equal- n design, η^2 s are additive; the sum of η^2 for all significant effects is the proportion of variation in the DV that is predictable from knowledge of the IVs.

This simple popular measure of effect size is flawed for two reasons. First, η^2 for a particular IV depends on the number and significance of other IVs in the design. η^2 for an IV tested in a one-way design is likely to be larger than η^2 for the same IV in a two-way design, where the other IV and the interaction increase the size of the total variance, especially if one or both of the additional effects are large. This is because the denominator of η^2 contains systematic variance for other effects in addition to error variance and systematic variance for the effect of interest.

Therefore, an alternative form of η^2 called partial η^2 is available in which the denominator contains only variance attributable to the effect of interest plus error.

$$\text{Partial } \eta^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{error}}} \quad (26)$$

¹²This is also called *strength of association* or *treatment magnitude*.

¹³All effect size values are associated with the particular levels of the IV used in the research and do not generalize to other levels.

With this alternative, η^2 s for all significant effects in the design do not sum to proportion of systematic variance in the DV. Indeed, the sum is sometimes greater than 1.00. It is imperative, therefore, to be clear in your report when this version of η^2 is used.

Second, η^2 describes proportion of systematic variance in a sample with no attempt to estimate proportion of systematic variance in the population. A statistic developed to estimate effect size between IV and DV in the population is $\hat{\omega}^2$ (omega squared).

$$\hat{\omega}^2 = \frac{SS_{\text{effect}} - (df_{\text{effect}})(MS_{\text{error}})}{SS_{\text{total}} + MS_{\text{error}}} \quad (27)$$

This is the additive form of $\hat{\omega}^2$, where the denominator represents total variance, not just variance due to effect plus error, and *is limited to between-subjects analysis of variance designs with equal sample sizes in all cells*. Forms of $\hat{\omega}^2$ are available for designs containing repeated measures (or randomized blocks), as described by Vaughn and Corballis (1969).

A separate measure of effect size is computed and reported for each main effect and interaction of interest in a design. Confidence intervals also may be developed around effect sizes using recent software (Smithson, 2003; Steiger & Fouladi, 1992). Effect sizes described can range from 0 to 1 because they are proportions of variance. Another type of effect size is Cohen's d , basically a difference between standardized means (i.e., means divided by their common standard deviation). That measure becomes less convenient in multivariate designs in which comparisons are more complex than simply the difference between a pair of means. Further, Cohen (1988) shows equations for converting d to η^2 . Therefore, the measures described in this chapter are based on η^2 .

A frequent question is "Do I have (or expect to find) a big effect?" The answer to this question depends on the research area and type of study. Simple experiments typically account for less variance than do nonexperimental studies (nature generally exhibits more control over people than we do in our roles as experimenters). Clinical/personality/social psychology and education tend to have smaller effects than found in sociology, economics, and perception/physiological psychology. Cohen (1988) has presented some guidelines for small ($\eta^2 = .01$), medium ($\eta^2 = .09$), and large ($\eta^2 = .25$) effects. These guidelines apply to experiments and social/clinical areas of psychology; larger values could be expected for nonexperimental research, sociology, and the more physiological aspects of psychology. Ultimately, the size of the effect desired or expected depends on the context of the research: Is it meaningful? Does it matter?

5 Bivariate Statistics: Correlation and Regression

Effect size as described in Section 4 is assessed between a continuous DV and discrete levels of an IV. Frequently, however, a researcher wants to measure the effect size between two continuous variables, where the IV–DV distinction is blurred. For instance, the association between years of education and income is of interest even though neither is manipulated and inferences regarding causality are not possible. Correlation is the measure of the size and direction of the

linear relationship between the two variables, and squared correlation is the measure of strength of association between them.

Correlation is used to measure the association between variables; regression is used to predict one variable from the other (or many others). However, the equations for correlation and bivariate regression are very similar, as indicated in what follows.

5.1 Correlation

The Pearson product-moment correlation coefficient, r , is easily the most frequently used measure of association and the basis of many multivariate calculations. The most interpretable equation for Pearson r is

$$r = \frac{\sum Z_X Z_Y}{N - 1} \quad (28)$$

where Pearson r is the average cross-product of standardized X and Y variable scores.

$$Z_X = \frac{X - \bar{X}}{S} \quad \text{and} \quad Z_Y = \frac{Y - \bar{Y}}{S}$$

and S is as defined in Equations 4 and 6.

Pearson r is independent of scale of measurement (because both X and Y scores are converted to standard scores) and independent of sample size (because of division by N). The value of r ranges between $+1$ and -1 , where values close to $.00$ represent no linear relationship or predictability between the X and Y variables. An r value of $+1.00$ or -1.00 indicates perfect predictability of one score when the other is known. When correlation is perfect, scores for all subjects in the X distribution have the same relative positions as corresponding scores in the Y distribution.¹⁴

The raw score form of Equation 28 also sheds light on the meaning of r :

$$r = \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{[N \sum X^2 - (\sum X)^2][N \sum Y^2 - (\sum Y)^2]}} \quad (29)$$

Pearson r is the covariance between X and Y relative to (the square root of the product of) X and Y variances. Only the numerators of variance and covariance equations appear in Equation 29, because the denominators cancel each other out.

¹⁴When correlation is perfect, $Z_X = Z_Y$ for each pair, and the numerator in Equation 28 is, in effect, $\sum Z_X Z_X$. Because $\sum Z_X^2 = N - 1$, Equation 28 reduces to $(N - 1)/(N - 1)$, or 1.00 .

5.2 Regression

Whereas correlation is used to measure the size and direction of the linear relationship between two variables, regression is used to predict a score on one variable from a score on the other. In a bivariate (two-variable) regression (simple linear regression) where Y is predicted from X , a straight line between the two variables is found. The best-fitting straight line goes through the means of X and Y and minimizes the sum of the squared distances between the data points and the line.

To find the best-fitting straight line, an equation is solved of the form

$$Y' = A + BX \quad (30)$$

where Y' is the predicted score, A is the value of Y when X is 0.00, B is the slope of the line (change in Y divided by change in X), and X is the value from which Y is to be predicted.

The difference between the predicted and the observed values of Y at each value of X represents errors of prediction or residuals. The best-fitting straight line is the line that minimizes the squared errors of prediction.

To solve Equation 30, both B and A are found.

$$B = \frac{N\sum XY - (\sum X)(\sum Y)}{N\sum X^2 - (\sum X)^2} \quad (31)$$

The bivariate regression coefficient, B , is the ratio of the covariance of the variables and the variance of the one from which predictions are made.

Note the differences and similarities between Equation 29 (for correlation) and Equation 31 (for the regression coefficient). Both have the covariance between the variables as a numerator but differ in denominator. In correlation, the variances of both are used in the denominator. In regression, the variance of the predictor variable serves as the denominator; if Y is predicted from X , X variance is the denominator, whereas if X is predicted from Y , Y variance is the denominator. To complete the solution, the value of the intercept, A , is also calculated.

$$A = \bar{Y} - B\bar{X} \quad (32)$$

The intercept is the mean of the observed value of the predicted variable minus the product of the regression coefficient times the mean of the predictor variable.

Figure 3 illustrates many of the relationships among slopes, intercepts, predicted scores, and residuals.

The intercept for the small data set is 2.16; at a value of zero on the X axis, the regression line crosses the Y axis at 2.16. The slope is .60; when the value on the X axis increases by 1 unit, the value on the Y axis goes up .60 units. The equation for the predicted Y score (Equation 30 above) is $Y' = 2.16 + .60X$. For an X score of 4, the predicted Y score is $[2.16 + .60(4) = 4.56]$. This is the value indicated by the up arrow Figure 3. But 5 is the actual Y score associated with an X score of 4 in the data set. The residual, then, for the X score of 4 is the difference between the actual Y score and the predicted Y score ($5 - 4.56 = .44$). The bracket indicates the residual for this value in the data set. Linear regression minimizes the squared residuals across the whole data set.

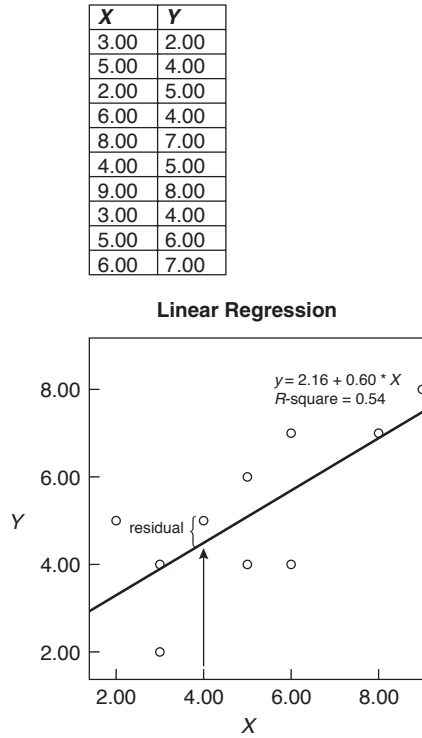


FIGURE 3 Slopes, intercepts, predicted scores, and residuals in bivariate regression; figure created using IBM SPSS graphs.

6 Chi-Square Analysis

Analysis of variance examines the relationship between a discrete variable (the IV) and a continuous variable (the DV); correlation and regression examine the relationship between two continuous variables; and the chi-square (χ^2) test of independence is used to examine the relationship between two discrete variables. If, for instance, one wants to examine a potential relationship between region of the country (Northeast, Southeast, Midwest, South, and West) and approval versus disapproval of current political leadership, χ^2 is the appropriate analysis.

In χ^2 analysis, the null hypothesis generates expected frequencies against which observed frequencies are tested. If the observed frequencies are similar to the expected frequencies, then the value of χ^2 is small and the null hypothesis is retained; if they are sufficiently different, then the value of χ^2 is large and the null hypothesis is rejected. The relationship between the size of χ^2 and the difference in observed and expected frequencies can be seen readily from the following computational equation for χ^2 :

$$\sum_{ij} \frac{(f_o - F_e)^2}{F_e} \quad (33)$$

where f_o represents observed frequencies and F_e represents the expected frequencies in each cell. Summation is over all the cells in a two-way table.

Usually, the expected frequencies for a cell are generated from the sum of its row and the sum of the column.

$$\text{Cell } F_e = \frac{(\text{row sum})(\text{column sum})}{N} \quad (34)$$

When this procedure is used to generate the expected frequencies, the null hypothesis tested is that the variable on the row (say, region of the country) is independent of the variable on the column (attitude toward current political leadership). If the fit to the observed frequencies is good (so that χ^2 is small), then one concludes that the two variables are independent; a poor fit leads to a large χ^2 , rejection of the null hypothesis, and the conclusion that the two variables are related.

The techniques discussed in this chapter for making decisions about differences, estimating population means, assessing association between two variables, and predicting a score on one variable from a score on another are important to, and widely used in, the social and behavioral sciences. They form the basis for most undergraduate—and some graduate—statistics courses. It is hoped that this brief review reminds you of material already mastered, so that, with common background and language, we begin in earnest the study of multivariate statistics.

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Cleaning Up Your Act

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Cleaning Up Your Act

Screening Data Prior to Analysis

This chapter deals with a set of issues that are resolved after data are collected but before the main data analysis is run. Careful consideration of these issues is time-consuming and sometimes tedious; it is common, for instance, to spend many days in careful examination of data prior to running the main analysis that, itself, takes about 5 min. But consideration and resolution of these issues before the main analysis are fundamental to an honest analysis of the data.

The first issues concern the accuracy with which data have been entered into the data file and consideration of factors that could produce distorted correlations. Second, missing data—the bane of (almost) every researcher—are assessed and dealt with. Next, many multivariate procedures are based on assumptions; the fit between your data set and the assumptions is assessed before the procedure is applied. Transformations of variables to bring them into compliance with the requirements of analysis are considered. Outliers—cases that are extreme (outlandish)—create other headaches because solutions are unduly influenced and sometimes distorted by them. Last, perfect or near-perfect correlations among variables can threaten a multivariate analysis.

This chapter deals with issues that are relevant to most analyses. However, the issues are not all applicable to all analyses all the time; for instance, multiway frequency analysis and logistic regression—procedures that use log-linear techniques—have far fewer assumptions than the other techniques. For these reasons, assumptions and limitations specific to each analysis are reviewed in the third section of the chapter describing that analysis.

There are differences in data screening for grouped and ungrouped data. If you are performing multiple regression, canonical correlation, factor analysis, or structural equation modeling, where subjects are not subdivided into groups, there is one procedure for screening data. If you are performing analysis of covariance, multivariate analysis of variance or covariance, profile analysis, discriminant analysis, or multilevel modeling where subjects are in groups, there is another procedure for screening data. Differences in these procedures are illustrated by example in Section 2. Other analyses (survival analysis and time-series analysis) sometimes have grouped data and often do not, so screening is adjusted accordingly.

You may want to read this chapter to get an overview of the tasks to be accomplished prior to the main data analysis.

1 Important Issues in Data Screening

1.1 Accuracy of Data File

The best way to ensure the accuracy of a data file is to proofread the original data against the computerized data file in the data window. In SAS, data are most easily viewed in the Interactive Data Analysis window. With a small data file, proofreading is highly recommended, but with a large data file, it may not be possible. In this case, screening for accuracy involves examination of descriptive statistics and graphic representations of the variables.

The first step with a large data set is to examine univariate descriptive statistics through one of the descriptive programs such as IBM SPSS FREQUENCIES, or SAS MEANS or UNIVARIATE or Interactive Data Analysis. For continuous variables, are all the values within range? Are means and standard deviations plausible? If you have discrete variables (such as categories of religious affiliation), are there any out-of-range numbers? Have you accurately programmed your codes for missing values?

1.2 Honest Correlations

Most multivariate procedures analyze patterns of correlation (or covariance) among variables. It is important that the correlations, whether between two continuous variables or between a dichotomous and continuous variable, be as accurate as possible. Under some rather common research conditions, correlations are larger or smaller than they should be.

1.2.1 *Inflated Correlation*

When composite variables are constructed from several individual items by pooling responses to individual items, correlations are inflated if some items are reused. Scales on personality inventories, measures of socioeconomic status, health indices, and many other variables in social and behavioral sciences are often composites of several items. If composite variables are used and they contain, in part, the same items, correlations are inflated; do not overinterpret a high correlation between two measures composed, in part, of the same items. If there is enough overlap, consider using only one of the composite variables in the analysis.

1.2.2 *Deflated Correlation*

Sample correlations may be lower than population correlations when there is restricted range in sampling of cases or very uneven splits in the categories of dichotomous variables.¹ Problems with distributions that lead to lower correlations are discussed in Section 1.5.

A falsely small correlation between two continuous variables is obtained if the range of responses to one or both of the variables is restricted in the sample. Correlation is a measure of the extent to which scores on two variables go up together (positive correlation) or one goes up while the other goes down (negative correlation). If the range of scores on one of the variables is narrow due to

¹A very small coefficient of determination (standard deviation/mean) was also associated with lower correlations when computers had less computational accuracy. However, computational accuracy is so high in modern statistical packages that the problem is unlikely to occur, unless, perhaps, to astronomers.

restricted sampling, then it is effectively a constant and cannot correlate highly with another variable. In a study of success in graduate school, for instance, quantitative ability could not emerge as highly correlated with other variables if all students had about the same high scores in quantitative skills.

When a correlation is too small because of restricted range in sampling, you can estimate its magnitude in a nonrestricted sample by using Equation 1 if you can estimate the standard deviation in the nonrestricted sample. The standard deviation in the nonrestricted sample is estimated from prior data or from knowledge of the population distribution.

$$\tilde{r}_{xy} = \frac{r_{t(xy)}[S_x/S_{t(x)}]}{\sqrt{1 + r_{t(xy)}^2[S_x/S_{t(x)}] - r_{t(xy)}^2}} \quad (1)$$

where $r_{t(xy)}$ = adjusted correlation
 $r_{t(xy)}^2$ = correlation between X and Y with the range of X truncated
 S_x = unrestricted standard deviation of X
 $S_{t(x)}$ = truncated standard deviation of X

Many programs allow analysis of a correlation matrix instead of raw data. The estimated correlation is inserted in place of the truncated correlation prior to analysis of the correlation matrix. (However, insertion of estimated correlations may create internal inconsistencies in the correlation matrix, as discussed in Section 1.3.3.)

The correlation between a continuous variable and a dichotomous variable, or between two dichotomous variables (unless they have the same peculiar splits), is also too low if most (say, over 90%) responses to the dichotomous variable fall into one category. Even if the continuous and dichotomous variables are strongly related in the population, the highest correlation that could be obtained is well below 1. Some recommend dividing the obtained (but deflated) correlation by the maximum it could achieve given the split between the categories and then using the resulting value in subsequent analyses. This procedure is attractive, but not without hazard, as discussed by Comrey and Lee (1992).

1.3 Missing Data

Missing data is one of the most pervasive problems in data analysis. The problem occurs when rats die, equipment malfunctions, respondents become recalcitrant, or somebody goofs. Its seriousness depends on the pattern of missing data, how much is missing, and why it is missing. Summaries of issues surrounding missing data are provided by Schafer and Graham (2002) and by Graham, Cumsille, and Elek-Fisk (2003).

The pattern of missing data is more important than the amount missing. Missing values scattered randomly through a data matrix pose less serious problems. Nonrandomly missing values, on the other hand, are serious no matter how few of them there are because they affect the generalizability of results. Suppose that in a questionnaire with both attitudinal and demographic questions, several respondents refuse to answer questions about income. It is likely that refusal to answer questions about income is related to attitude. If respondents with missing data on income are deleted, the sample values on the attitude variables are distorted. Some method of estimating income is needed to retain the cases for analysis of attitude.

Missing data are characterized as MCAR (missing completely at random), MAR (missing at random, called ignorable nonresponse), and MNAR (missing not at random or nonignorable).

The distribution of missing data is unpredictable in MCAR, the best of all possible worlds if data must be missing. The pattern of missing data is predictable from other variables in the data set when data are MAR. In NMAR, the missingness is related to the variable itself and, therefore, cannot be ignored.

If only a few data points—say, 5% or less—are missing in a random pattern from a large data set, the problems are less serious and almost any procedure for handling missing values yields similar results. If, however, a lot of data are missing from a small to moderately sized data set, the problems can be very serious. Unfortunately, there are as yet no firm guidelines for how much missing data can be tolerated for a sample of a given size.

Although the temptation to assume that data are missing randomly is nearly overwhelming, the safest thing to do is to test it. Use the information you have to test for patterns in missing data. For instance, construct a dummy variable with two groups, cases with missing and nonmissing values on income, and perform a test of mean differences in attitude between the groups. If there are no differences, decisions about how to handle missing data are not so critical (except, of course, for inferences about income). If there are significant differences and η^2 is substantial, care is needed to preserve the cases with missing values for other analyses, as discussed in Section 1.3.2.

IBM SPSS MVA (Missing Values Analysis) is specifically designed to highlight patterns of missing values as well as to replace them in the data set. Table 1 shows syntax and output for a data set with missing values on ATTHOUSE and INCOME. A TTEST is requested to see if missingness is related to any of other variables, with $\alpha = .05$ and tests done only for variables with at least 5 PERCENT of data missing. The EM syntax requests a table of correlations and a test of whether data are missing completely at random (MCAR).

The Univariate Statistics table shows that there is one missing value on ATTHOUSE and 26 missing values on INCOME. Separate Variance *t* Tests show no systematic relationship between missingness on INCOME and any of the other variables. ATTHOUSE is not tested because fewer than 5% of the cases have missing values. The Missing Patterns table shows that Case number 52, among others, is missing INCOME, indicated by an S in the table. Case number 253 is missing ATTHOUSE. The last table shows EM Correlations with missing values filled in using the EM method, to be discussed. Below the table is Little's MCAR test of whether the data are missing completely at random. A statistically nonsignificant result is desired: $p = .76$ indicates that the probability that the pattern of missing diverges from randomness is greater than .05, so that MCAR may be inferred.

MAR can be inferred if the MCAR test is statistically significant but missingness is predictable from variables (other than the DV) as indicated by the Separate Variance *t* Tests. MNAR is inferred if the *t* test shows that missingness is related to the DV.

The decision about how to handle missing data is important. At best, the decision is among several bad alternatives, several of which are discussed in the subsections that follow.

1.3.1 *Deleting Cases or Variables*

One procedure for handling missing values is simply to drop any cases with them. If only a few cases have missing data and they seem to be a random subsample of the whole sample, deletion is a good alternative. Deletion of cases with missing values is the default option for most programs in the IBM SPSS and SAS packages.²

²Because this is the default option, numerous cases can be deleted without the researcher's knowledge. For this reason, it is important to check the number of cases in your analyses to make sure that all of the desired cases are used.

TABLE 1 IBM SPSS MVA Syntax and Output for Missing Data

MVA VARIABLES=timedrs attdrug atthouse income emplmnt mstatus race
 /TTEST NO PROB PERCENT=5
 /MPATTERN
 /EM(TOLERANCE=0.001 CONVERGENCE=0.0001 ITERATIONS=25).

Univariate Statistics

	N	Mean	Std. Deviation	Missing		No. of Extremes ^{a,b}	
				Count	Percent	Low	High
TIMEDRS	465	7.90	10.948	0	.0	0	34
ATTDUG	465	7.69	1.156	0	.0	0	0
ATTHOUSE	464	23.54	4.484	1	.2	4	0
INCOME	439	4.21	2.419	26	5.6	0	0
EMPLMNT	465	.47	.500	0	.0	0	0
MSTATUS	465	1.78	.416	0	.0	.	.
RACE	465	1.09	.284	0	.0	.	.

a. Indicates that the interquartile range (IQR) is zero.

b. Number of cases outside the range ($Q1 - 1.5 \cdot IQR$, $Q3 + 1.5 \cdot IQR$)

Separate Variance t Tests^a

		TIMEDRS	ATTDUG	ATTHOUSE	INCOME	MSTATUS	RACE	EMPLMNT
INCOME	t	.2	-1.1	-.2	.	-1.0	-.4	-1.1
	df	32.2	29.6	28.6	.	29.0	27.3	28.0
	# Present	439	439	438	439	439	439	439
	# Missing	26	26	26	0	26	26	26
	Mean (Present)	7.92	7.67	23.53	4.21	1.77	1.09	.46
	Mean (Missing)	7.62	7.88	23.69	.	1.85	1.12	.58

For each quantitative variable, pairs of groups are formed by indicator variables (present, missing).

a. Indicator variables with less than 5% missing are not displayed.

TABLE 1 Continued

Missing Patterns (cases with missing values)									
Case	# Missing	% Missing	Missing and Extreme Value Patterns ^a						
			TIMEDRS	ATTDRUG	EMPLMNT	MSTATUS	RACE	ATTHOUSE	INCOME
52	1	14.3				-	-		S
64	1	14.3	+			-	-		S
69	1	14.3				-	-		S
77	1	14.3				-	-		S
118	1	14.3				-	-		S
135	1	14.3				-	-		S
161	1	14.3				-	-		S
172	1	14.3				-	-		S
173	1	14.3				-	-		S
174	1	14.3				-	-		S
181	1	14.3				-	-		S
196	1	14.3				-	+		S
203	1	14.3	+			-	-		S
236	1	14.3				-	-		S
240	1	14.3				-	-		S
258	1	14.3				-	+		S
304	1	14.3				-	-		S
321	1	14.3				-	-		S
325	1	14.3				-	-		S
352	1	14.3				-	-		S
378	1	14.3				-	-		S
379	1	14.3				-	-		S
409	1	14.3	+			-	-		S
419	1	14.3				-	-		S
421	1	14.3				-	-		S
435	1	14.3				-	+	-	S
253	1	14.3				-	+	S	

- Indicates an extreme low value, and + indicates an extreme high value. The range used is (Q1 - 1.5*IQR, Q3 + 1.5*IQR).

a.Cases and variables are sorted on missing patterns.

(continued)

TABLE 1 Continued

EM Correlations ^a							
	timeds	attdrug	athouse	income	emplmnt	mstatus	race
timeds	1						
attdrug	.104	1					
athouse	.128	.023	1				
income	.050	-.005	.002	1			
emplmnt	.059	.085	-.023	-.006	1		
mstatus	-.065	-.006	-.030	-.466	.234	1	
race	-.035	.019	-.038	.105	-.081	-.035	1

a. Little's MCAR test: Chi-Square = 19.550, DF = 12, Sig. = 0.76.

If missing values are concentrated in a few variables and those variables are not critical to the analysis, or are highly correlated with other, complete variables, the variable(s) with missing values are profitably dropped.

But if missing values are scattered throughout cases and variables, deletion of cases can mean substantial loss of subjects. This is particularly serious when data are grouped in an experimental design because loss of even one case requires adjustment for unequal *n*. Further, the researcher who has expended considerable time and energy collecting data is not likely to be eager to toss some out. And as previously noted, if cases with missing values are not randomly distributed through the data, distortions of the sample occur if they are deleted.

1.3.2 Estimating Missing Data

A second option is to estimate (impute) missing values and then use the estimates during data analysis. There are several popular schemes for doing so: using prior knowledge; inserting mean values; using regression; expectation-maximization; and multiple imputation.

Prior knowledge is used when a researcher replaces a missing value with a value from an educated guess. If the researcher has been working in an area for a while, and if the sample is large and the number of missing values small, this is often a reasonable procedure. The researcher is often confident that the value would have been about at the median, or whatever, for a particular case. Alternatively, the researcher can downgrade a continuous variable to a dichotomous variable (e.g., "high" vs. "low") to predict with confidence into which category a case with a missing value falls. The discrete variable replaces the continuous variable in the analysis, but it has less information than the continuous variable. An option with longitudinal data is to apply the last observed value to fill in data missing at a later point in time. However, this requires the expectation that there are no changes over time.

Remaining options for imputing missing data are available through software. Table 2 shows programs that implement missing data procedures.

TABLE 2 Missing Data Options Available in Some Computer Programs

Strategy		Program						
		SPSS MVA	SOLAS MDA	SPSS REGRESSION	NORM	SAS STANDARD	SAS MI and MIANALYZE	AMOS, EQS, and LISREL
Mean substi- tution	Grand mean	No	Group Means ^a	MEAN SUBSTITUTION	No	REPLACE	No	No
	Group mean	No	Group Means	No	No	No	No	No
Regression		Regression	No	No	No	No	No	No
Expectation Maximization (EM/FIML)		EM	No	No	Yes ^c	No	PROC MI with NIMPUTE =	Yes
Multiple imputation		No ^b	Multiple Imputation	No	Yes	No	Yes	No

^aWith omission of group identification.

^bMay be done by generating multiple files through the EM method and computing additional statistics.

^cFor preparation of data prior to multiple imputation; provides missing values for one random imputation.

Mean substitution has been a popular way to estimate missing values, although it is less commonly used now that more desirable methods are feasible through computer programs. Means are calculated from available data and are used to replace missing values prior to analysis. In the absence of all other information, the mean is the best guess about the value of a variable. Part of the attraction of this procedure is that it is conservative; the mean for the distribution as a whole does not change and the researcher is not required to guess at missing values. On the other hand, the variance of a variable is reduced because the mean is closer to itself than to the missing value it replaces, and the correlation the variable has with other variables is reduced because of the reduction in variance. The extent of loss in variance depends on the amount of missing data and on the actual values that are missing.

A compromise is to insert a group mean for the missing value. If, for instance, the case with a missing value is a Republican, the mean value for Republicans is computed and inserted in place of the missing value. This procedure is not as conservative as inserting overall mean values and not as liberal as using prior knowledge. However, the reduction in within-group variance can make differences among groups spuriously large.

Many programs have provisions for inserting mean values. SAS STANDARD allows a data set to be created with missing values replaced by the mean on the variable for complete cases. SOLAS MDA, a program devoted to missing data analysis, produces data sets in which group means are used to replace missing values. IBM SPSS REGRESSION permits MEANSUBSTITUTION. And, of course, transformation instructions can be used with any program to replace any defined value of a variable (including a missing code) with the mean.

Regression is a more sophisticated method for estimating missing values. Other variables are used as IVs to write a regression equation for the variable with missing data

serving as DV. Cases with complete data generate the regression equation; the equation is then used to predict missing values for incomplete cases. Sometimes, the predicted values from a first round of regression are inserted for missing values and then all the cases are used in a second regression. The predicted values for the variable with missing data from round two are used to develop a third equation, and so forth, until the predicted values from one step to the next are similar (they converge). The predictions from the last round are the ones used to replace missing values.

An advantage of regression is that it is more objective than the researcher's guess but not as blind as simply inserting the grand mean. One disadvantage to the use of regression is that the scores fit together better than they should; because the missing value is predicted from other variables, it is likely to be more consistent with them than a real score is. A second disadvantage is reduced variance because the estimate is probably too close to the mean. A third disadvantage is the requirement that good IVs be available in the data set; if none of the other variables is a good predictor of the one with missing data, the estimate from regression is about the same as simply inserting the mean. Finally, estimates from regression are used only if the estimated value falls within the range of values for complete cases; out-of-range estimates are not acceptable. Using regression to estimate missing values is convenient in IBM SPSS MVA. The program also permits adjustment of the imputed values so that overconsistency is lessened.

Expectation maximization (EM) methods are available for randomly missing data. EM forms a missing data correlation (or covariance) matrix by assuming the shape of a distribution (such as normal) for the partially missing data and basing inferences about missing values on the likelihood under that distribution. It is an iterative procedure with two steps—expectation and maximization—for each iteration. First, the E step finds the conditional expectation of the “missing” data, given the observed values and current estimate of the parameters, such as correlations. These expectations are then substituted for the missing data. Second, the M step performs maximum likelihood estimation (often referred to as FIML—full information maximum likelihood) as though the missing data had been filled in. Finally, after convergence is achieved, the EM variance–covariance matrix may be provided and/or the filled-in data saved in the data set.

However, as pointed out by Graham et al. (2003), analysis of an EM-imputed data set is biased because error is not added to the imputed data set. Thus, analyses based on this data set have inappropriate standard errors for testing hypotheses. The bias is greatest when the data set with imputed values filled in is analyzed, but bias exists even when a variance–covariance or correlation matrix is used as input. Nevertheless, these imputed data sets can be useful for evaluating assumptions and for exploratory analyses that do not employ inferential statistics. Analysis of EM-imputed data sets can also provide insights when amounts of missing data are small if inferential statistics are interpreted with caution.

IBM SPSS MVA performs EM to generate a data set with imputed values as well as variance–covariance and correlation matrices, and permits specification of some distributions other than normal. IBM SPSS MVA also is extremely helpful for assessing patterns of missing data, providing *t* tests to predict missingness from other variables in the data set, and testing for MCAR, as seen in Table 1.

Structural equations modeling (SEM) programs typically have their own built-in imputation procedures which are based on EM. The programs do not produce data sets with imputed values but utilize appropriate standard errors in their analyses.

SAS, NORM, and SOLAS MDA can be used to create an EM-imputed data set by running the MI (multiple imputation, see below) procedure with *m* (number of imputations) = 1, but analysis